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SGCP's

Guru Nanak Khalsa College
of Arts, Science & Commerce (Autonomous)

DEPARTMENT OF BIOINFORMATICS

CERTIFICATE

This is to certify that Ms. Prarthi Hrishit Kothari (Roll No: 115) of M. Sc. Bioinformatics (Part II) has satisfactorily completed the practical for Mandatory Paper I: Mass Spectrometry, Clinical Trial & Drug Discovery (GNKPSBIMJ1P504) for Semester IV course prescribed by the University of Mumbai during the academic year 2024-2025.

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PRACTICAL NO. 1
DEMONSTRATION OF LC-MS

AIM:

To demonstrate the working and interpretation of LC-MS (Liquid Chromatography Mass Spectrometry)

THEORY:

The Liquid Chromatography (LC) the separation of components can be done and then the sample eluents from LC are transferred into Mass Spectrometry (MS). The detection, identification, and determination of masses of components can be done in the presence of other components. LC-MS is used in the determination of pharmaceutical drug substances, intermediates, and their related compounds for quantitative and qualitative purposes. LC-MS is used most significantly in in-vitro dissolution, bioequivalence, bioavailability, and metabolite studies. Also, LC-MS is used in basic research, agrochemicals, forensic laboratories, and food industries.

INTRODUCTION:

LC-MS is an analytical technique that involves the physical separation of target compounds (or analytes) followed by their mass-based detection. Liquid Chromatography/Mass Spectrometry (LC/MS) is a powerful analytical technique that combines the resolving power of liquid chromatography with the detection specificity of mass spectrometry. Liquid chromatography (LC) separates the sample components and introduces them to the mass spectrometer (MS). The MS creates and detects charged ions. The LC/MS data may be used to provide information about the molecular weight, structure, identity, and quantity of specific sample components. LC and LC/MS-based methods can be applied to most organic compounds. Sample types range from small pharmaceutical compounds to large proteins. Because it is a much more widely applicable method than GC/MS, LC/MS is suitable for analyzing large, polar, ionic, thermally unstable and involatile compounds. Some of these compounds can be made amenable to GC/MS by derivatization, but LC/MS eliminates the need for time-consuming chemical modifications. This permits MS analysis of non-volatile, thermally labile, or charged molecules.

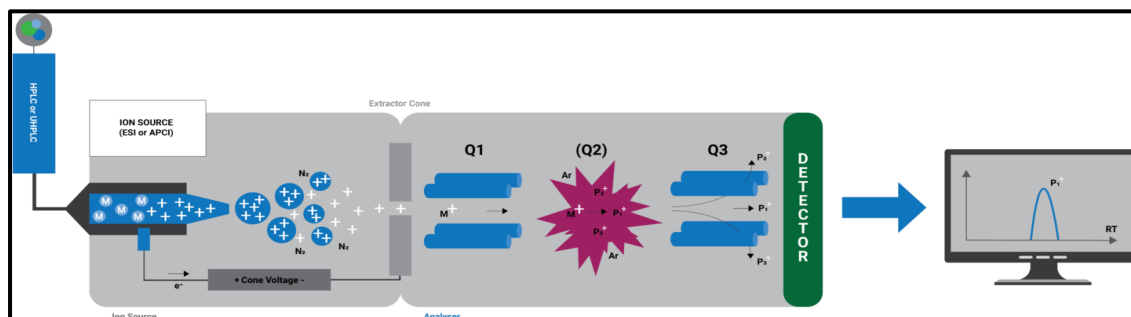


Fig 1: The working of Mass Spectrometry

PRINCIPLE:

Liquid chromatography-mass spectrometry (LC-MS) stands as a cornerstone in modern analytical chemistry, offering unparalleled capabilities in the separation, identification, and quantification of chemical compounds across scientific disciplines. The brilliance of LC-MS lies in its amalgamation of two potent techniques: liquid chromatography (LC) and mass spectrometry (MS), each bringing its own set of strengths to the analytical table.

Beginning with liquid chromatography, this facet of the technique operates on the fundamental principle of selective partitioning. Within a chromatographic system, the sample of interest is introduced into a column where it encounters a stationary phase. This stationary phase, typically comprised of a finely packed material, interacts with the sample components as they traverse through the column under the influence of a mobile phase, usually a solvent. This interaction between the sample constituents and the stationary phase that governs their respective retention times and ultimately dictates the degree of separation achieved. Compounds with higher affinity for the stationary phase will experience reduced mobility and, thus, longer retention times, whereas those with lesser affinity will elute more swiftly.

Mass spectrometry, in its essence, serves as a molecular fingerprinting tool. Once the compounds have been separated via liquid chromatography, they are directed into the mass spectrometer. They undergo ionization process wherein they are bombarded with high-energy particles, leading to the generation of ions. These ions are then subjected to electromagnetic fields, causing them to traverse a specified path based on their mass-to-charge ratio (m/z). A mass spectrum is generated by precisely measuring the m/z ratios and intensities of these ions, revealing crucial information about the molecular composition and structure of the compounds.

INSTRUMENTATION:

The combination of liquid chromatography and mass spectrometry thus forms a symbiotic relationship, wherein the separation capabilities of LC are seamlessly integrated with the molecular characterization prowess of MS. This synergy enables the elucidation of complex mixtures with unprecedented precision and accuracy. From pharmaceutical analysis to environmental monitoring, LC-MS has become indispensable in the arsenal of analytical chemists, driving advancements in fields ranging from drug discovery to metabolomics. Through its ability to unravel the intricacies of chemical compositions, LC-MS continues to propel scientific inquiry forward, unlocking new frontiers in our understanding of the molecular world.

The basic components of a LC unit consist of:

1. Pump – delivers the mobile phase at a required flow rate,
2. Autosampler – injects the samples,
3. Column – for separation of sample,
4. Detector – for the analysis of the separated components in a sample.

For a LCMS system the instrumentation comprises of:

1. A LC unit
2. An interface between the LC and MS
3. An ion source that ionizes samples (e.g. API unit)
4. An ion guide (an electrostatic lens that efficiently introduces the generated ions into the MS)
5. A mass analyzer unit that separates the ions based on their mass-to-charge (m/z)
6. A detector unit that detects the separated ions.

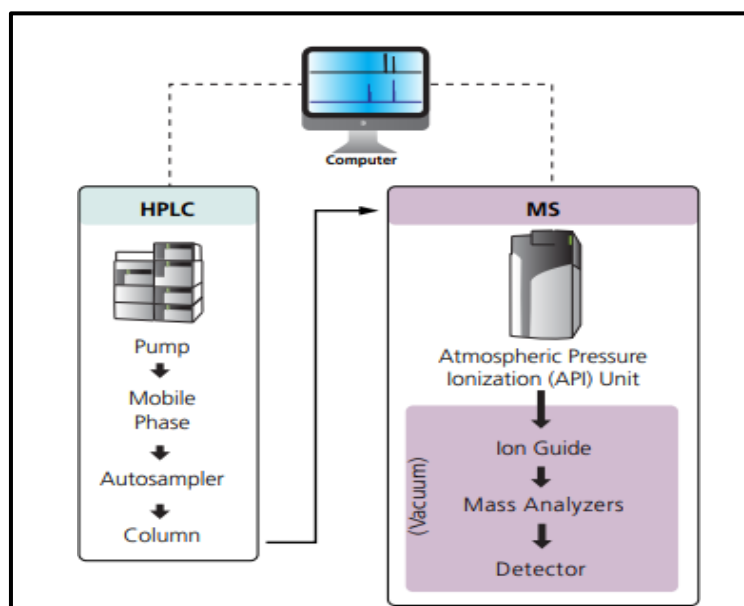


Fig 2: Basic instrumentation of LC-MS

There are two main parts of an LC-MS system: the ionization source and the interfaces. The many parts of a mass spectrometer are detailed below:

- Ionization Sources and Interfaces
- Mass Analyzers

a. Ionization Sources and Interfaces: Liquid chromatography is a technique used to separate liquid components, typically consisting of methanol, acetonitrile, and water. The mixture of components in liquid form is introduced into the ion source of the mass spectrometer. The ion source operates under high vacuum conditions. Vaporizing liquid drops while preserving the mixture of components is challenging due to pressure differentials. Interfaces are utilized to address this issue. Various interface types utilized in mass spectrometry are outlined below.

i. Direct liquid Introduction (DLI):

DLI ionization is achieved through chemical ionization and reagent gas vaporization of the solvent. Both solvent systems, normal and reverse phase, have been utilized. Methanol/water and acetonitrile/water (up to 60% water) are commonly employed as reverse-phase solvents. Buffers containing salts are generally not recommended due to the risk of capillary plugging during heating. DLI operates through the combined effects of thermal energy and liquid flow rate. The flow rate of the liquid entering the interface is restricted. The analyte ions generated by thermal energy are subsequently conveyed to the ion source via a capillary inlet or pinhole diaphragm.

ii. Atmospheric-Pressure Ionization (API):

Nebulization, evaporation, and ionization comprise atmospheric-pressure ionization (API). When a solvent containing a sample is fed through a thin capillary tube and nebulized in a large chamber, atmospheric pressure ionization (API) produces a mist of minute droplets. Ionization occurs and certain droplets have a positive or negative electric charge. The solvent evaporates in massive heating chambers. Droplets shrink as the solvent evaporates. Ions and molecules

collide. Ions entered the mass analyzer via capillary. The Atmospheric-Pressure Ionization (API) technique is utilized for a diverse array of moderate molecular weight analytes, both polar and non-polar in nature.

iii. Electrospray Ionization (ESI):

Electrospray Ionization (ESI) involves passing the liquid sample through a stainless steel capillary tube held at a high positive or negative electric potential of approximately 3-5kV. Charged droplets are formed at the capillary tip and subsequently undergo vaporization. The evaporation of the solvent from droplets results in a reduction in their size and an increase in their surface charge. The collision ceases upon the conversion of highly charged droplets into gas phase ions. The gas-phase ions traverse the capillary sampling orifice and enter the ion source's low-pressure region. The main benefit of ESI is that the ions are multiply charged, increasing by 1 to 3 for molecules 1000 Da or over 50000 Da. The m/z ratio is always below 2000. ESI-LC-MS measures the molecular weight of peptides, proteins, biological samples, polymers, nucleotides, sugars, and organometallics. Biomedical research and analysis also use it.

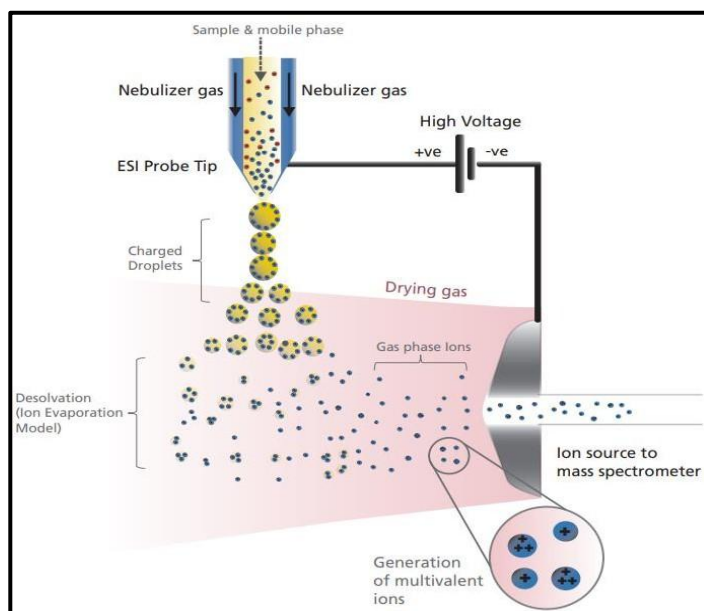


Fig 3: Schematic of ionization and desolvation process in ESI positive (+) mode

iv. Atmospheric Pressure Chemical Ionization (APCI):

APCI involves two steps: analyte evaporation/desolvation and charged transfer reactions in the vapor phase to produce vapor phase ions. The Atmospheric Pressure Chemical Ionization (APCI) method involves nebulizing a liquid sample-containing solvent through a narrow capillary tube into a large chamber. Solvent evaporation occurs at atmospheric pressure in a sizable heating chamber, forming small droplets. Ionization occurs. Ionization typically occurs at temperatures ranging from 250 to 400 °C. The ions transfer charges to molecules via chemical reactions. The ions are passed through a capillary orifice in the mass analyzer. This technique is commonly employed for moderate molecular weight analytes with low polarity or non-polarity.

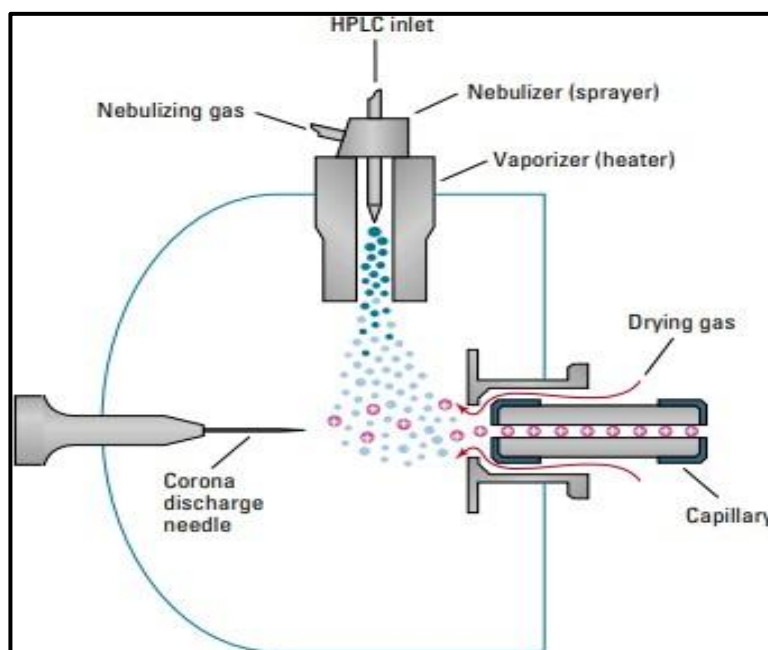


Fig 4: Schematic of the ion-molecular reaction in APCI

v. Thermo spray and Plasma spray Ionization(TSPI):

Thermo spray involves heating a capillary tube through which a liquid sample solution is passed, resulting in solvent evaporation. Charged droplets are created. The droplets decrease in size as a result of solvent evaporation. The surface charge density of droplets increases. The ions are subsequently introduced into a mass analyzer equipped with an electrostatic voltage system. Plasma spray does not produce ions, although corona discharge or plasma can enhance ions in thermal spray. Electric discharge ionizes neutral substances more. Enhanced molecular ionization. Clinical and medical analysis uses plasma spray because it is more sensitive.

vi. Atmospheric pressure photo Ionization (APPI):

APPI utilizes photons to ionize and excite molecules. APPI involves two primary stages: exciting and ionizing the analyte from the eluent. APPI uses LC eluent vaporization like APCI. APPI uses a Kr lamp for photon production. Kr lamp excites and ionizes molecules with high- energy photons. Energy range minimizes analyte ionization. Ionized analytes go into a mass analyzer. This methodology is advantageous for analytes with non-polar characteristics, making it challenging to ionize through Electrospray Ionization (ESI) and Atmospheric Pressure Chemical Ionization (APCI) techniques.

vii. Particle Beam Ionization:

Browner et al. have devised a particle beam interface that enables the separation of solvents from solutes with minimal solute loss. The nebulization and evaporation processes bear resemblance to Thermo spray (TSP), Atmospheric pressure chemical ionization (APCI), and Electrospray ionization (ESI). The eluent obtained from HPLC or LC is passed through a narrow tube for separation. Helium gas injection generates high-velocity liquid droplet spray. The nebulizer's liquid droplets undergo evaporation in the heating chamber, reducing their size. The liquid droplets exit the heating chamber as a particle beam. The beam subsequently traverses an ionization chamber, analogous to Electro Spray Ionization (ESI) and Atmospheric Pressure Chemical Ionization (APCI).

viii. Continuous Flow Fast Atom Bombardment (FAB):

FAB is a technique for interface detection that is both simple and highly sensitive. Fast atoms, such as Argon (Ar) or Xenon, are used to bombard the target in FAB liquid. The sample is dissolved in glycerol and applied onto a thin metal plate or probe. The probe is introduced into the mass spectrometer, where ionization of the samples occurs upon bombardment by a beam of fast-moving atoms. The resulting ions are subsequently analyzed based on their mass-to-charge ratio (m/z). FAB is employed for the analysis of bulky and thermally labile molecules. It is utilized for surfactants and proteins.

- b. Mass Analyzer:** Following ionization, the ions are directed to a mass analyzer, where they are separated based on their m/z ratio. Mass analyzers are typically evaluated based on speed, time, rate, and reaction. The major components of the mass analyzer are discussed below:

1. Quadrupole mass analyzer:

It is the most popular and practical mass analyzer available today. It has two rows of parallel rods between an ion source and a detector. The mass analyzer can separate ions in space or time based on their m/z . The linear Quadrupole mass Analyzer has four parallel hyperbolic or cylindrical rods in a radial array. Opposite rods are charged with a DC potential and an oscillating RF voltage. DC and RF applied to rods stabilize ions of one m/z , which are transmitted to the detector. The rods discharge unstable m/z ions. Low-accelerating potential introduces ions to Quadrupole. As they trend through the Quadrupole filter, ions oscillate perpendicular to rod length. Applying DC and RF voltage at a constant ratio drives m/z -carrying ions toward the detector. DC/RF ratio determines resolution. The Quadrupole typically scans at 1000 m/z at <4000 m/z . Mass accuracy rarely exceeds 0.1 m/z due to unit mass resolution. RF values are usually 1-2 MHz. 1000V DC and 6000V RF are possible.

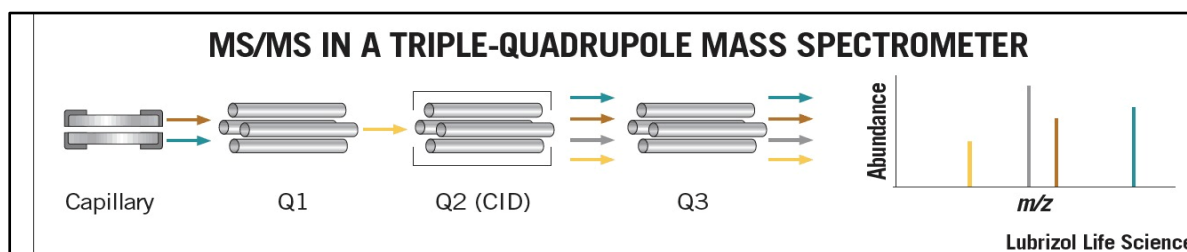


Fig 5: Triple Quadrupole Mass Analyzer

2. Time of flight analyzer:

Time of flight is a versatile method for various ion sources and inlet systems. The system lacks a magnetic field and requires only basic electrostatic maintenance and calibration. The ions are extracted and accelerated by a voltage. The duration of travel along the drift or flight path is contingent on the ion's mass and charge. The time of flight for single-charged ions ($z=1$, $m/z=w$) is directly proportional to their mass. Lighter ions will reach the detector before heavier ions as they move towards it. Simultaneous detection of all ions is achieved through scanning. The mass range scanning is fast and suitable for high m/z values.

3. Ion trap mass analyzer:

The Ion Trap Mass Analyzer exhibits high resolution, sensitivity, and the ability to perform multiple product ion scans. The Quadrupole ion trap is a 3D ion trap. The apparatus comprises a cylindrical ring electrode subjected to a Quadrupole field. Two additional types of electrodes are end-capped electrodes. One electrode of the end cap features a solitary central aperture for the introduction of electrons or ions into the trap, while the other electrode has multiple apertures or holes for the passage of ions to a detector. The ion trajectories are stabilized by the presence of Helium bath gas in the trap. The collision occurs between ions and a helium bathgas. The motion of ions enhances the trapping efficiency of the analyzer. The mass spectrum is generated by selectively releasing ions from the trap based on their mass-to-charge (m/z) ratios.

4. Fourier Transfer Ion Cyclotron Resonance (FT-ICR):

The Fourier transform ion cyclotron resonance (FT-ICR) is a highly significant mass analyzer. The ions generated by the ionization source are subsequently introduced into the mass analyzer, where they undergo separation based on their respective mass-to-charge ratios (m/z). The ions introduced into the chamber are confined within circular trajectories. The acceleration of ions is facilitated by the combined influence of electric and magnetic fields. As a result of this phenomenon, the ions become stimulated and produce a current that varies over time. The ions are segregated based on their mass-to-charge (m/z) ratios while being trapped.

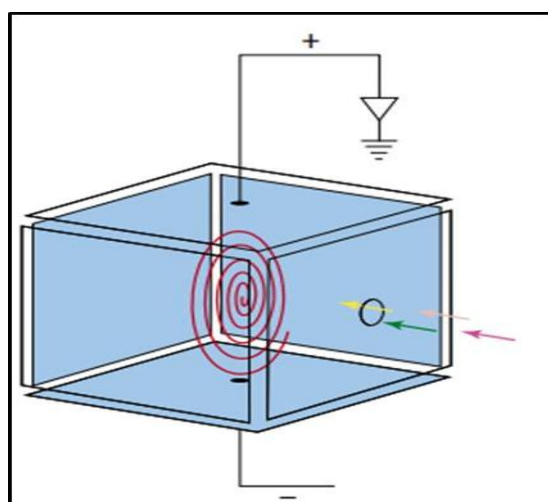


Fig 6: Fourier Transfer Ion Cyclotron Resonance

c. Detectors:

The detector is a crucial component of the mass spectrometer, as it generates an electrical signal directly proportional to the quantity of ions that impact it. After the ions have been passed through the analyzer, it is necessary to detect them and convert them into a signal.

1. Point Ion Collectors Detector:

The ion collectors are situated at specific points within the mass spectrometer. All ions are directed toward a singular detector location. The detection of ions may be captured by measuring electric current and subsequently documented for analysis.

2. Array detector:

An array detector is a planar arrangement of point collectors. An array detector detects the ions at a specific location or across a plane. The ions are separated based on their mass-to-charge ratio (m/z) and detected using a planar ion collector.

- d. Scan and SIM:** Scan and SIM Mass spectrometers can be operated in either a scan mode or a selected ion monitoring (SIM) mode.

1. Scan Mode:

In the scan mode, the instrument detects signals over a mass range (e.g. from 50–2000 m/z) during a short period (e.g. 2 sec). During this scan period, the MS electronics sequentially read the signals detected within narrower mass intervals until the complete mass range is covered. The spectra that are stored represent the detected signal for the full mass range. Since full mass spectra are recorded, this mode of operation is typically selected for qualitative analysis or quantitation when all analyte masses are not known in advance. Samples may be introduced into a mass spectrometer by infusion or through an HPLC. In the latter, it is important to match the peak width and the scan range. The narrower the peaks, the shorter the total scan time must be in order to get proper peak definition. To get a short total scan time, it may be necessary to reduce the scan range.

2. SIM Mode:

Mass spectrometers can also operate in the selected ion monitoring (SIM) mode. Rather than scanning continuously, they can be set to only monitor a few mass-to-charge ratios (m/z). As a result, the quadrupole can spend significantly more time sampling each of the m/z values, with a concomitant and large increase in sensitivity. Moreover, because the cycle time between data points is often shorter than in scan mode, quantitative precision and accuracy are improved through optimal peak-shape profiling. Since the m/z values to be sampled must be set in advance, SIM is most often used for target compound analysis. For analyses of multiple target compounds, SIM ion sampling choices can be time programmed to match compound elution time windows. No data is collected at m/z values other than those specifically sampled, so SIM is rarely used in qualitative analysis.

The general methodology for LC-MS:

1. Extract the analytes of interest from the sample matrix using suitable extraction techniques such as solid-phase extraction (SPE), liquid-liquid extraction (LLE), or protein precipitation.
2. Remove impurities and unwanted matrix components to enhance the selectivity and sensitivity of the analysis. This may involve additional sample clean-up steps such as filtration or solid phase microextraction.
3. After preparation: The samples are separated and concentrated in a liquid chromatography instrument (HPLC).
4. In the mass spectrometer's ion source, molecules become charged ions. Different techniques for ionization have been developed based on the type of samples the MS will detect. The most common are:
 - a) Electrospray ionization (ESI)
 - b) Atmospheric Pressure Chemical Ionization (APCI) Source
5. In the mass spectrometer, the quadrupoles filter separates ions based on their mass-to-charge ratios (m/z). Within the triple quadrupole configuration, the first quadrupole isolates analyte molecules, the second quadrupole fragments select molecular ions, and the third quadrupole selectively isolates the ions for measurement via a detector.
6. Ions and ion fragments reach the electrostatic detector and create a current that dedicated software translates into a mass spectrum.
7. The mass spectrum plots the different mass-to-charge ratios (m/z) against their abundances (occurrence of a specific ion divided by the occurrence of the most plentiful ion) within the sample.

APPLICATIONS:

1. **Pharmaceutical Analysis:** LC-MS is extensively used in pharmaceutical research and development for drug discovery, quality control, and pharmacokinetic studies due to its ability to separate, identify, and quantify drug compounds with high sensitivity and accuracy.
2. **Environmental Monitoring:** LC-MS is employed in environmental analysis to detect and quantify pollutants, pesticides, and other contaminants in air, water, soil, and biological samples, aiding in regulatory compliance and environmental protection efforts.
3. **Clinical Diagnostics:** LC-MS plays a crucial role in clinical laboratories for the analysis of biomarkers, metabolites, and therapeutic drug monitoring, facilitating disease diagnosis, treatment optimization, and patient care management.
4. **Food and Beverage Testing:** LC-MS is utilized in food safety and quality control to detect residues of pesticides, veterinary drugs, mycotoxins, and other contaminants, ensuring the safety and integrity of food products consumed by the public.
5. **Metabolomics and Proteomics:** LC-MS is instrumental in metabolomics and proteomics research, enabling comprehensive analysis of metabolites and proteins in biological samples to elucidate disease mechanisms, biomarker discovery, and personalized medicine applications.

LIMITATIONS:

1. Matrix co-extractants can suppress or enhance ionization potential. It may need to employ matrix matched calibrants for quantitation.
2. Without the use of stable-labeled internal standards, the reproducibility is often not as good as LC-UV or other LC detectors.
3. Limited use for large molecules ($> 4,000$ m/z)

OBSERVATIONS:

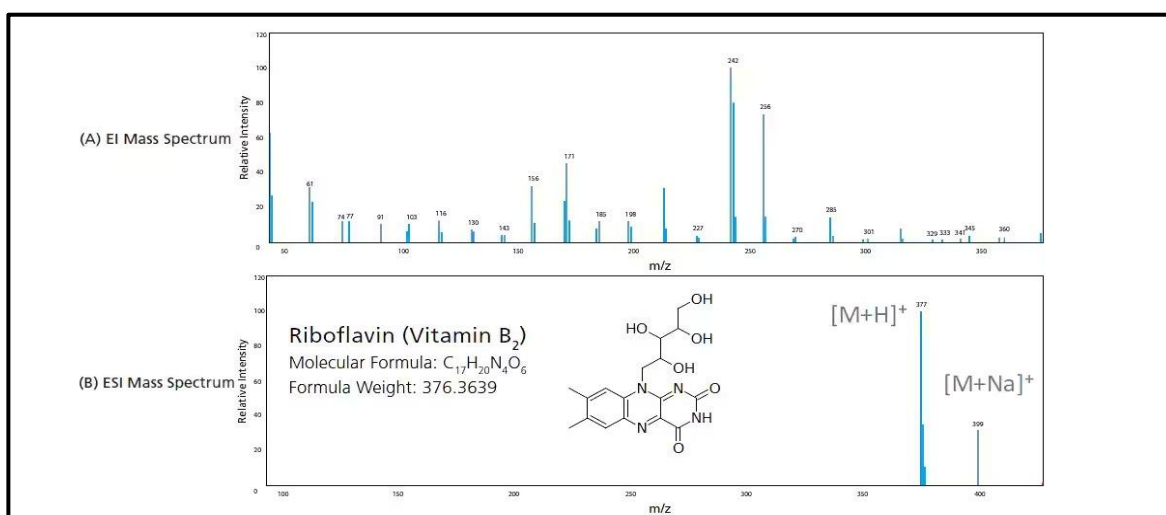


Fig 7: Mass spectra produced (A) EI (hard ionization) and (B) ESI (soft ionization)

CONCLUSION:

The LC-MS is a hyphenated technique used in combination with the separation power of HPLC with the detection power of mass spectrometry. LC-MS enables detailed analysis of complex mixtures providing invaluable insights into the composition and structure of compounds. By identifying the base peak of the total ion content, LC-MS offers precise and accurate information about the relative abundance of ions, enhancing the understanding of sample composition. Its versatility makes LC-MS an essential tool in various fields, including biochemistry, pharmaceuticals and environmental analysis.

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Date: 24/02/2025

PRACTICAL NO: 2

Clinical Trial Simulation: The Effect of "FocusBoost" on Reaction Time

Objective:

To study the effect of "FocusBoost," on reaction time in participants.

1. Inclusion and Exclusion Criteria

Inclusion Criteria:

- Age between 18 and 25 years.
- No known neurological or cognitive impairments.
- No prior consumption of stimulants (e.g., energy drinks) 2 hours before the test.
- Willing to provide informed consent and participate in the full trial duration.

Exclusion Criteria:

- History of epilepsy, severe anxiety, or other conditions affecting reaction time.
- Current use of medications that may alter reaction speed (e.g., sedatives, stimulants).
- Participants with vision or motor impairments that could affect test results.
- Allergy to the study material (candies used as placebo and FocusBoost).

2. Informed Consent Form

Title: Participation in the "FocusBoost" Clinical Trial Simulation

Introduction: You are invited to participate in a classroom clinical trial simulation designed to study the effects of "FocusBoost" on reaction time. Participation is voluntary, and you may withdraw at any time.

Purpose of the Study: The goal of this study is to determine whether "FocusBoost" has an effect on participants' reaction time compared to a placebo.

Procedures:

- You will be randomly assigned to receive either "FocusBoost" or a placebo.
- You will complete a reaction time test before and after taking the assigned treatment.
- Your results will be recorded and analyzed as part of the study.
- Some participants may experience mild side effects (e.g., dizziness, nausea) as part of the simulated adverse event reporting.

Risks & Benefits:

- There are no real risks, as this is a simulated trial.
- You may learn about clinical trials and data collection.
- Your participation helps in understanding the placebo effect and reaction time variability.

Confidentiality: Your personal data will remain anonymous. Results will be used for educational purposes only.

Voluntary Participation: Participation is entirely voluntary. You may stop at any time without any consequences.

Consent Statement: I have read and understood the information above. I voluntarily agree to participate in this simulation.

Participant Name: _____ **Participant Signature:** _____

Date: _____

Participant Demographics & Randomization Sheet

Participant ID	Age	Gender	Group (A/B)	Notes
P001	22	Male	B	Alert and responsive; met all inclusion criteria.
P002	21	Female	A	No caffeine intake reported in past 2 hours.
P003	23	Male	B	Slightly anxious but fit to proceed.
P004	22	Female	A	Calm and cooperative; no issues reported.
P005	23	Male	B	Energetic mood; eager to participate.
P006	22	Female	A	Slight fatigue reported; otherwise, eligible.
P007	20	Male	B	Compliant with pre-test instructions; appeared focused.
P008	22	Female	A	No stimulant use; ready for baseline test.
P009	22	Male	A	Quiet but attentive; fit for participation.

2. Baseline Reaction Time Test

Participant ID	Reaction Test Type	Baseline Time (seconds) - B1		
		T ₁	T ₂	T ₃
P001	FocusBoost	1.41	1.43	1.41
P002	Placebo	1.30	1.38	1.37
P003	FocusBoost	1.61	1.65	1.58
P004	Placebo	1.20	1.26	1.26
P005	FocusBoost	0.67	0.27	0.09
P006	Placebo	0.10	0.18	0.11
P007	FocusBoost	0.90	0.12	0.14
P008	Placebo	1.21	1.47	1.41
P009	Placebo	1.62	1.61	1.66

3. Post-Treatment Reaction Time Test

Participant ID	Reaction Test Type	Post Reaction Time (seconds)		
		T ₁	T ₂	T ₃
P001	FocusBoost	0.15	0.16	0.11
P002	Placebo	-	0.12	0.16
P003	FocusBoost	0.15	0.16	0.17
P004	Placebo	0.10	0.11	0.12
P005	FocusBoost	0.20	0.23	0.22
P006	Placebo	0.16	0.16	0.17
P007	FocusBoost	0.09	0.09	0.10
P008	Placebo	0.12	0.09	0.10
P009	Placebo	0.14	0.15	0.10

4. Adverse Events Reporting Sheet

Participant ID	Adverse Event	Severity (Mild/Moderate/Severe)	Notes
P001	Itchy Throat	Mild	Reported slight throat irritation post-treatment; did not affect testing.
P002	No Adverse Events	N/A	No symptoms observed or reported post-treatment.
P003	Nausea	Mild	Mild queasiness after dosage; participant completed test comfortably.
P004	No Adverse Events	N/A	No symptoms observed or reported post-treatment.
P005	Giggles	Severe	Uncontrollable laughter disrupted focus post testing.
P006	Concentration Lapse	Moderate	Participant experienced loss of focus and was unable to catch the ball, resulting in incomplete test.
P007	No Adverse Events	N/A	Participant reported feeling well and experienced no adverse effects.
P008	No Adverse Events	N/A	Participant reported feeling well and experienced no adverse effects.
P009	Dizziness	Mild	Felt lightheaded briefly after treatment; no interruption to testing.

5. Data Analysis & Summary:

Data Cleaning: P002 is missing post-treatment T_1 value ("-"). Since, 2/3 values are present, we calculated the mean using just those.

Calculating the Mean-Difference:

1. Mean= $(x_1+x_2+x_3+\dots+x_n)/\text{number of values}$
2. Mean Difference: Pre-treatment Mean–Post-treatment Mean

Summary of each participant's average pre- and post-treatment reaction times, along with the calculated average improvement, i.e., mean difference.

Participant ID	Group	Pre-Test Avg (s)	Post-Test Avg (s)	Mean Difference (s)
P001	B	1.417	0.140	1.277
P002	A	1.350	0.133	1.217
P003	B	1.613	0.160	1.453
P004	A	1.240	0.110	1.130
P005	B	0.343	0.217	0.127
P006	A	0.130	0.163	-0.033
P007	B	0.387	0.093	0.293
P008	A	1.363	0.103	1.260
P009	A	1.630	0.130	1.500

Calculating the Standard Deviation:

Summary of the mean and standard deviation (SD) of reaction times between Group A (Placebo) and Group B (FocusBoost).

Group	Pre-Mean	Pre SD	Post Mean	Post SD	Diff Mean	Diff SD	Improvement? Yes/No
A	1.143	0.584	0.128	0.024	1.015	0.602	Yes
B	0.94	0.669	0.152	0.051	0.788	0.674	Yes

OBSERVATIONS:

1. Both groups improved significantly, as shown by large mean differences (> 0.75 s).
2. Surprisingly, Group A (Placebo) showed a slightly greater average improvement than Group B.
3. However, standard deviations are high, especially for Pre-treatment times, which shows a lot of variability in initial performance.

Graphical Analysis:

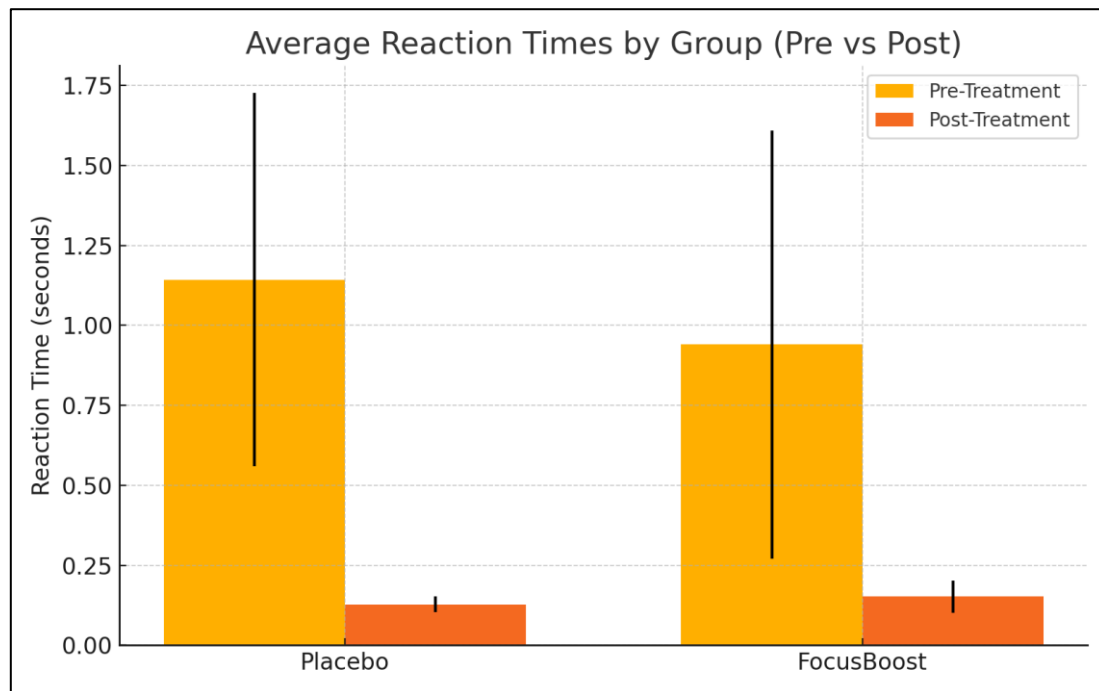


Figure 1: Average reaction times (with standard deviation) before and after treatment for both groups.

The bar graph shows a clear reduction in average reaction time for both the Placebo and FocusBoost groups. Group A (Placebo) improved from 1.143 seconds to 0.128 seconds, while Group B (FocusBoost) improved from 0.94 seconds to 0.152 seconds.

The Placebo group showed slightly more improvement, suggesting the possibility of a placebo effect. The changes in both groups may also be influenced by practice or learning effects due to repeated testing.

Testing Hypothesis (T-Test):

To determine whether FocusBoost had a statistically significant effect compared to the Placebo group, an independent two-sample t-test was performed using the mean differences in reaction time.

Null Hypothesis (H_0): There is no difference in the mean improvement in reaction time between the FocusBoost and Placebo groups.

Alternative Hypothesis (H_1): There is a significant difference in the mean improvement in reaction time between the FocusBoost and Placebo groups.

CALCULATION:

Group A (Placebo):

1. Sample Size (n_1) = 5
2. Mean Difference (X_1) = 1.015 seconds

3. Standard Deviation (S_1) = 0.602

Group B (FocusBoost)

1. Sample Size (n_2) = 4
2. Mean Difference($\bar{X}_2$) = 0.788 seconds
3. Standard Deviation (S_2) = 0.674

Using the formula for Welch's t-test (for unequal variances): $t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

$$t = \frac{1.015 - 0.788}{\sqrt{\frac{0.602^2}{5} + \frac{0.674^2}{4}}} \approx 0.527$$

T-TEST RESULTS:

1. T-statistic: 0.527
2. P-value: 0.617

INTERPRETATION:

Since the p-value is greater than 0.05, we fail to reject the null hypothesis. This indicates that there is no statistically significant difference in reaction time improvement between the two groups.

6. Discussion:

1. **Was there a significant improvement in reaction time or was there a real effect?**

Yes, individually both groups showed improvement in reaction time as reflected in the average differences. However, the statistical test (p-value = 0.617) indicates that this difference is not statistically significant. Therefore, although improvement occurred, it cannot be confidently attributed to FocusBoost. The changes may result from factors unrelated to the supplement, such as practice effects, environmental influence, or psychological expectations.

2. **Could there be a Placebo Effect?**

Yes, the Placebo group's improvement suggests a strong placebo effect. Participants may have believed the placebo was effective, leading to improved performance through psychological mechanisms.

3. **Did any participants experience a placebo effect?**

Yes. The Placebo group had a strong average improvement (mean = 1.015s), which was even higher than that of the FocusBoost group (0.788 seconds). This suggests that participants who believed they were taking an active substance may have experienced psychological

improvements in focus and performance. This classic placebo effect is important to account for in clinical trial analysis.

Participant ID	Group	Pre Avg (s)	Post Avg (s)	Improvement (s)
P002	Placebo	1.350	0.133	1.217
P008	Placebo	1.363	0.103	1.260

Both participants showed over 1-second improvement without receiving FocusBoost, indicating a strong placebo effect likely due to expectation, motivation, or test repetition.

4. What other factors might have influenced results?

Several confounding variables may have affected the outcome:

- Practice/Learning Effect: Performing the test a second time often results in naturally faster responses.
- Fatigue/Distraction: Participant P006, for instance, could not complete the post-test due to a concentration lapse.
- Random Variation: Reaction times had large pre-treatment variability.
- Small Sample Size: With only 9 participants, even large differences in means may not reach statistical significance.

5. Some participants randomly experience "side effects" (e.g., dizziness, giggles, "enhanced superpowers"). Discuss how trials monitor and report side effects.

AE stands for Adverse event; It is an unfavorable change or any untoward medical occurrence (including a symptom/ disease or an abnormal laboratory finding) in the health of a participant, during a clinical study or within a certain amount of time after the study has ended. This change may or may not be caused by the intervention/treatment being studied.

Participants experienced simulated side effects such as itchy throat, nausea, dizziness, giggles, and concentration lapse. These were recorded in an Adverse Events Reporting Sheet. Each report included participant ID, type of side effect, severity, and notes. For instance, P005 experienced uncontrollable giggles that interfered with testing, and P006 was unable to catch the ball. This system mimics real clinical trials where monitoring and documentation of adverse effects are critical for participant safety and data interpretation.

In reality, trials use a structured system for monitoring side effects, typically involving:

- Observation during and after treatment
- Self-reporting by participants
- Regular check-ins or questionnaires
- Designated staff to record and assess the impact of any symptoms

These entries help identify patterns of concern, determine if the side effect is likely related to the treatment and evaluate whether the trial should be modified, paused, or stopped.

Adverse Event Reporting is important because it ensures transparency in trial results, protects participants from harm, helps determine treatment safety and tolerability and supports regulatory review and ethical compliance

6. What were the limitations of this trial?

- a) Small sample size (n=9), reducing statistical power.
- b) Lack of blinding; participants may have guessed what they received.
- c) The trial was simulated in a classroom, which may not reflect real-world testing rigor.
- d) No control for the practice effect.
- e) Large variability in baseline measurements.

7. Were there any unexpected trends in the data?

Yes. Surprisingly, the Placebo group had a higher average improvement than the FocusBoost group. Participant P006 worsened post-treatment due to distraction, and P005's unusual baseline and side effect (giggles) disrupted normal performance. These anomalies suggest that individual variability and side effects significantly influence test outcomes.

PRACTICAL NO: 3

CLINICAL DATA MANAGEMENT CASE STUDY

Introduction:

Clinical Data Management (CDM) plays a pivotal role in the successful execution of clinical trials. It involves the collection, validation, cleaning, and analysis of clinical trial data. In this case study, we explore the challenges and strategies involved in managing data for a clinical trial of a new investigational drug designed to treat hypertension.

Background:

- **Study Overview:** The trial aims to evaluate the safety and efficacy of Drug Bioinformine, a novel antihypertensive medication. The study is a multi-center, Phase III randomized controlled trial (RCT) with 500 participants.
 - **Data Types:** The data collected includes patient demographics, medical history, adverse events, vital signs, laboratory results, and efficacy measures like blood pressure readings.
 - **Data Management Team:** The team includes Clinical Data Managers, Statisticians, Data Entry Clerks, Clinical Research Coordinators, and Medical Monitors.
-

Objectives:

1. To assess the process of data collection and validation.
 2. To highlight the data management challenges in a multi-center trial.
 3. To discuss strategies for data cleaning and discrepancy resolution.
 4. To understand the role of clinical data management software in the trial.
-

Case Study Scenario:

1. Data Collection:

- **Patient Enrollment:** 500 patients are enrolled across 10 sites. Each site is responsible for entering the data into the Electronic Data Capture (EDC) system.
- **Data Capture:** Data points such as demographic details, baseline measurements, and medical history are captured at baseline. Ongoing visits collect data on vital signs, adverse events, and lab results.

2. Data Cleaning and Validation:

- **Initial Validation:** Upon data entry, the EDC system performs automated checks for range errors (e.g., a blood pressure reading of 200/150 mmHg) and missing data fields.
- **Discrepancy Management:** Clinical Data Managers regularly review the data, identify discrepancies (e.g., inconsistent lab results), and generate queries. They send these queries to the clinical research coordinators at the respective sites for clarification.

3. Data Integration:

- **Merge Data from Multiple Sites:** The data collected from all sites must be combined into a single database. This requires careful mapping of variables (e.g., consistent units of measurement, patient identifiers).
- **Third-Party Data:** The trial collects external data such as ECG reports from a third-party provider. Data integration from this provider requires coordination with the external company and matching data formats.

4. Statistical Analysis:

- **Clean Dataset:** Once the data is cleaned and validated, it is prepared for statistical analysis. The dataset is handed over to the biostatistics team.
- **Analysis Plan:** A statistical analysis plan (SAP) is developed, specifying how to assess the primary and secondary endpoints of the study. The analysis is done using software like SAS or R.

5. Reporting and Final Submission:

- **Final Report Preparation:** After the analysis, a clinical study report (CSR) is created, summarizing the findings. The report includes the results of efficacy and safety analysis, along with any adverse events.
 - **Regulatory Submission:** The final cleaned data and reports are submitted to regulatory bodies like the FDA or EMA for review.
-

Key Challenges:

1. **Data Consistency:** With multiple sites involved, maintaining consistent data quality can be challenging. Variations in how data is recorded and entered can lead to discrepancies.
 2. **Missing Data:** Incomplete or missing data for certain patients or visits can complicate data analysis. Handling this involves methods like imputation or sensitivity analysis.
 3. **Timeliness:** Ensuring that data is cleaned, validated, and ready for statistical analysis within tight timelines is crucial, especially when multiple sites are involved.
 4. **Compliance with Regulations:** The data management process must comply with regulatory guidelines such as GxP (Good Clinical Practice) and 21 CFR Part 11 (electronic records and signatures).
 5. **Data Security:** Protecting patient data privacy and security, especially when handling large volumes of sensitive information.
-

Solutions Implemented:

1. **Automated Data Entry and Validation:** Implementation of an EDC system with automated checks helped minimize errors during data entry and reduce manual review workload.
 2. **Centralized Data Monitoring:** A centralized system allowed for real-time monitoring of data quality across all sites, enabling faster identification and resolution of discrepancies.
 3. **Data Imputation:** Missing data was handled through imputation strategies (e.g., last observation carried forward, or LOCF) to minimize biases in the analysis.
 4. **Training and SOPs:** Regular training sessions and standardized operating procedures (SOPs) ensured that all sites adhered to consistent data management practices.
-

Conclusion:

Clinical data management is a critical part of any clinical trial. Proper planning, rigorous data validation, and timely issue resolution ensure that the data collected is reliable and usable for making important decisions. By addressing challenges such as data consistency, missing data, and regulatory compliance, CDM professionals help ensure that clinical trials provide credible and accurate results.

Materials and Methods:

1. Study Design and Participants

Study Design:

- The study is a **multi-center, Phase III randomized controlled trial (RCT)** conducted across **10 clinical sites**. The trial will evaluate the safety and efficacy of **Drug X** in patients with hypertension.
- The study duration is **12 months**, with participant follow-ups occurring every 3 months.

Inclusion Criteria:

- Adults aged 18-70 years.
- Diagnosed with moderate to severe hypertension (systolic blood pressure >160 mmHg).
- Willing to provide informed consent.

Exclusion Criteria:

- Pregnant or breastfeeding women.
- Patients with significant comorbidities (e.g., renal failure, cardiovascular disease).

Study Population:

- A total of **500 participants** will be enrolled across the 10 sites.
- **Randomization:** Participants will be randomly assigned to either the treatment group (Drug Bioinfomine) or placebo group, in a 1:1 ratio.

2. Data Collection Tools and Software

Electronic Data Capture (EDC) System:

- Data from all study sites will be collected using an **Electronic Data Capture (EDC) system**.
- The EDC system selected for this study is **Medidata Rave**, a commonly used platform in clinical trials for data entry, validation, and storage.
- The system provides real-time data access for the monitoring and management team, ensuring the data's integrity and accuracy.

Case Report Forms (CRFs):

- CRFs will be designed for data entry on the EDC platform, including sections for:
 - Demographics (age, gender, ethnicity, etc.)
 - Medical History
 - Baseline and follow-up measurements (e.g., blood pressure readings)
 - Laboratory Test Results (e.g., serum creatinine, electrolytes)
 - Adverse Events (AEs) and Serious Adverse Events (SAEs)
 - Concomitant Medications
 - Efficacy Endpoints (e.g., reduction in blood pressure)

Data Management Software:

- **SAS (Statistical Analysis System)** will be used for statistical analysis, allowing the biostatistics team to generate outputs for final reporting.
 - **Excel** will also be used for internal data validation, discrepancy management, and intermediate data processing.
-

3. Data Collection and Monitoring

Data Collection Process:

- **At Enrollment (Visit 1):**
 - Participants will undergo baseline assessments, including demographic information, medical history, and baseline vital signs (blood pressure, heart rate).
 - Laboratory samples (blood, urine) will be collected for analysis.
- **Follow-up Visits (Visit 2, Visit 3, etc.):**
 - Participants will return every 3 months for follow-up visits. At each visit, blood pressure will be measured and laboratory tests will be repeated.
 - Any **adverse events** or changes in health status will be recorded.
 - Data will be entered into the EDC system by Clinical Research Coordinators (CRCs) at each site immediately following each participant's visit.

Site Monitoring:

- The clinical trial will include regular site visits by a **Clinical Research Associate (CRA)** to ensure proper implementation of protocols.
- The CRA will verify the accuracy of data entry, conduct data reconciliation, and ensure patient safety through adverse event documentation.

Data Validation:

- **Automated Validation:** The EDC system will automatically flag discrepancies (e.g., out-of-range values such as excessively high blood pressure readings) and missing data.
 - **Manual Validation:** Clinical Data Managers (CDMs) will perform manual checks, including:
 - **Cross-checking of data** (e.g., lab results against source documents).
 - **Query Generation:** Any discrepancies will trigger queries that are sent to the respective sites for clarification.
-

4. Data Cleaning and Discrepancy Management

Data Cleaning Process:

- **Discrepancy Detection:** After data collection, the CDM team will perform rigorous data cleaning:
 - Identification of missing data, inconsistent data entries, or entries that don't meet predefined ranges (e.g., abnormal lab values).
 - Duplicate data points will be flagged for review.
- **Data Queries:** For any discrepancies found, the CDM team will generate queries to the site coordinators, asking for confirmation or correction. Common examples include:
 - Query for missing blood pressure readings.
 - Query for unclear adverse event severity reporting.
- **Data Imputation:** If the missing data cannot be resolved, imputation techniques such as **Last Observation Carried Forward (LOCF)** will be applied for continuous variables (e.g., blood pressure). For categorical data, **multiple imputation** may be used to address missing responses.

Data Monitoring and Interim Analysis:

- An **Interim Data Monitoring Committee (DMC)** will review the data periodically (e.g., after every 100 enrolled participants) to ensure patient safety and assess the progress of the trial.
 - The DMC may recommend changes to the study design or early termination based on safety or efficacy concerns.
-

5. Statistical Analysis

Analysis Plan:

- **Primary Endpoint:** The primary endpoint of the trial is the **change in systolic blood pressure** from baseline to 12 months.
- **Secondary Endpoints:** Secondary endpoints include changes in diastolic blood pressure, the incidence of adverse events, and laboratory abnormalities.

Statistical Methods:

- Descriptive statistics will be used to summarize baseline characteristics, including means, standard deviations, and frequencies for continuous and categorical variables.
- **Analysis of Variance (ANOVA)** will be applied to compare the blood pressure reduction between the treatment group and placebo group.
- **Kaplan-Meier survival analysis** will be conducted for time-to-event data such as the time to the first adverse event.
- **Logistic regression** will be applied to assess the risk factors associated with adverse events or severe hypertension.

Software Used:

- **SAS** software will be used for all statistical analyses, ensuring that the results are reliable and meet regulatory standards.
-

6. Data Security and Regulatory Compliance

Data Security:

- The EDC system is compliant with **21 CFR Part 11** (Electronic Records and Signatures), ensuring that all data entries are secure, auditable, and traceable.
- Data encryption techniques are implemented to protect patient confidentiality and prevent unauthorized access.

Regulatory Compliance:

- All data management activities will adhere to **Good Clinical Practice (GCP)** guidelines to ensure the integrity, reliability, and ethical conduct of the study.
- The data management process will also comply with **Health Insurance Portability and Accountability Act (HIPAA)** regulations to safeguard patient privacy.

Sheet 1: Patient Information & Enrollment

Field	Entry
Patient ID	
Visit ID	
Date of Enrollment (DD/MM/YYYY)	
Full Name	
Gender	
Date of Birth	
Age (years)	
Ethnicity	
Weight (kg)	
Height (cm)	
Smoking Status	
Medical History	
Primary Physician	

Sheet 2: Baseline Visit (Visit 1)

Field	Entry
Diagnosis of Hypertension	Yes / No
Comorbid Conditions	
Blood Pressure (Systolic/Diastolic)	/
Heart Rate (bpm)	
Other Relevant Lab Results	

Sheet 3: Laboratory Tests

Lab Test	Result	Unit	Normal Range	Date
Serum Creatinine		mg/dL	0.6 - 1.2	// ____
Serum Electrolytes		mEq/L	Normal Range	// ____
Other Relevant Tests				// ____

Sheet 4: Treatment Assignment & Randomization

Field	Entry
Randomization Date	// ____
Treatment Group	Drug X / Placebo
Concomitant Medications	

Sheet 5: Follow-Up Visit (Visit 2, Visit 3, etc.)

Field	Entry
Visit Date (DD/MM/YYYY)	// ____
Blood Pressure (Systolic/Diastolic)	/
Heart Rate (bpm)	
Adverse Events	Yes / No
Lab Results (Serum Creatinine, etc.)	

Sheet 6: Adverse Events (AE)

Adverse Event	Severity	Onset Date	Resolution Date	Action Taken	Outcome

Sheet 7: Final Visit (Visit 4, End of Study)

Field	Entry
Final Blood Pressure (Systolic/Diastolic)	/
Final Serum Creatinine	
Final Lab Results	
Final Adverse Event	
Final Outcome	

Sheet 8: Study Conclusion

Field	Entry
Study Completion Date	// ____
Study Outcome	Treatment Success / Failure
Comments / Notes	

Sheet 9: Signature Page (Informed Consent)

Field	Entry
Participant Signature	
Date	// ____
Investigator Signature	
Date	// ____

Observations:

Data: Clinical Trial of Drug Bioinfomine for Hypertension

1. Demographic Data:

Patient ID	Age	Gender	Ethnicity	Weight (kg)	Height (cm)	Smoking Status	Medical History
001	58	Male	Caucasian	85	175	Non-Smoker	Hypertension, Diabetes Type 2
002	63	Female	Hispanic	72	160	Smoker	Hypertension, Hyperlipidemia
003	47	Male	Asian	78	170	Non-Smoker	Hypertension, No Comorbidities
004	54	Female	African-American	90	162	Non-Smoker	Hypertension, Asthma

2. Baseline Blood Pressure (Systolic/Diastolic):

Patient ID	Visit Date	Systolic BP (mmHg)	Diastolic BP (mmHg)
001	01/01/2025	160	100
002	01/01/2025	155	98
003	01/01/2025	102	162
004	01/01/2025	158	96

3. Laboratory Test Results (Baseline):

Patient ID	Lab Test	Result	Unit	Normal Range	Date
001	Serum Creatinine	1.2	mg/dL	0.6 - 1.2	01/01/2025
002	Serum Creatinine	0.9	mg/dL	0.6 - 1.2	01/01/2025
003	Serum Creatinine	1.0	mg/dL	0.6 - 1.2	01/01/2025
004	Serum Creatinine	0.01	mg/dL	0.6 - 1.2	01/01/2025

4. Adverse Events (AE) Reported:

Patient ID	Adverse Event	Severity	Start Date	End Date	Action Taken	Outcome
001	Dizziness	Mild	01/15/2025	01/16/2025	Dose Reduced	Resolved
002	Nausea	Moderate	01/20/2025	01/21/2025	Drug Discontinued	Resolved
003	Fatigue	Mild	01/22/2025	01/23/2025	No Action	Resolved
004	Chest Pain	Severe	01/25/2025	01/25/2025	Drug Discontinued	Hospitalized

5. Treatment Group Assignment:

Patient ID	Group	Randomization Date
001	Drug Bioinfomine	01/01/2025
002	Placebo	01/01/2025
003	Drug Bioinfomine	01/01/2025
004	Placebo	01/01/2025

6. Follow-Up Visit Data (Month 3):

Patient ID	Visit Date	Systolic BP (mmHg)	Diastolic BP (mmHg)	Lab Results (Creatinine)	Adverse Event
001	04/01/2025	155	95	1.1	None
002	04/01/2025	150	92	0.8	Nausea (resolved)
003	04/01/2025	148	00	00	None
004	04/01/2025	160	98	1.2	Chest pain (resolved)

Notes on Data Privacy and Security:

- The data is anonymized, and each patient has an ID (e.g., Patient 001), which ensures that personal details are not directly linked to the data in Medidata Rave.
- Data access is restricted to authorized personnel only, with audit trails being maintained for any changes or updates to the data.



SGCP's

Guru Nanak Khalsa College
of Arts, Science & Commerce (Autonomous)

REPORT

INDUSTRIAL VISIT: SILVASSA, DADRA AND NAGAR HAVELI AND DAMAN AND DIU, INDIA

Submitted By:
Ms. Prarthi Hrishit Kothari
M. Sc. Bioinformatics (Part - II)
Roll No.: 115

Submitted On:
17th February, 2025



SGCP's
Guru Nanak Khalsa College
of Arts, Science & Commerce (Autonomous)

DEPARTMENT OF BIOINFORMATICS

CERTIFICATE

This is to certify that **Ms. Prarthi Hrishit Kothari** (Roll No: **115**) of M. Sc. Bioinformatics (Part II) has satisfactorily completed the report on the Industrial Visit to Silvassa (9th-10th February, 2025) for Semester IV course during the academic year 2024 - 2025.

Teacher In-charge
(Prof. Sermarani Nadar
Prof. Aparna Patil Kose)

Head of the Department
(Dr. Gursimran Kaur Uppal)

ACKNOWLEDGEMENT

I would like to express my sincere appreciation to everyone who contributed to the success of our industrial visit to Nakshatra Garden, Parle-G Factory (Man Foods Pvt. Ltd.), and IPCA Laboratories. This immersive experience has significantly enhanced our understanding of the practical applications of our studies.

Firstly, I extend my sincere thanks to Dr. Ratna Sharma (The Principal, of Guru Nanak Khalsa College of Arts, Science & Commerce (Autonomous), Dr. Gursimran Kaur Uppal (Head of Department of Bioinformatics), Prof. Sermarani Nadar and Prof. Aparna Patil Kose for fostering an environment that encourages exploration and creativity. Your leadership and vision have created a space where students can thrive and pursue their academic interests with passion.

Furthermore, I am grateful to the management and staff of Nakshatra Garden, Parle-G Factory (Man Foods Pvt. Ltd.), and IPCA Laboratories for warmly welcoming us and providing informative tours of their facilities. Their willingness to share their expertise and demonstrate their operational processes was truly enlightening.

I would also like to thank my fellow students for their active participation and collaborative spirit throughout the industrial visit. The shared learning experience and insightful discussions have enriched our understanding of the industries we explored.



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DAY 1 – NAKSHATRA GARDEN

Nakshatra Garden, located in Silvassa, India, is a unique garden that blends nature with astrological concepts¹²³. The garden is named after the 27 Nakshatras, or lunar mansions, in Hindu astrology, and it embodies a serene retreat with a cosmic alignment.

Visiting Information

- Hours:** The garden is open from 6:30 am to 7 pm daily. The best times to visit are early morning or in the evening.
- Location and Access:** Nakshatra Garden is located in Silvassa and is approximately 14 kilometers from Bhilad and 18 kilometers from Vapi. The nearest airport is in Surat and Mumbai, and the nearest railway station is Vapi on the Western Railway.



Key Aspects of Nakshatra Garden

1. Design and Layout

Nakshatra Garden, an astro-themed garden, spans a vast area and is designed according to the Indian system of astrology, linking plants with the motion of different planets. The garden represents the 12 Zodiac signs, aligning with the Indian system of planetary influences. Each area within the garden is carefully landscaped to reflect the characteristics of its corresponding Zodiac sign. The garden also includes plants associated with the Nav Grahas, or nine planets worshiped by Hindus. It features a circular arrangement with 27 spokes, with 27 plants associated with each Nakshatra.

2. Botanical Collection

The garden features a wide array of flora, including:

- Medicinal Herbs:** These are plants known for their therapeutic properties, used in traditional medicine systems like Ayurveda. Examples include Neem (*Azadirachta indica*), known for its antiseptic and antibacterial qualities and Tulsi (*Ocimum sanctum*), revered for boosting immunity and reducing stress.
- Ayurvedic Plants:** The garden emphasizes plants integral to Ayurvedic practices, such as Amla (*Phyllanthus emblica*), which is celebrated for its high vitamin C content, antioxidant properties and known for improving digestion and immunity and Ashwagandha (*Withania somnifera*), used for reducing stress and increasing vitality.
- Spice Trees:** It includes trees that provide spices used in cooking and traditional remedies, such as Cardamom (*Elettaria cardamomum*), known for digestive and aromatic properties and Clove (*Syzygium aromaticum*), valued for its antiseptic and pain-relieving properties.

Informative Nameplates

Each plant in Nakshatra Garden is accompanied by informative nameplates that serve multiple purposes:

a. **Educational Insights:** These plates provide details about the plant species, including their scientific names, local names, and descriptions of their medicinal uses. This educational aspect helps visitors appreciate the relationship between flora and traditional practices.

b. **Cultural Connection:** The nameplates often highlight the astrological significance of each plant, linking them to the 27 Nakshatras (lunar mansions) in Hindu astrology. This connection enriches the visitor's understanding of how these plants are integrated into cultural beliefs and practices.



c. **Scientific and Cultural Insights:** Including botanical and local names, descriptions, and medicinal uses.

d. **Astrological Connections:** Highlighting the linkage between plants and their corresponding Nakshatras.

e. **Mythological Context:** Offering stories from Indian mythology that connect specific plants to gods and celestial bodies.



Medicinal Value and Learning Center

The garden is designed not only for aesthetic enjoyment but also as a resource for learning about the medicinal value of various plants:

- Healing Properties:** Many plants are recognized for their specific healing properties. For example, Aloe Vera (*Aloe barbadensis miller*) is known for its skin-soothing benefits, while Ginger (*Zingiber officinale*) is often used to aid digestion.
- Traditional Remedies:** Visitors can learn about how these plants are used in traditional remedies, contributing to holistic health practices.
- Ayurvedic Park:** A dedicated section displaying plants commonly used in Ayurveda with descriptions of their uses.
- Educational Workshops:** Regular workshops and guided tours to promote knowledge about Ayurveda and plant-based healing.

Promoting Awareness

Nakshatra Garden plays a vital role in promoting awareness about biodiversity and the importance of preserving medicinal plants:

- a. Conservation Efforts:** By showcasing these plants, the garden encourages visitors to understand the significance of conserving native species and their habitats.
- b. Cultural Heritage:** The garden serves as a reminder of the rich cultural heritage associated with plants in Indian traditions, fostering respect for nature and traditional knowledge.

3. Water Features

Nakshatra Garden is known for its stunning waterfalls and fountains, which are designed to take advantage of the natural terrain. The garden incorporates several water bodies, including a large lake and smaller ponds, enhancing its overall charm and tranquility. Small bridges connect the ponds, which serve as a home for various species of ducks.

The garden features natural and man-made water attractions, including:



- a. Waterfalls and Fountains:** Cascading water features add tranquility to the landscape.
- b. Lakes and Ponds:** Scenic water bodies that host ducks and other aquatic life.
- c. Bridges and Pathways:** Connecting different areas of the garden, enhancing the exploration experience.

4. Additional Attractions

Nakshatra Garden offers various attractions, including an arena for cultural performances, a meditation center for yoga and meditation, walking trails, playgrounds, and a mini-train for garden tours.

- a. Cultural Arena:** A space for performances and traditional events.
- b. Meditation Center:** A peaceful zone for yoga and meditation.
- c. Walking Trails:** Scenic pathways for leisurely strolls.
- d. Playgrounds:** Dedicated areas for children.
- e. Mini-Train:** Offering tours for visitors to explore the garden conveniently.

5. Family and Leisure

Nakshatra Garden is suitable for families, offering play areas for children, food stalls, and restaurants. The garden provides a peaceful environment for relaxation, picnics, and leisurely walks.

6. Learning and Knowledge

Visitors can learn about Indian astrology, mythology, and various plant species through plaques and signboards. The garden also features an Ayurvedic park with information on the uses of different plants in Ayurveda.

a. **Biodiversity Conservation:**

Showcases the importance of preserving native and medicinal plant species.

b. **Cultural Education:**

Promotes knowledge about Indian traditions, astrology, and plant-based healing.

c. **Eco-Tourism:**

Encourages sustainable tourism practices.



Nakshatra Garden in Silvassa is more than just a park; it is a fusion of nature, culture, and education. Visitors can immerse themselves in the rich tapestry of Indian astrology, explore diverse plant species, and experience the healing power of nature. The garden offers something for everyone—from families to nature enthusiasts and spiritual seekers.

DAY 2 – PARLE-G FACTORY

Man Foods Pvt. Ltd., located in Silvassa, Dadra & Nagar Haveli, is a prominent manufacturing unit under the umbrella of Parle Products Pvt. Ltd., India's leading biscuit and confectionery producer. Established in 1929 in Mumbai's Vile Parle suburb, Parle has expanded its operations nationwide, with Man Foods Pvt. Ltd. Being a significant contributor to its production capabilities.

Records and Milestones

One of the notable achievements in Parle's manufacturing history is the installation of Asia's largest oven, measuring approximately 260 feet in length. This massive oven is integral to the baking process, ensuring consistent quality and efficiency in biscuit production. The oven operates at temperatures around 340°C, facilitating the baking of vast quantities of biscuits daily.



Brief History and Overview

Parle Products began as a small confectionery company in 1929, focusing on sweets and toffees. Over the decades, it has evolved into a household name, especially known for its flagship product, Parle-G biscuits. The Silvassa facility, operated by Man Foods Pvt. Ltd., plays a crucial role in meeting the high demand for Parle's diverse product range.

1929	First factory established.
1939	Began manufacturing biscuits. Parle Gluco biscuit introduced.
1947	Launched ad campaign for Gluco biscuits as an Indian alternative to British biscuits.
1956	Launched Cheeslings.
1963	Kismi toffee introduced.
1966	Poppins introduced.
1971	Received first Monde Selection Award.
1972	Krackjack introduced.
1982	First TV commercial released.
1983	Melody launched.
1985	Parle Gluco became Parle-G.

1986	Mango Bite launched.
1996	Hide & Seek introduced.
2013	Parle-G was India's first FMCG brand to cross ₹50 billion in retail sales.

Parle Products

Parle's product portfolio is extensive, encompassing a variety of biscuits, confectioneries, snacks, and more. Some of the well-known products include:

- 1. Biscuits:** Parle-G, Krackjack, Monaco, 20-20 Cookies, Parle Marie, Parle Creams
- 2. Confectioneries:** Melody Chocolate, Mango Bite, Kacha Mango Bite, Poppins, Kismi
- 3. Snacks:** Full Toss

Machinery and Production Rate

The Silvassa plant is equipped with state-of-the-art machinery to handle large-scale production. While specific daily production rates for this facility are not publicly disclosed, other units, such as the Raipur plant, have reported production capacities of over 70 tons per day. This includes various products, with Parle-G being a significant contributor.

Production Procedure

The manufacturing process at Parle involves several meticulous steps to ensure product quality and consistency:

STEP 1: Raw Material Testing

Upon arrival at the manufacturing facility, all ingredients—such as wheat flour, sugar, edible oils, and milk solids—undergo rigorous quality checks. This testing ensures that each ingredient meets Parle's stringent standards for safety and quality. The quality control team conducts these tests to prevent any substandard materials from entering the production line.



STEP 2: Mixing

Once the raw materials pass inspection, they are combined in high-capacity mixers. This step is crucial as it forms a homogeneous dough that serves as the foundation for the final product. The mixing process is carefully timed and monitored to achieve the desired consistency and texture of the dough.



STEP 3: Moulding

The prepared dough is then fed into moulding machines. Here, it is shaped into individual biscuits or confectionery pieces. This automated process ensures that each piece is uniform in size and shape, which is vital for both aesthetic appeal and consumer satisfaction.



STEP 4: Baking

Shaped dough pieces are baked in large industrial ovens. One notable example is a 260-foot-long oven that operates at approximately 340°C (644°F). This high-temperature baking ensures that the products are cooked evenly, resulting in a consistent texture and flavor across batches.



STEP 5: Cooling

After baking, the products are transferred to extensive conveyor systems for cooling. This cooling phase allows the biscuits or confectionery pieces to reach ambient temperatures before they proceed to packaging. Proper cooling is essential to maintain product integrity and prevent moisture buildup.



STEP 6: Packing

Finally, cooled products are packaged using automated systems designed to maintain hygiene and extend shelf life. The packaging process includes sealing and labeling, which helps ensure that the products remain fresh during distribution.



Production Processes of Select Parle Products

Each product line at Parle follows specific manufacturing processes tailored to its unique formulation.

Product	Ingredients	Process
Parle-G Biscuits	Wheat flour, sugar, milk powder, salt, palm oil.	The ingredients are mixed to form a dough, shaped into biscuits, baked in a long oven at high temperatures (up to 280°C), cooled, and then packaged.
Melody Chocolate	Cocoa, sugar, milk solids.	Ingredients are mixed and cooked until achieving the desired consistency. The mixture is then moulded into individual pieces, cooled, and wrapped for sale.
Mango Bite and Kacha Mango Bite	Mango pulp, sugar.	The ingredients are cooked to create a candy mixture that is poured into moulds, cooled until firm, and individually wrapped for distribution.
20-20 Cookies	Flour, sugar, butter, flavorings.	A dough is prepared from these ingredients, shaped into cookies, baked until golden brown, cooled down appropriately, and packaged.
Parle Marie and Parle Kreams Biscuits	Specific formulations based on flour and other components.	The dough is moulded into classic round shapes; for Kreams biscuits, flavored cream is added between layers after baking.
Poppins	Sugar and flavourings.	A syrup is created by cooking sugar with flavorings; this syrup is then moulded into small discs that are cooled and packaged in rolls.
Kismi	Caramelized sugar and milk solids.	These ingredients are mixed together with flavors, moulded into rectangular pieces, cooled thoroughly, and wrapped for sale.
Full Toss	Flour, spices, seasonings.	Ingredients are mixed to form a dough that is shaped into bite-sized pieces; these can be fried or baked until crispy before cooling and packaging.

At Parle's Silvassa facility, stringent quality control measures are implemented throughout the production process. This ensures that every product meets Parle's legacy of taste and trust. Regular inspections at each stage—from raw material testing to final packaging—help maintain high standards of quality across all product lines. Each product line at the Silvassa facility adheres to stringent quality control measures, ensuring that consumers receive products that meet Parle's legacy of taste and trust.

DAY 2 – IPCA LABORATORIES

Ipca Laboratories Limited is an Indian multinational pharmaceutical company headquartered in Mumbai. Established in 1949, Ipca manufactures various formulations and active pharmaceutical ingredients (APIs), serving over 120 countries.

Brief History and Overview

Ipca Laboratories Ltd. was founded on October 19, 1949, in Mumbai, originally named "The Indian Pharmaceutical Combine Association Limited". In 2004, Forbes recognized Ipca as one of Asia's top 200 "Best under a billion companies" for the second consecutive year.



Ipca has multiple manufacturing facilities across India, including locations at Athal (Dadra & Nagar Haveli), Ratlam, Kandla, Silvassa, Dehradun, Indore (SEZ), Sikkim, Pithampur, and Tarapur. These facilities are equipped to produce a wide range of pharmaceutical products. The Silvassa plant specializes in tablets and capsules

and has received certifications such as WHO-GMP, TGA-Australia, and Health-Canada.

Drug Manufacturing Process at Ipca Laboratories

STEP 1: Weighing

Raw materials, including active pharmaceutical ingredients (APIs) and excipients, are precisely weighed according to the batch manufacturing record (BMR). Accurate measurement ensures the correct ratio of ingredients for each batch.

Machinery used:

- **Weighing Balance Machinery:** Used to precisely measure raw materials. These are available in bulk and electronic systems.



STEP 2: Granulation

Granulation is a process of converting powdered materials into granules. This improves the flowability and compression characteristics of the powder, which is essential for efficient tablet formation. There are three main methods of granulation:

- a. **Wet Granulation:** A liquid binder is added to the powder mixture while mixing continuously to create granules. The wet mass is then agitated to promote the formation of small granules. Rapid mixer granulators (RMG) are commonly used for wet granulation.
- b. **Dry Granulation:** This method is used for materials that are sensitive to moisture or heat. The powder is compressed into large compacts or slugs, which are then broken down into granules.

- c. **Direct Compression:** Some materials can be directly compressed into tablets without granulation. This method is suitable for ingredients with good flow and compression properties.

Machinery used:

- **Rapid Mixer Granulator (RMG):** Used for mixing ingredients and wet granulation.
- **Fluid Bed Dryer:** Dries wet powder or granules at a desired temperature.
- **Granulators:** Creates wet or dry tablet granules. Types include rotating shape granulators, mechanical agitator granulators, high-shear granulators, and fluidized-bed granulators.



STEP 3: Material Transfer

Materials are transferred between different stages of the manufacturing process through airlocks. This includes man-air locks for personnel and material-air locks for the transfer of materials. Airlocks help to maintain cleanliness and prevent contamination of the materials.

Machinery used:

- This process often involves **conveyor systems** and **pneumatic transfer systems** to move materials while maintaining containment.



STEP 4: Compression

In this step, granules are compressed into tablets using tablet compression machines. The tablet press applies controlled pressure to the granules, compacting them into the desired tablet shape. Key components include the die, which determines the tablet's size and shape, and punches, which apply the pressure. Different types of tablet presses exist, such as single-punch, rotary, and high-speed tablet presses. The compression process affects the tablet's density, hardness, and disintegration properties.



Machinery used:

- **Tablet Press Machines:** Compresses granules into tablets of uniform shape, size, and weight. Types include:
 - Single Sided Rotary Tablet Press Machine
 - Double Sided Rotary Tablet Press Machine
 - High-Speed Rotary Tablet Press Machine
 - Multi-layer rotary tablet presses for quality tablet compression

**STEP 5: Tablet Coating**

Tablets are often coated for various reasons, such as taste masking, ease of swallowing, controlling the release of the drug, and improving appearance. The coating process typically involves several steps:

- a. **Pre-heating and Tablet Loading:** Tablets are pre-heated and loaded into the coating machine.
- b. **Spraying:** A coating solution is sprayed onto the tablets as they move within the coating pan or fluidized bed.
- c. **Drying:** The coated tablets are dried to remove the solvent from the coating solution.
- d. **Cooling:** Finally, the tablets are cooled to room temperature.

Machinery used:

- **Conventional Coating Pan:** Used for masking the smell and taste of the drug.
- **Automatic Tablet Coater:** Used for tablet coating, especially with increasing demands.
- **Spray Coating Machines:** Used for spray coating liquid onto a powder.

**STEP 6: Quality Checks**

Quality control checks are performed throughout the entire manufacturing process. These checks ensure that the tablets meet the required standards for identity, strength, quality, and purity. Parameters such as size, weight, hardness, thickness, and disintegration time are tested. Tablets are also inspected for any visual defects such as cracking, chipping, or variations in color.

Machinery used:

- **Tablet and Capsule Inspection Machine:** Used for inspecting tablets and capsules.
- **Tablet De Duster:** Removes dust from tablets and often polishes them.
- **Weighing balance equipment:** Used to get the precise weight of tablets.
- **Metal detectors:** Detect tramp metal that may have contaminated the product.

**STEP 7: Batch Unloading**

Once the tablets have been coated and have passed all quality control checks, they are unloaded from the coating machines. They are then prepared for packaging and distribution.

Quality Control in Tablet Manufacturing

Importance of Quality Control

Quality control is essential to guarantee that pharmaceutical products are safe for consumption and effective in treating conditions. The objectives of QC include:

1. **Ensuring Compliance:** Tablets must comply with regulatory standards set by pharmacopoeias, such as the United States Pharmacopoeia (USP) and European Pharmacopoeia (PhEur).
2. **Preventing Defects:** QC processes are designed to identify and eliminate defects before products reach consumers.
3. **Maintaining Consistency:** Regular monitoring ensures that each batch of tablets maintains consistent quality, potency, and efficacy.



In-Process Quality Control (IPQC)

In-process quality control involves monitoring and testing during the manufacturing process to ensure that the tablets meet predefined specifications. Key aspects include:

1. **Monitoring Parameters:** Critical parameters such as temperature, humidity, pressure, and mixing times are continuously monitored.
2. **Sampling:** Regular sampling of materials at different stages allows for timely adjustments in the manufacturing process.
3. **Testing:** Various tests are performed on samples to assess attributes like granule size, moisture content, and flow properties.

Finished Product Quality Control (FPQC)

After manufacturing is complete, finished product quality control tests are conducted to verify that the tablets meet all specifications. This includes:

1. **Physical Tests:** Assessing attributes such as size, shape, color, hardness, and weight uniformity.
2. **Chemical Tests:** Ensuring that the active ingredient content is within acceptable limits.
3. **Dissolution Testing:** Evaluating how quickly and effectively the tablet dissolves in a specified medium.

Defect Identification

Defect identification is crucial for ensuring that only high-quality tablets are released into the market. Common defects include:

1. **Cracking:** Cracking can occur during compression or storage. It affects the integrity of the tablet and can lead to inconsistent drug release.
2. **Chipping:** Chipping refers to small pieces breaking off from the edges of tablets. This defect can occur due to improper handling or excessive pressure during compression.
3. **Variations in Color or Size:** Inconsistencies in color or size may indicate issues with formulation or manufacturing processes. Such variations can affect patient acceptability and dosage accuracy.
4. **Incorrect Markings:** Tablets must have accurate markings for identification purposes. Incorrect markings can lead to medication errors.

Rejection of Defective Tablets

Tablets that do not meet quality standards are systematically rejected during both IPQC and FPQC stages. The rejection process includes:

1. **Visual Inspection:** Trained personnel visually inspect tablets for defects.
2. **Automated Inspection Systems:** Advanced technology may be employed to detect defects using imaging systems.
3. **Documentation:** All rejected products are documented for traceability and further investigation into the cause of defects.

In conclusion, Ipca Laboratories Ltd. has evolved from its beginnings in 1949 to become a significant multinational pharmaceutical company. With a broad portfolio of APIs and formulations, Ipca serves numerous therapeutic segments across more than 100 countries. The company's commitment to quality is reflected in its approvals from leading drug regulatory authorities and its continuous efforts to expand its manufacturing capabilities and market reach. Ipca's journey from a struggling pharmaceutical company to a major player in the global market underscores its resilience and strategic vision.

CONCLUSION

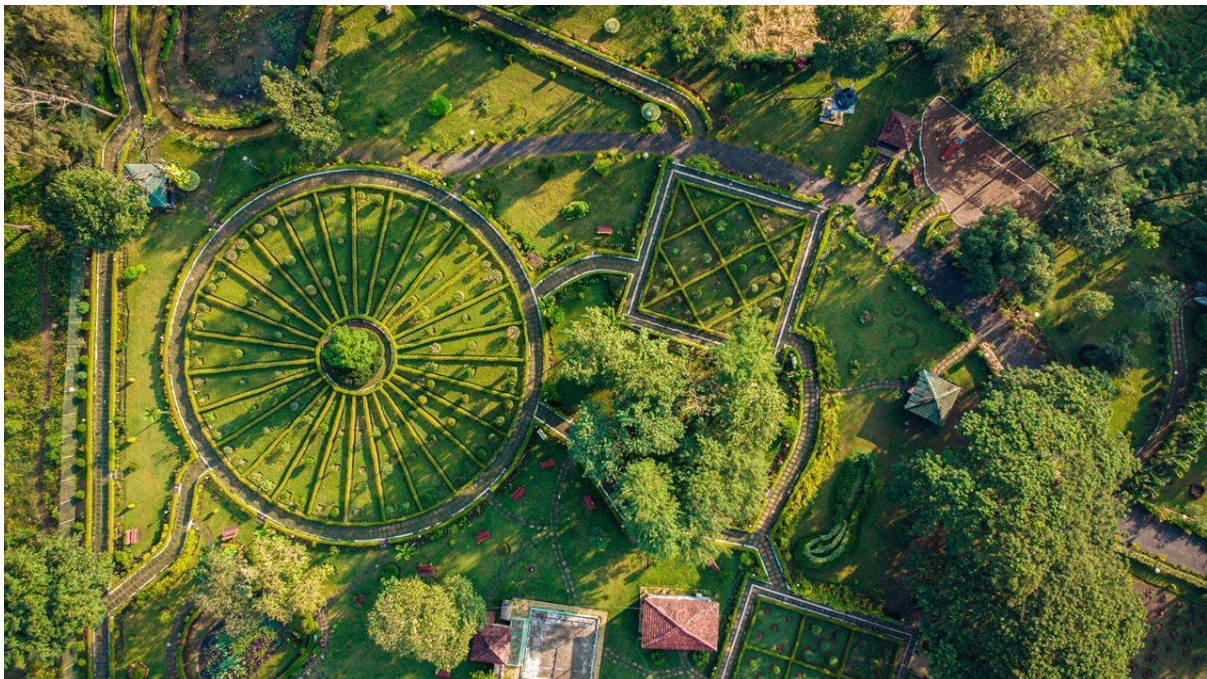
This industrial visit provided valuable insights into diverse sectors, including tourism and culture (Nakshatra Garden), food manufacturing (Parle-G Factory), and pharmaceuticals (IPCA Laboratories). Each location offered unique learning experiences, highlighting the practical applications of theoretical knowledge and the importance of quality control, innovation, and cultural integration in various industries.

Key Takeaways

Nakshatra Garden

The Nakshatra Garden visit underscored the intersection of nature, culture, and education. The garden's design, based on Indian astrology and the alignment of plants with the 27 Nakshatras, demonstrated a unique approach to blending tourism with cultural heritage. Key takeaways include:

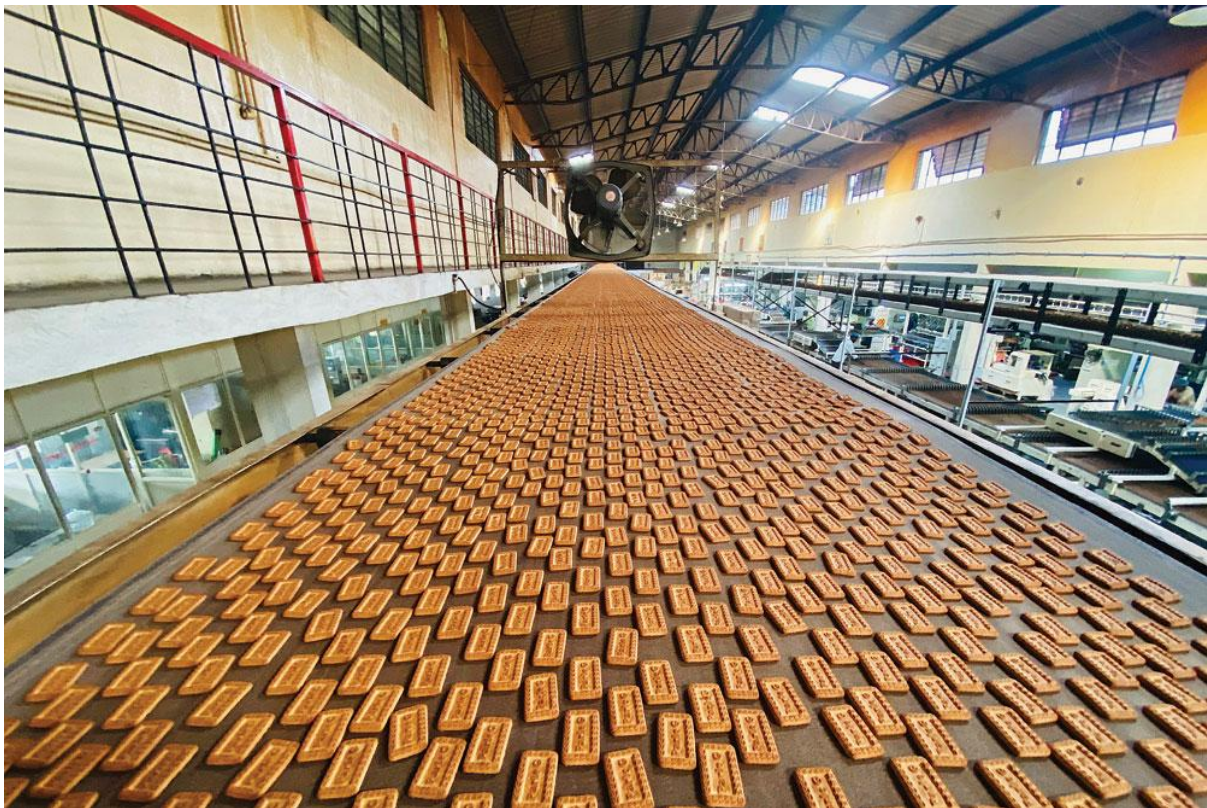
- The importance of integrating cultural themes into tourism to enhance visitor experience and promote cultural awareness.
- The role of botanical gardens in conserving biodiversity and educating the public about medicinal plants and their traditional uses.
- The significance of thoughtful landscape design in creating tranquil and educational spaces for families and individuals.



Parle-G Factory (Man Foods Pvt. Ltd.)

The visit to the Parle-G factory showcased the intricacies of large-scale food manufacturing and the importance of quality control in ensuring product consistency and safety. Key takeaways include:

- The critical role of rigorous raw material testing in maintaining product quality and preventing substandard materials from entering the production line.
- The significance of automation in achieving uniformity and efficiency in processes like mixing, moulding, baking, cooling, and packaging.
- The importance of adhering to stringent quality control measures at every stage of production, ensuring that products meet the company's standards of taste and trust.



IPCA Laboratories

The visit to IPCA Laboratories provided insights into the pharmaceutical manufacturing process, emphasizing the precision and quality control required in drug production. Key takeaways include:

- The importance of accurate measurement of raw materials in pharmaceutical manufacturing to ensure the correct ratio of ingredients for each batch.
- The role of granulation in improving the flowability and compression characteristics of powders for efficient tablet formation.
- The necessity for sterile environments and advanced equipment in pharmaceutical production to prevent contamination and ensure product safety.

- The significance of strict adherence to regulatory guidelines and certifications like WHO-GMP, TGA-Australia, and Health-Canada to ensure product quality and compliance with international standards.



In conclusion, these industrial visits provided a holistic learning experience, demonstrating the practical applications of theoretical concepts and the importance of innovation, quality control, and cultural integration across different industries. The insights gained from these visits will undoubtedly contribute to a better understanding of industry practices and future career paths.

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WEBLEM: 1
Swiss Target Prediction Tool

AIM:

Introduction to study In-silico Target fishing and optimization using computational methods.

INTRODUCTION:

Insilico Target Prediction:

Insilico target prediction refers to computational methods used to identify potential biological targets for small molecules, which is crucial in drug discovery and development. These methods can be categorized into structure-based and ligand-based approaches. Structure-based predictions rely on the three-dimensional structures of proteins to identify potential binding sites, while ligand-based predictions utilize the known bioactivity of compounds to infer target interactions based on chemical similarity. The process often involves virtual screening, where thousands of small molecules are docked against specific protein targets to identify those with the highest likelihood of binding and affecting protein activity. Recent advancements have incorporated machine learning (ML) techniques, enhancing the predictive capabilities of these methods by analyzing large datasets and identifying patterns that may not be apparent through traditional approaches. For instance, ML models can predict drug-target interactions by comparing a query compound with existing compounds associated with known targets, utilizing various descriptors and scoring systems to estimate binding affinities. Network-based approaches have emerged, leveraging biological networks to elucidate complex interactions between drugs and targets, facilitating the identification of essential molecular processes. Insilico target prediction is particularly valuable for drug repurposing, as it allows researchers to explore new uses for existing drugs by identifying potential off-target interactions or novel therapeutic targets. The integration of these computational methods has significantly streamlined the drug discovery process, reduced costs and time while increasing the likelihood of successful outcomes.

SwissTargetPrediction:

SwissTargetPrediction is a web-based tool launched in 2014 that predicts the most probable protein targets of bioactive small molecules using a similarity principle through reverse screening. The tool was significantly updated in 2019 with data from ChEMBL version 23, expanding its dataset to over 376,000 compounds and 3,068 macromolecular targets, which improved predictive accuracy and broadened the biological and chemical spaces explored. Predictions are made by comparing the query molecule to known active compounds, focusing on those with high similarity scores (0.65 for 2D and 0.85 for 3D), ensuring meaningful results. It features an intuitive interface for inputting molecules via SMILES notation or a graphical editor and supports predictions for multiple organisms, including humans, mice, rats, cows, and horses. Designed for efficiency, it returns results within 15-20 seconds for drug-like molecules and maintains high predictive performance, achieving at least one correct target in the top 15 predictions for over 70% of external compounds tested. Freely accessible online, SwissTargetPrediction is valuable for researchers in drug discovery to understand molecular mechanisms, rationalize potential side effects, and assess opportunities for repurposing existing therapeutics.

For both 2D and 3D similarity measures, the principle is that two similar molecules are represented by analogous vectors, which exhibit a quantified similarity close to 1. The

SwissTargetPrediction model was trained by applying multiple logistic regression on various size-related subsets of known actives in order to weight 2D and 3D similarity parameters in a so-called Combined-Score. A Combined-Score higher than 0.5 predicts that the molecules are likely to share a common protein target. In reverse screening, the Combined-Score allows the calculation of the probability that a given protein is targeted by a query molecule assumed to be bioactive. As 2D and 3D descriptions of molecules are complementary, this dual-scoring ligand-based reverse screening demonstrated high performance in predicting macromolecular targets across various test sets.

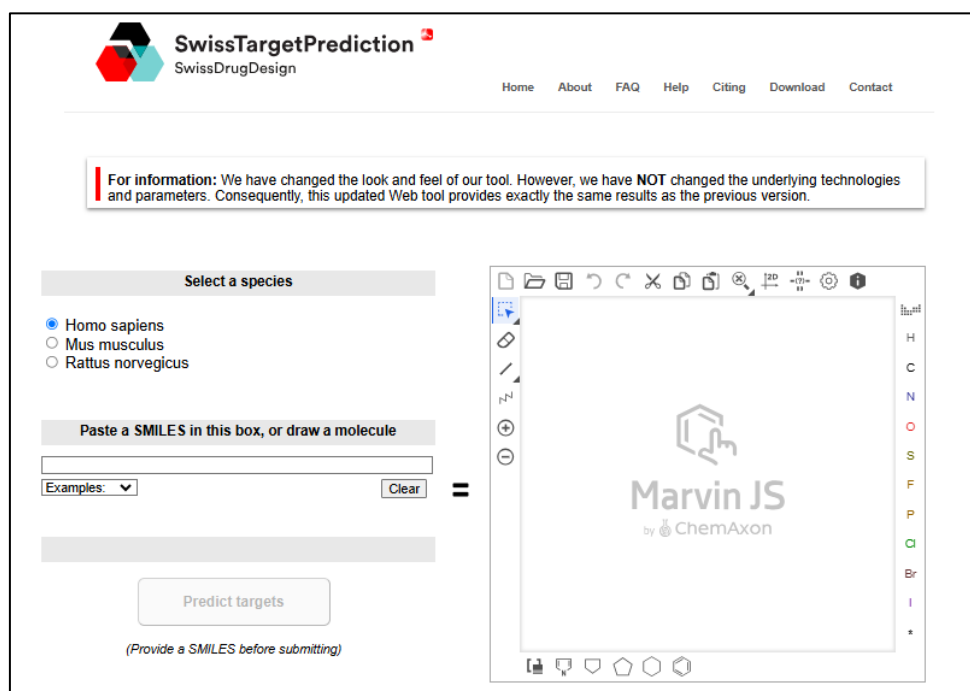


Figure 1: SwissTargetPrediction Tool Homepage

Applications:

1. SwissTargetPrediction aids researchers in identifying potential protein targets for new or existing small molecules, facilitating the understanding of their mechanisms of action. By predicting targets based on structural similarities, it helps narrow down the list of possible interactions that a compound may have within biological systems.
2. The tool is particularly valuable for drug repurposing efforts, allowing researchers to explore new therapeutic uses for established drugs by identifying secondary targets or off-target interactions. This can lead to faster development timelines and reduced costs compared to discovering entirely new drugs.
3. SwissTargetPrediction is part of the SwissDrugDesign initiative, which provides a suite of online tools for computer-aided drug design. It supports virtual screening processes where large libraries of compounds can be rapidly evaluated for potential activity against specific targets, thereby streamlining the early stages of drug development.
4. By mapping potential off-target interactions, SwissTargetPrediction helps in predicting possible side effects associated with small molecules. This information is crucial for assessing the safety profiles of drugs during the development process.
5. The platform allows seamless integration with other online computer-aided drug design tools developed by the SIB Swiss Institute of Bioinformatics, enhancing its utility in comprehensive drug discovery workflows.

6. The tool supports predictions across various organisms, including humans, mice, rats, cows, and horses. This feature enables researchers to map predictions by homology within and between different species, which is essential for translational research

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-

Date: 20/02/2025

WEBLEM: 1(A)

Swiss Target Prediction Tool

(URL: <http://www.swisstargetprediction.ch/>)

AIM:

To identify potential biological target for query 'Probuphine' using Swiss target prediction tool.

INTRODUCTION:

SwissTargetPrediction is an online tool designed to predict the most probable protein targets of small bioactive molecules.

Prediction Methodology: The tool employs a ligand-based approach based on the similarity principle, which posits that similar molecules are likely to have similar biological activities. It utilizes both 2D and 3D similarity measures to identify potential targets from a large dataset .

User-Friendly Interface: SwissTargetPrediction features an intuitive graphical interface that allows users to input molecular structures easily, either through SMILES notation or by drawing them directly.

Organism Selection: Users can select predictions for various organisms, including humans, mice, rats, cows, and horses. This flexibility supports comparative studies across different species.

In addition to the basic ligand-based approach, SwissTargetPrediction uses a variety of computational techniques to refine its predictions. These include:

1. **2D Similarity:** This method compares molecules based on their 2D structural features, such as functional groups and connectivity. It allows the identification of compounds that share similar molecular scaffolds, which are likely to interact with the same or similar protein targets.
2. **3D Similarity:** This method accounts for the three-dimensional arrangement of atoms in the molecule. By considering the spatial geometry and physicochemical properties, this approach helps identify targets based on the shape and electrostatic interactions of the molecule, enhancing the accuracy of predictions, especially for complex molecules.
3. **Database Integration:** SwissTargetPrediction taps into large, curated datasets of protein-ligand interactions from various databases, such as ChEMBL and DrugBank, ensuring that the predictions are grounded in experimental data and existing biological knowledge.
4. **Scoring System:** The tool assigns a score to each predicted protein target, reflecting the likelihood of the small molecule binding to that target. This score is based on structural similarity, known biological activity, and statistical validation from existing data.

Probuphine:

Probuphine is a buprenorphine implant used for the maintenance treatment of opioid dependence in patients who have achieved stability on low-to-moderate doses of buprenorphine. It provides a constant, low-level dose of buprenorphine for six months, reducing the need for daily medication. The implant requires specialized insertion and removal due to potential complications like implant migration. Probuphine is part of a

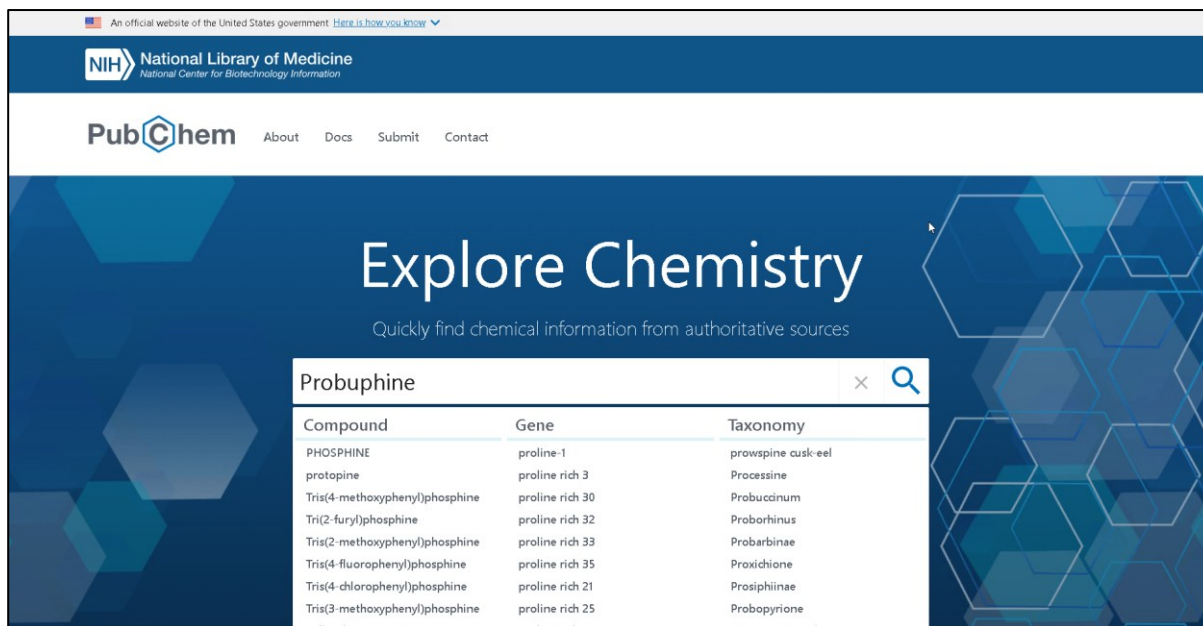
comprehensive treatment program that includes counseling and psychosocial support. However, it is no longer actively marketed in the U.S.

METHODOLOGY:

Steps for Swiss target prediction tool

1. Open PubChem (<https://pubchem.ncbi.nlm.nih.gov/>) and search for 'Probuphine'.
2. Copy the SMILES representation of the query 'Probuphine'.
3. Select species as homo sapiens in Swiss target prediction tool.
4. Paste the SMILES string into the Swiss target prediction tool (<http://www.swisstargetprediction.ch/>).
5. Click on predict targets and wait for the results to be processed.

OBSERVATIONS:



The screenshot shows the PubChem website interface. At the top, there is a header for the National Library of Medicine (NIH) and the National Center for Biotechnology Information. Below this, the PubChem logo is visible along with navigation links: About, Docs, Submit, and Contact. The main heading is "Explore Chemistry" with the tagline "Quickly find chemical information from authoritative sources". A search bar contains the text "Probuphine". Below the search bar, a table displays search results organized into three columns: Compound, Gene, and Taxonomy.

Compound	Gene	Taxonomy
PHOSPHINE	proline-1	prowspine cusk-eel
protopine	proline rich 3	Processine
Tris(4-methoxyphenyl)phosphine	proline rich 30	Probuccinum
Tri(2-furyl)phosphine	proline rich 32	Proborhinus
Tris(2-methoxyphenyl)phosphine	proline rich 33	Probarbinae
Tris(4-fluorophenyl)phosphine	proline rich 35	Proxichione
Tris(4-chlorophenyl)phosphine	proline rich 21	Prosiphiinae
Tris(3-methoxyphenyl)phosphine	proline rich 25	Probopyrione
probuphine	proline rich 27	Protophine vetula

Figure 1: Drug Probuphine has been searched in PubChem

PubChem Buprenorphine (Compound)

Computed by InChI 1.0.6 (PubChem release 2021.10.14)

PubChem

2.1.4 SMILES  

C[C@]1([C@H]1C[C@@]23CC[C@@]1([C@H]4[C@@]25CCN([C@@H]3CC6=C5C(=C(C=C6)O4)CC7CC7)OC)(C(C)(C)C)O

Computed by OEChem 2.3.0 (PubChem release 2024.12.12)

PubChem

2.2 Molecular Formula  

C₂₉H₄₁NO₄

Computed by PubChem 2.2 (PubChem release 2021.10.14)

PubChem

2.3 Other Identifiers  

2.3.1 CAS  

52485-79-7

CAS Common Chemistry; ChemIDplus; DrugBank; EPA DSSTox; European Chemicals Agency (ECHA); FDA Global Substance Registrati...

2.3.2 Related CAS  

CONTENTS

- Title and Summary
- 1 Structures
- 2 Names and Identifiers
- 3 Chemical and Physical Properties
- 4 Spectral Information
- 5 Related Records
- 6 Chemical Vendors
- 7 Drug and Medication Information
- 8 Pharmacology and Biochemistry
- 9 Use and Manufacturing
- 10 Safety and Hazards
- 11 Toxicity
- 12 Associated Disorders and Diseases
- 13 Literature
- 14 Patents
- 15 Interactions and Pathways
- 16 Biological Test Results
- 17 Taxonomy
- 18 Classification
- 19 Information Sources

Figure 2: Smiles format of query

SwissTargetPrediction
SwissDrugDesign

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For information: We have changed the look and feel of our tool. However, we have **NOT** changed the underlying technologies and parameters. Consequently, this updated Web tool provides exactly the same results as the previous version.

Select a species

☒ Homo sapiens
☐ Mus musculus
☐ Rattus norvegicus

Paste a SMILES in this box, or draw a molecule

Examples: Clear

Predict targets

(Can take up to one minute)

Chemical structure visualization showing the molecule structure and a legend for atom types (H, C, N, O, S, F, P, Cl, Br, I).

Figure 3: Smile format of the query pasted

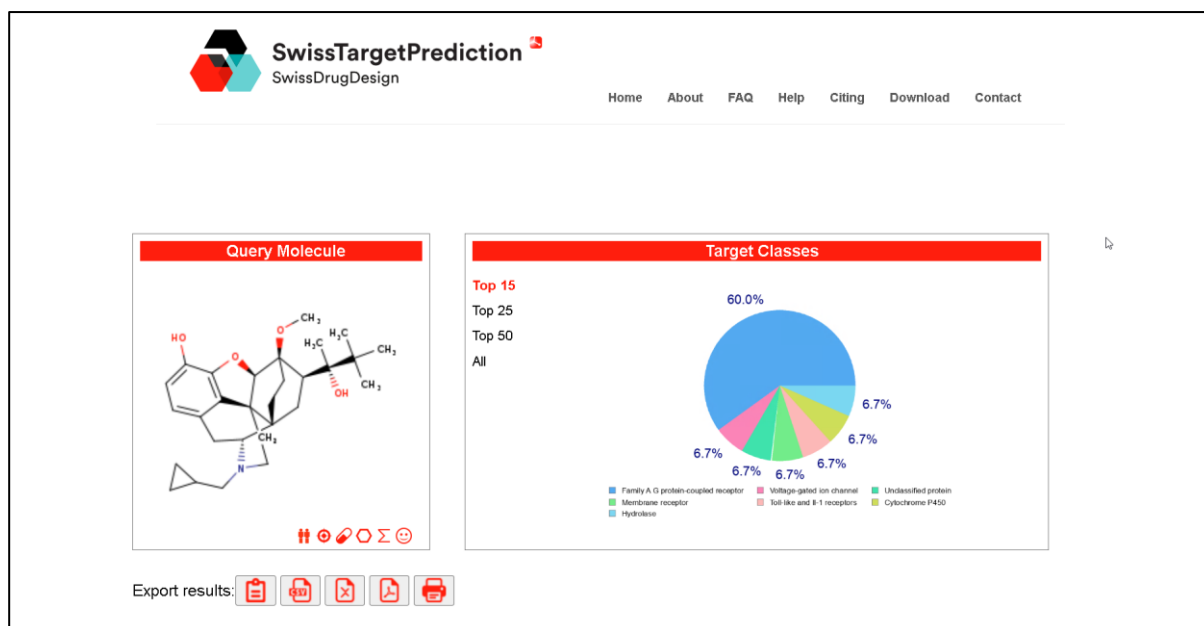


Figure 4: Query molecule and target classes obtained

Show entries

Search:

Target	Common name	Uniprot ID	ChEMBL ID	Target Class	Probability*	Known actives (3D/2D)
Mu opioid receptor	OPRM1	P35372	CHEMBL233	Family A G protein-coupled receptor		1137 / 1381
Delta opioid receptor	OPRD1	P41143	CHEMBL236	Family A G protein-coupled receptor		930 / 1402
Kappa Opioid receptor	OPRK1	P41145	CHEMBL237	Family A G protein-coupled receptor		733 / 1008
Nociceptin receptor	OPRL1	P41146	CHEMBL2014	Family A G protein-coupled receptor		286 / 37
HERG	KCNH2	Q12809	CHEMBL240	Voltage-gated ion channel		444 / 18
Opioid growth factor receptor-like protein 1	OGFRL1	Q5TC84	CHEMBL3638334	Unclassified protein		15 / 13
Sigma opioid receptor	SIGMAR1	Q99720	CHEMBL287	Membrane receptor		387 / 245
Toll-like receptor 4 (by homology)	TLR4	O00206	CHEMBL5255	Toll-like and IL-1 receptors		13 / 11
Serotonin 2a (5-HT2a) receptor (by homology)	HTR2A	P28223	CHEMBL224	Family A G protein-coupled receptor		376 / 185

Figure 5: Results page for the predicted target against the ligand

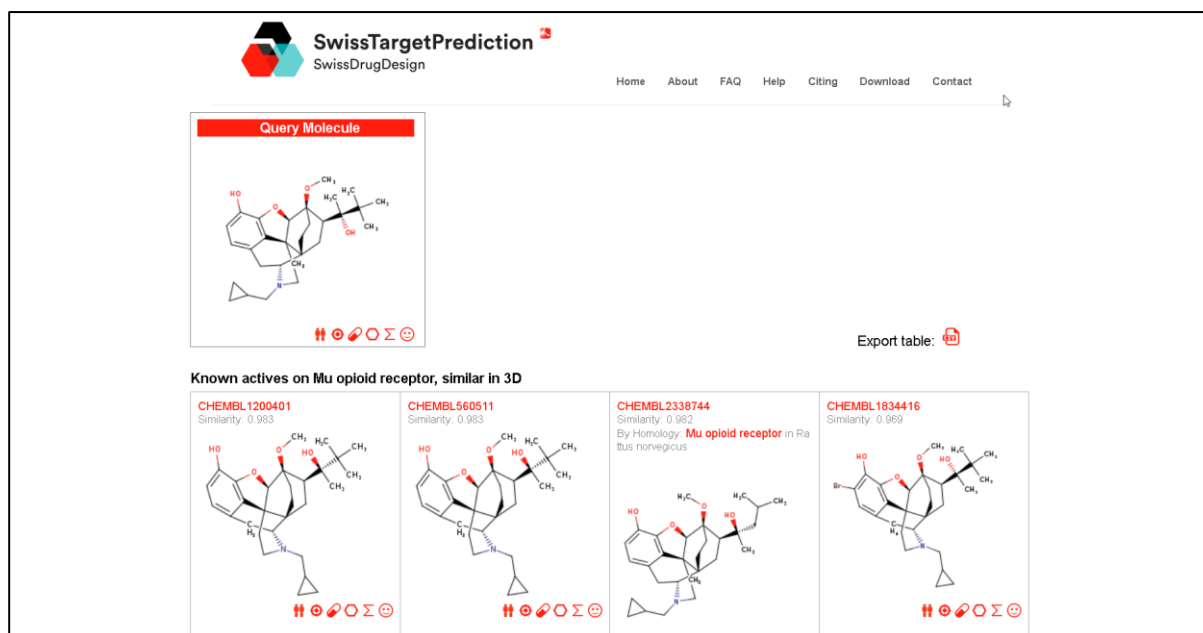


Figure 6: Results page for known actives on Mu Opiod receptor , similar in 3D

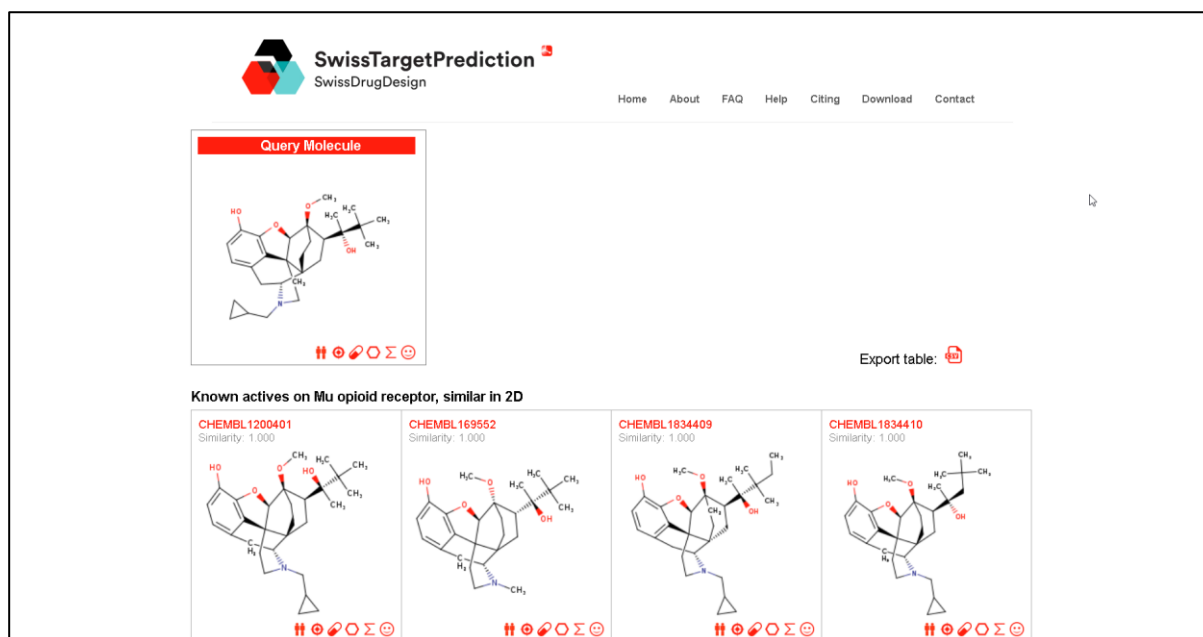


Figure 7: Results page for known actives on Mu Opiod receptor , similar in 2D

RESULTS:

Using the Swiss target prediction tool, the biological targets of Probuphine (SMILES: C[C@]([C@H]1C[C@@]23CC[C@@]1([C@H]4[C@@]25CCN([C@@H]3CC=C5C(=C(C=C6)O)O4)CC7CC7)OC)(C(C)(C)C)O) as a ligand molecule was analyzed for Homo sapiens. The analysis identified 100 target entries, with the Mu Opiod receptor (UniProt ID: P35372) being the most significant. The receptor is categorized as a nuclear receptor, with a ChEMBL ID of CHEMBL233. The results show 3D and 2D known actives, suggesting that Probuphine interacts effectively with nuclear receptors, indicating its potential biological relevance.

CONCLUSION:

The Swiss target prediction tool effectively identified the Mu Opioid receptor as a biological target for Probuphine, demonstrating its reliability in predicting drug-target interactions. This highlights the tool's capability in accurately predicting potential interactions for drug discovery.

REFERENCES:

1. Daina, A., Michielin, O., & Zoete, V. (2019). SwissTargetPrediction: updated data and new features for efficient prediction of protein targets of small molecules. *Nucleic Acids Research*, 47(W1), W357-W364. <https://doi.org/10.1093/nar/gkz382>
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 3. Probuphine: package insert / prescribing information. (n.d.). Drugs.com. <https://www.drugs.com/pro/probuphine.html>
 4. Probuphine Patient Tips: 7 things you should know. (n.d.). Drugs.com. <https://www.drugs.com/tips/probuphine-patient-tips>
-

WEBLEM: 2

High Throughput Virtual Screening Techniques For Drug Discovery

AIM:

Introduction to High throughput virtual screening techniques for drug discovery: Technological advancement in ligand discovery.

INTRODUCTION:

SwissSimilarity: A Comprehensive Tool for Drug Discovery

SwissSimilarity is a powerful web tool developed by the Molecular Modelling Group of the SIB Swiss Institute of Bioinformatics and the University of Lausanne. It is designed to facilitate low-to-high-throughput virtual screening of multiple chemical libraries to identify molecules similar to a compound of interest. This tool plays a crucial role in rational drug discovery by supporting key tasks such as

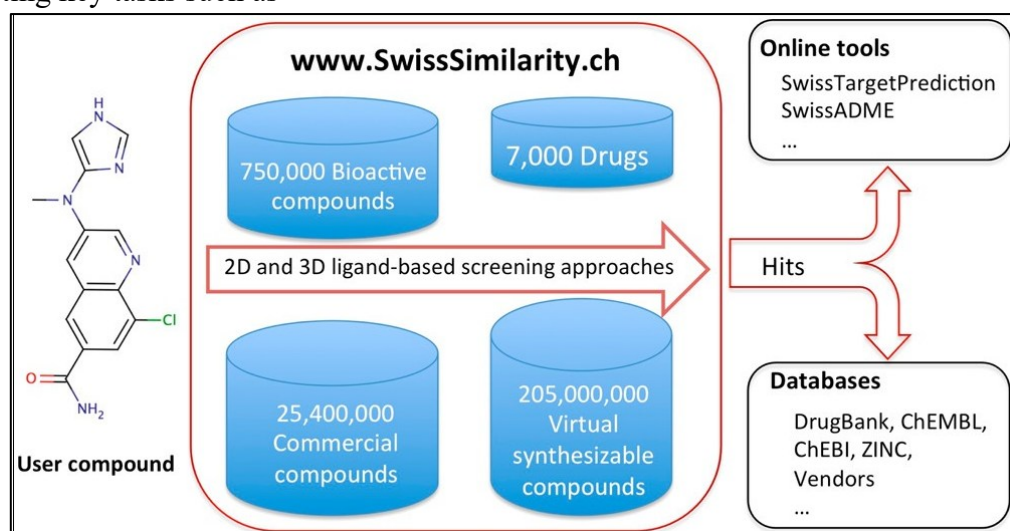


Figure 1: Overview of Swiss similarity Database

SwissSimilarity is freely available online at <http://www.swisssimilarity.ch>. The tool has undergone significant updates, including new molecular descriptors and an improved user interface, making it a valuable resource for drug discovery scientists.

SwissSimilarity offers a wide range of compound libraries for screening, categorized into four main groups:

1. Drugs and Clinical Candidates:

- Approved drugs
- Experimental drugs
- Investigational drugs
- Withdrawn drugs
- Illicit drugs
- Nutraceutical compounds

2. Bioactive Compounds:

Known bioactive molecules from databases like LigandExpo and ChEMBL.

3. Commercially Available Compounds:

Compounds available from vendor catalogs and ZINC subsets.

4. Synthesizable Molecules:

Over 200 million virtual compounds that can be readily synthesized from commercially available reagents.

Key Features and Applications:

1. Compound Libraries

SwissSimilarity offers a diverse range of compound libraries for screening, including:

- a. Approved Drugs: FDA-approved and experimental drugs.
- b. Bioactive Compounds: Known bioactive molecules from databases like LigandExpo and ChEMBL.
- c. Commercial Compounds: Available from vendor catalogs and ZINC subsets.
- d. Synthesizable Compounds: Over 200 million virtual compounds readily synthesizable from commercially available reagents.

2. Molecular Similarity Methods

The tool utilizes various 2D and 3D molecular descriptors to calculate similarity:

- a. 2D Methods: FP2, ECFP4, MHFP6, 2D pharmacophore, ErG fingerprints, extended-connectivity, MinHash, and extended reduced graph.
- b. 3D Methods: ES5D and E3FP fingerprints. It also allows searching for compounds with the same Murcko or generic Murcko scaffold.

3. Applications in Drug Discovery

- a. Hit Finding: Identifying novel compounds that exhibit similar bioactivity to a known molecule.
- b. Scaffold Hopping: Discovering compounds with different chemical cores but similar bioactivity.
- c. Structure-Activity Relationship (SAR) Studies: Finding easily accessible compounds for initial SAR studies.
- d. Drug Repurposing: Searching for similar compounds among marketed drugs or clinical candidates to repurpose them for new therapeutic uses.

Benefits and Usage:

- a. User Interface: The tool provides a user-friendly interface, making it accessible to both specialists and non-experts in the field.
- b. Usage Statistics: As of 2021 SwissSimilarity had performed over 150,000 screenings for nearly 50,000 unique users worldwide.
- c. Integration with Other Tools: It is part of the SwissDrugDesign suite which includes tools like SwissDock and SwissTargetPrediction, enhancing its utility in drug discovery workflows.

DataWarrior:

DataWarrior is an open-source cheminformatics software developed by OpenMolecules. It is widely used for chemical structure visualization, data analysis, and molecular modelling. The software is designed to help researchers in drug discovery, bioinformatics, and computational chemistry by providing tools for handling and analyzing chemical datasets.

DataWarrior is commonly used for chemical structure analysis, allowing users to visualize, edit, and search molecular structures. It supports data handling and visualization, enabling

researchers to manage large datasets through interactive charts and tables. The software is also used for structure-activity relationship analysis, which helps in identifying correlations between chemical structures and biological activity. Additionally, DataWarrior includes physicochemical property prediction features, allowing users to calculate molecular weight, logP, and topological polar surface area. It also supports clustering and pattern recognition to identify molecular similarities using various algorithms.

One of the main advantages of DataWarrior is that it is free and open source, making it accessible to both academic and industrial users. It has a user-friendly interface that provides intuitive visualization and filtering options. The software includes built-in cheminformatics tools for molecular descriptors, similarity searching, and structure-activity relationship analysis. It is also compatible with multiple operating systems, including Windows, macOS, and Linux.

Despite its benefits, DataWarrior has some limitations. It is primarily limited to two-dimensional molecular representations and lacks advanced three-dimensional molecular modeling features. Compared to commercial software like Schrödinger or MOE, it has fewer high-end capabilities. Additionally, new users may find it challenging to use without prior knowledge of cheminformatics concepts.

Overall, DataWarrior remains a valuable tool for cheminformatics and bioinformatics research, especially for those looking for a cost-effective and versatile solution. Let me know if you need guidance on a specific feature or workflow.

Features of DataWarrior

DataWarrior is an open-source cheminformatics software developed by OpenMolecules. It combines chemical structure visualization, data analysis, and molecular modeling, making it a valuable tool for researchers in drug discovery, bioinformatics, and computational chemistry.

1. Chemical Structure Handling

- Allows users to visualize, edit, and search chemical structures.
- Supports SMILES, InChI, and SDfile formats for importing and exporting molecular data.
- Provides substructure and similarity searches for compound identification.

2. Data Analysis and Visualization

- Offers interactive tables, charts, and scatter plots to explore large datasets.
- Supports Principal Component Analysis (PCA), clustering, and pattern recognition for data interpretation.

3. Structure-Activity Relationship (SAR) Analysis

- Helps in identifying correlations between chemical structures and biological activity.
- Includes activity cliffs detection to highlight significant changes in molecular properties.

4. Physicochemical Property Calculations

- Computes molecular weight, logP, topological polar surface area (TPSA), hydrogen bond donors/acceptors, and rotatable bonds.
- Predicts drug-likeness and bioavailability scores to assess molecular properties.

5. Clustering and Similarity Searching

- Groups molecules based on structural similarity or physicochemical properties.
- Supports Tanimoto similarity searching for compound comparison.

6. Multi-Platform Compatibility

- Works on Windows, macOS, and Linux without requiring extensive setup.

7. Free and Open Source

- Available without licensing restrictions, making it accessible for both academic and industrial use.

Steps for DataWarrior installation:

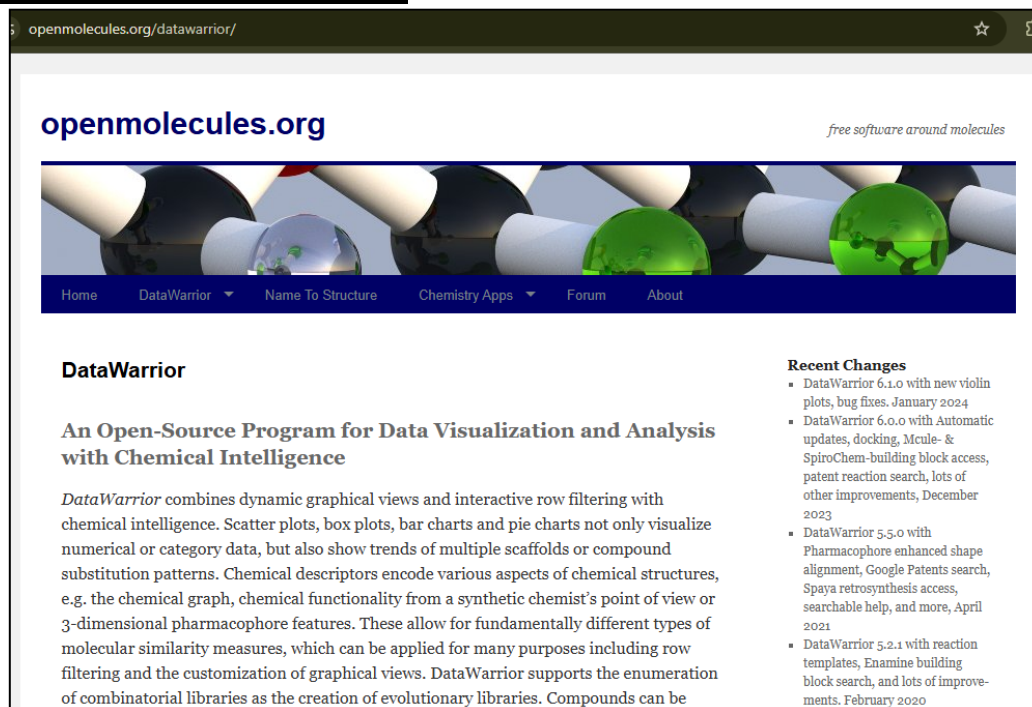


Figure 2: Open Homepage of openmolecules.org

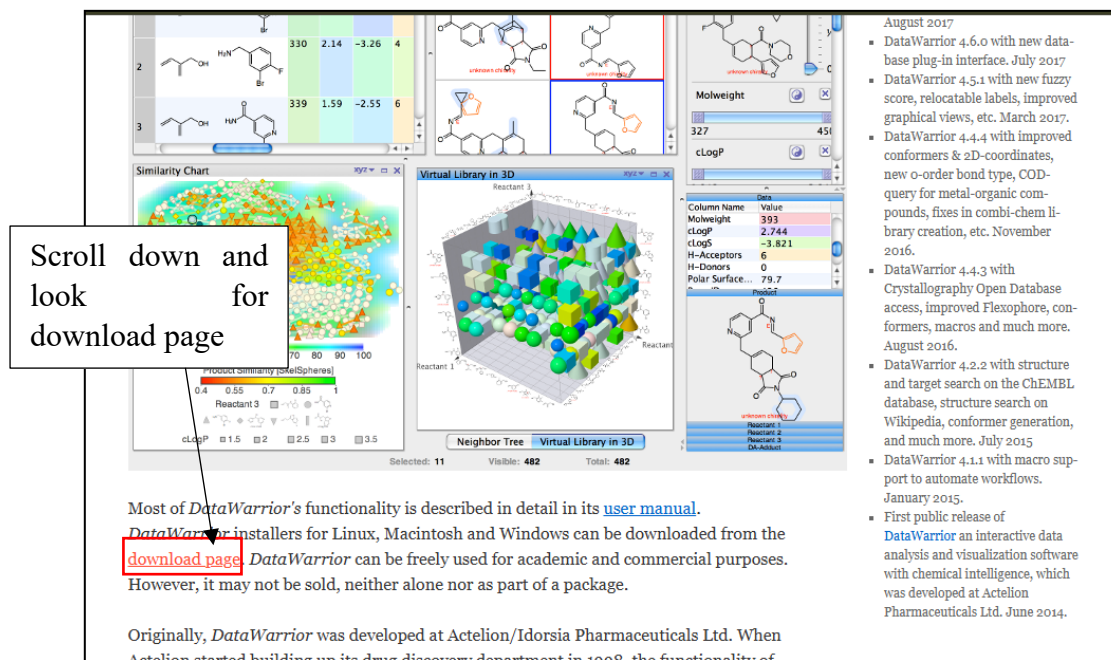


Figure 3: Click on the Download page to download DataWarrior



Figure 4: Download option for DataWarrior

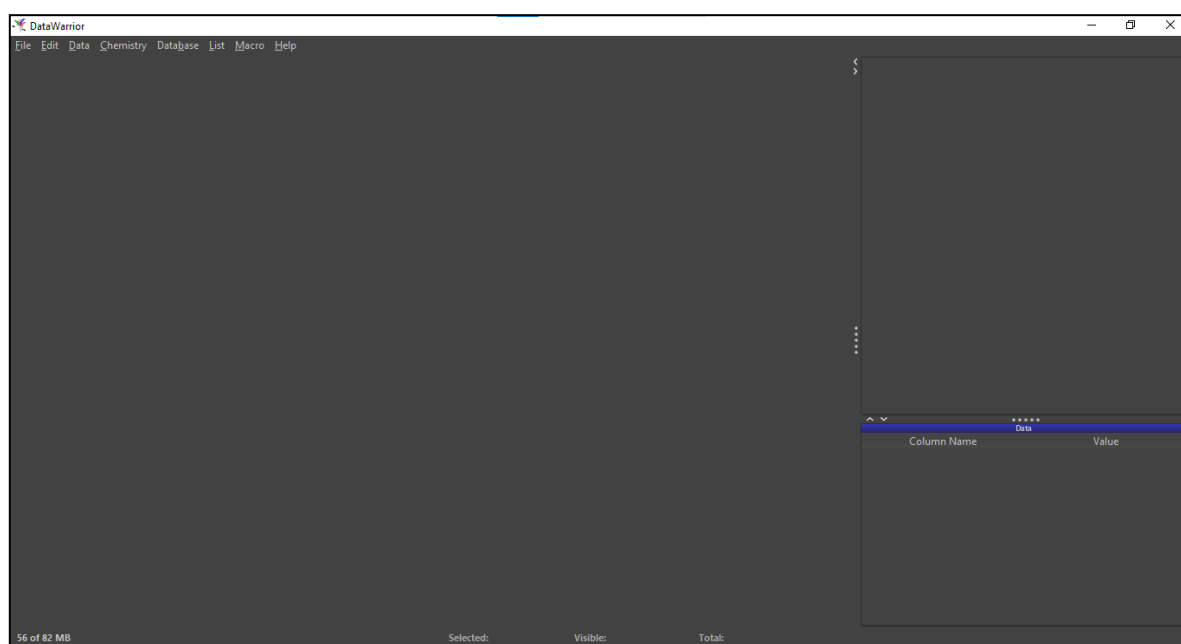


Figure 5: Open the DataWarrior application from desktop

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1. Sander, T., Freyss, J., Von Korff, M., & Rufener, C. (2015). DataWarrior: An open-source program for chemistry-aware data visualization and analysis. *Journal of Chemical Information and Modeling*, 55(2), 460-473. <https://doi.org/10.1021/ci500588j>
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 6. Daina, A., & Zoete, V. (2019). Application of the SwissDrugDesign online resources in virtual screening. International Journal of Molecular Sciences, 20(18), 4612. <https://doi.org/10.3390/ijms20184612>
 7. Molecular Modelling Group. (n.d.). <https://www.molecular-modelling.ch/swiss-drug-design.html>
-

Date: 20/02/2025

WEBLEM: 2(A)
Swiss Similarity Tool
(URL: <http://www.swiss similarity.ch/>)

AIM:

To create a library of compounds similar to query Azithromycin via high throughput screening using Swiss Similarity tool.

INTRODUCTION:

SwissSimilarity is an online tool developed by the Swiss Institute of Bioinformatics (SIB) for ligand-based virtual screening. It allows users to find compounds structurally or functionally similar to a given query molecule making it a valuable resource in drug discovery and chemoinformatics. By leveraging multiple similarity approaches such as fingerprint-based, shape-based and pharmacophore-based methods, SwissSimilarity can screen large chemical libraries including DrugBank, ChEMBL and ZINC to identify potential drug candidates. The tool is designed for ease of use, requiring only a molecule's SMILES or structural representation to generate a ranked list of similar compounds.

This tool is widely used in lead compound identification, drug repurposing and hit expansion in pharmaceutical research. Researchers can explore alternative bioactive molecules, assess potential drug candidates and refine chemical structures for improved efficacy. SwissSimilarity provides a web-based interface that requires no installation, making it accessible for both academic and industry users. By offering a free and open-access platform, it facilitates early-stage drug discovery and computational chemistry applications.

Azithromycin

Azithromycin is a macrolide antibiotic used to treat various bacterial infections. It works by inhibiting bacterial protein synthesis, binding to the 50S ribosomal subunit, which stops bacterial growth.

Key Points:

1. **Clinical Use:** Effective against respiratory, urogenital, dermal, and sexually transmitted infections, as well as MAC infections in HIV patients.
2. **Pharmacokinetics:** Well-absorbed orally with a long half-life, allowing for short treatment courses.
3. **Immunomodulatory Effects:** Reduces inflammation, making it useful in chronic conditions like cystic fibrosis and COPD.
4. **Safety and Interactions:** Generally safe with fewer drug interactions than other macrolides. Common side effects include gastrointestinal issues and allergic reactions.

METHODOLOGY:

1. Search for your query in the PubChem database.
2. Retrieve canonical SMILES of the best match.
3. Open Swiss similarity and enter your smiles string.
4. Select class compound.

5. Select compound library and screening method
6. View results obtained (i.e. Run parameters, Query molecule and its Reference targets)
7. Download in csv format.
8. View it in excel file format.

OBSERVATIONS:

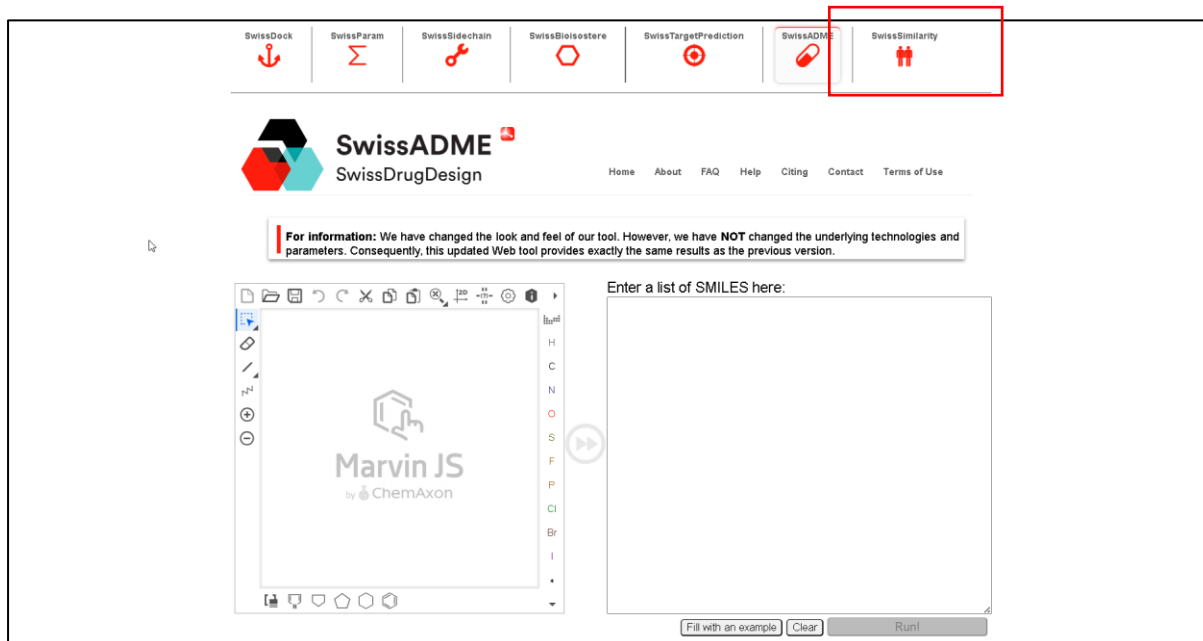


Figure 1: Homepage of SwissADME

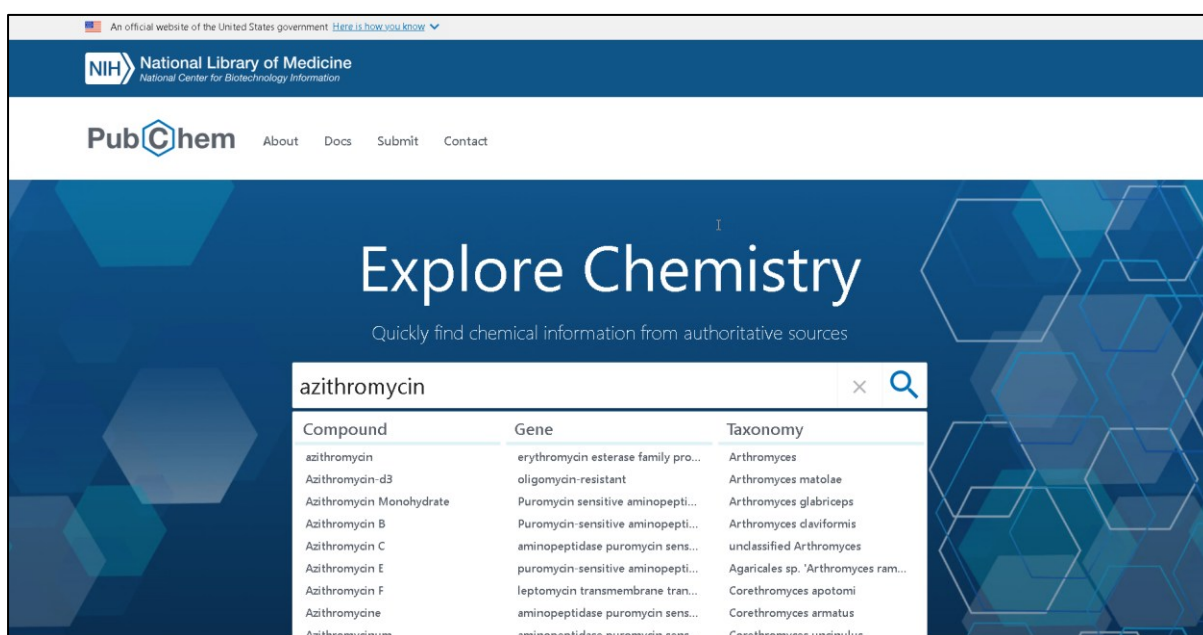


Figure 2: Searched for Drug Azithromycin in PubChem database



SwissSimilarity
 SwissDrugDesign

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Don't know where to start? Try with an example [Diclofenac](#), [Propranolol](#) or [Nilotinib](#).

1 - Enter a molecule in SMILES format

No SMILES available? Draw a molecule **using the sketcher**

2 - Select a class of compounds

Please, select a class of compounds here:

3 - Select compound library and screening method

Put the mouse over a radio button to see the corresponding computation time

	2D								3D	2D & 3D
	FP2	ECFP4	MHFP6	Pharmacophore	ERG	Scaffold	Generic Scaffold	Electroshape	E3FP	Combined
LigandExpo	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
ChEBI	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Figure 4A: Select class compound

2 - Select a class of compounds

Please, select a class of compounds here:

3 - Select compound library and screening method

Put the mouse over a radio button to see the corresponding computation time

	2D								3D	2D & 3D
	FP2	ECFP4	MHFP6	Pharmacophore	ERG	Scaffold	Generic Scaffold	Electroshape	E3FP	Combined
LigandExpo	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
ChEBI	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
GPCR Glass	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
HMDB	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
ChEMBL (full database; 29)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒
ChEMBL (actives only)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
ChEMBL (GPCRs)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
ChEMBL (kinases)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
ChEMBL (proteases)	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

4 - Submit

Will use method **Combined** to screen the library **CHEMBL**
 (Computation time, excluding queueing delay: 64 seconds)



Figure 4B: Select class compound



SwissSimilarity
 SwissDrugDesign

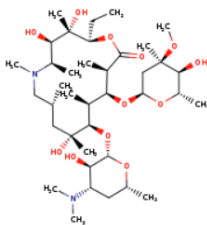
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Run parameters

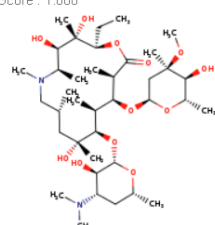
Library screened: CHEMBL
 Screening method: Combined
 Date: March 14, 2025, 1:37 pm UTC

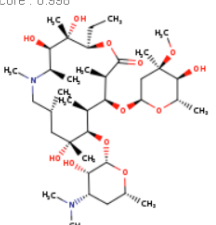
If you publish these results, please, cite the following paper:

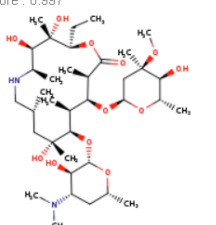
Zoete, V., Daina, A., Bovigny, C., & Michielin, O. SwissSimilarity: A Web Tool for Low to Ultra High Throughput Ligand-Based Virtual Screening., *J. Chem. Inf. Model.*, **2016**, 56(8), 1399-1404.

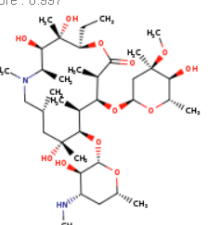
Query Molecule


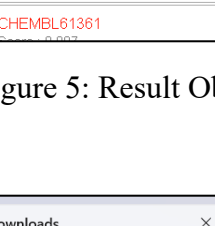
Results

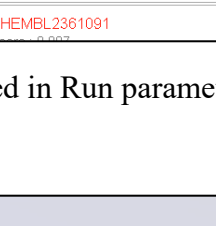
CHEMBL529
 Score : 1.000


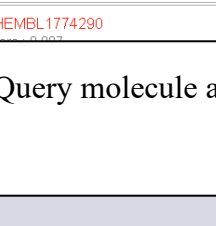
CHEMBL3348928
 Score : 0.998


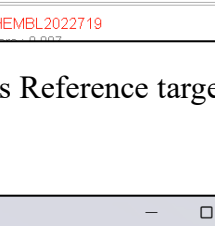
CHEMBL1222181
 Score : 0.997


CHEMBL2022716
 Score : 0.997


CHEMBL61361
 Score : 0.997


CHEMBL2361091
 Score : 0.997


CHEMBL1774290
 Score : 0.997


CHEMBL2022719
 Score : 0.997


Retrieve data:
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Figure 5: Result Obtained in Run parameters, Query molecule and its Reference targets

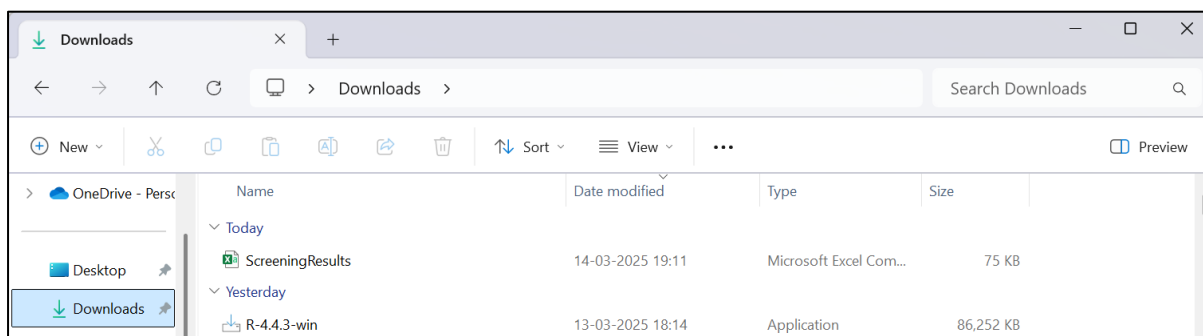


Figure 6: Results downloaded in CSV format

A1	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	CHEMBL529	1.000	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
2	CHEMBL3348928	0.998	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
3	CHEMBL122181	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
4	CHEMBL2022716	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
5	CHEMBL61361	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
6	CHEMBL2361091	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
7	CHEMBL1774290	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
8	CHEMBL2022719	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
9	CHEMBL1200502	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
10	CHEMBL58145	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
11	CHEMBL301375	0.997	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
12	CHEMBL274365	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
13	CHEMBL2367783	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
14	CHEMBL294380	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
15	CHEMBL2114273	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
16	CHEMBL300373	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
17	CHEMBL1774453	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
18	CHEMBL96349	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
19	CHEMBL441792	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
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21	CHEMBL273931	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
22	CHEMBL2115261	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
23	CHEMBL300372	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
24	CHEMBL311220	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
25	CHEMBL2115255	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
26	CHEMBL2115259	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
27	CHEMBL4082834	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
28	CHEMBL293343	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
29	CHEMBL2114267	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
30	CHEMBL3038222	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			
31	CHEMBL2114272	0.996	CC[C@H]1OC(=O)[C@H](C)[C@H](O)[C@H]2C[C@@H](C)OC[C@H]1O																			

Figure 7: Obtained results with its CHEMBL id, energy and smiles format

RESULTS:

Using Swiss Similarity, structurally and functionally similar compounds to Azithromycin were identified. CHEMBL529 & CHEMBL3348928 showed highest similarity to our target query. The results were retrieved and downloaded in the .csv format and was studied and examined via excel.

CONCLUSION:

By utilizing different similarity approaches and screening diverse chemical libraries the tool provided valuable insights into potential drug candidates. The results obtained highlight the importance of computational methods in modern drug discovery allowing for rapid identification of bioactive molecules with therapeutic potential.

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Date: 23/02/2025

WEBLEM: 2(B)

DataWarrior Program

(URL: <https://openmolecules.org/datawarrior/>)

AIM:

To filter the SwissSimilarity results for 'Lapatinib' based on druglikeness using DataWarrior.

INTRODUCTION:

DataWarrior is an open-source cheminformatics software developed by OpenMolecules. It is widely used for chemical structure visualization, data analysis, and molecular modelling. The software is designed to help researchers in drug discovery, bioinformatics, and computational chemistry by providing tools for handling and analyzing chemical datasets.

DataWarrior is commonly used for chemical structure analysis, allowing users to visualize, edit, and search molecular structures. It supports data handling and visualization, enabling researchers to manage large datasets through interactive charts and tables. The software is also used for structure-activity relationship analysis, which helps in identifying correlations between chemical structures and biological activity. Additionally, DataWarrior includes physicochemical property prediction features, allowing users to calculate molecular weight, logP, and topological polar surface area. It also supports clustering and pattern recognition to identify molecular similarities using various algorithms.

One of the main advantages of DataWarrior is that it is free and open source, making it accessible to both academic and industrial users. It has a user-friendly interface that provides intuitive visualization and filtering options. The software includes built-in cheminformatics tools for molecular descriptors, similarity searching, and structure-activity relationship analysis. It is also compatible with multiple operating systems, including Windows, macOS, and Linux.

Lapatinib

Lapatinib, known by the brand name Tykerb, is a dual tyrosine kinase inhibitor targeting the epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (HER2). It is used to treat HER2-positive advanced or metastatic breast cancer, often in combination with other medications like capecitabine or letrozole. Lapatinib works by inhibiting tumor cell growth through the blockade of EGFR and HER2 signaling pathways. Common side effects include rash and gastrointestinal issues. It is administered orally and requires caution with certain drug interactions, particularly with strong CYP3A4 inhibitors.

METHODOLOGY:

1. Open data warrior application
2. Click on "file" option select "open"
3. Upload the CSV file generated using Swiss Similarity
4. Apply the filters to filter the data

5. Interpret the results.

OBSERVATIONS:

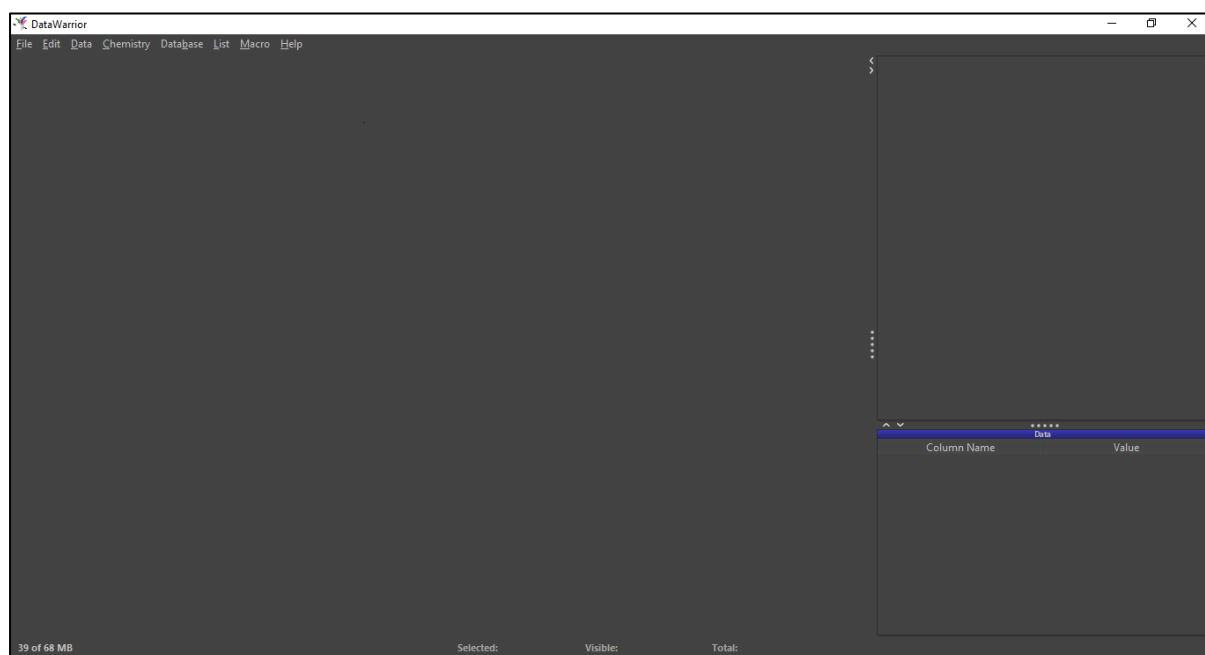


Figure 1: Open the application

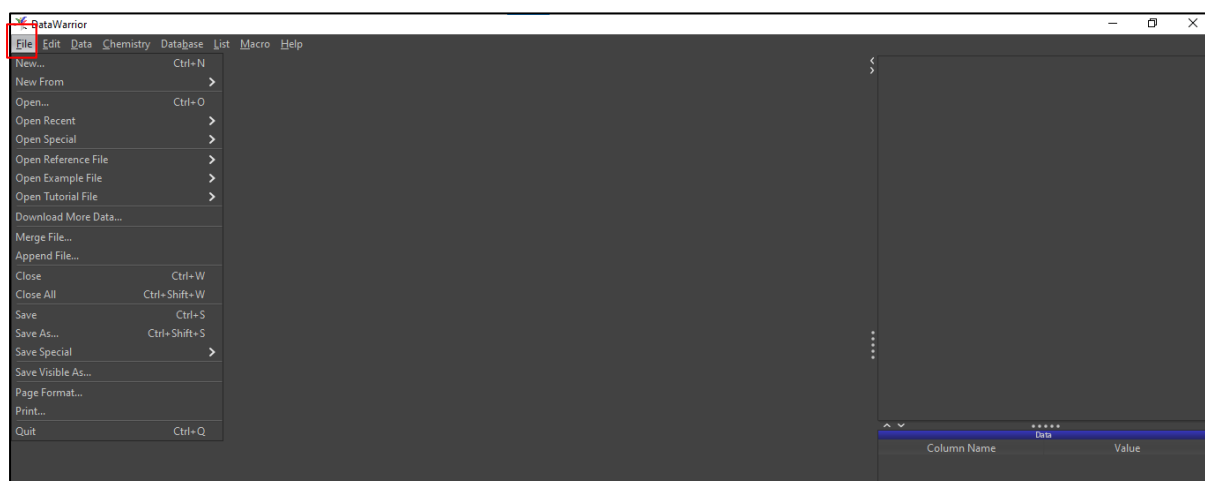


Figure 2: Click on “File” option

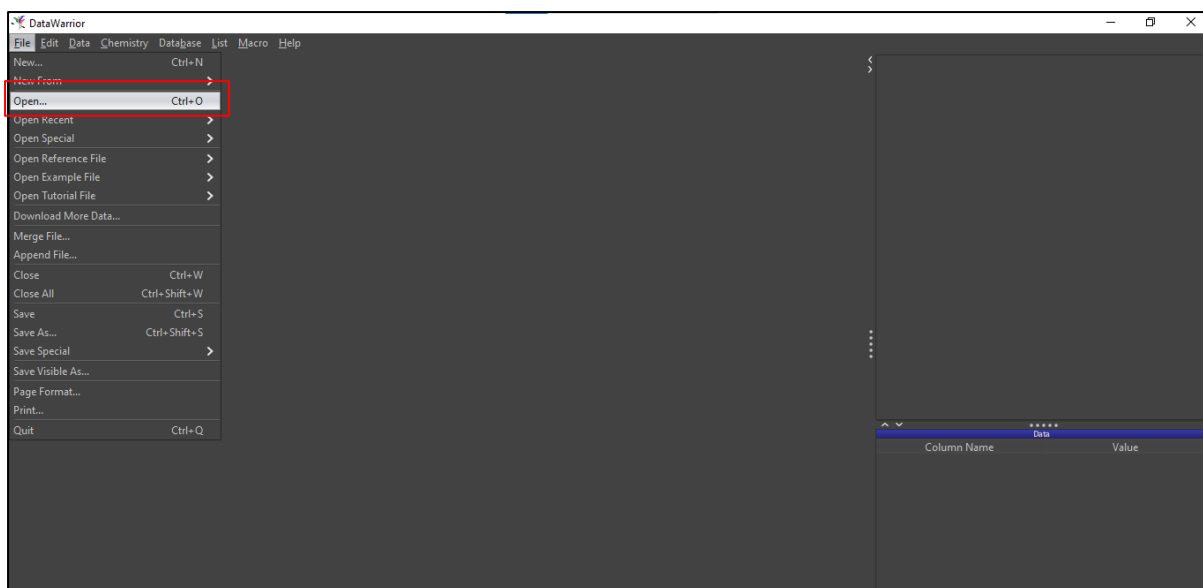


Figure 3: Click “open file” to access the CSV File

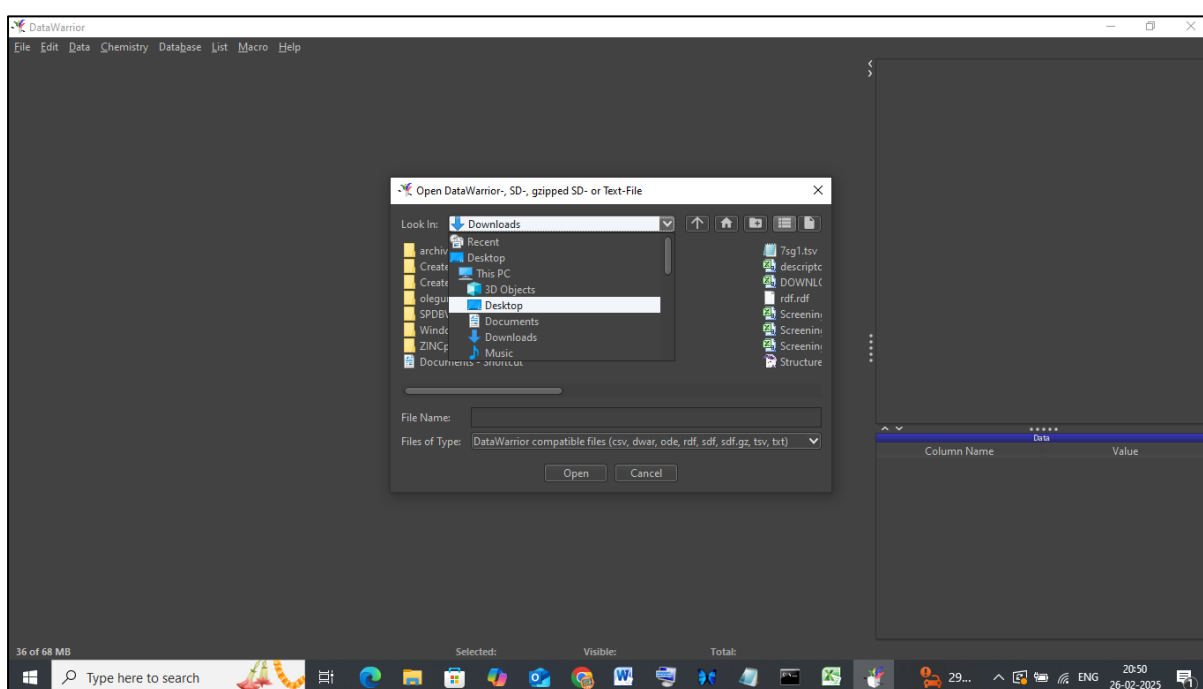
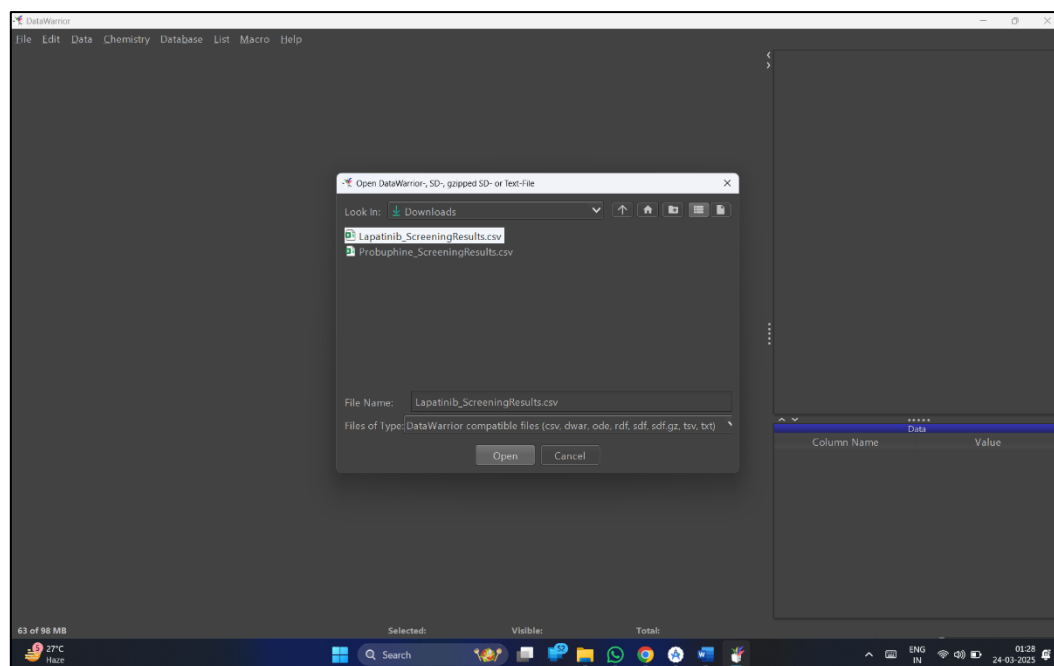


Figure 3a: Upload your csv file



3b: Select the file and open it

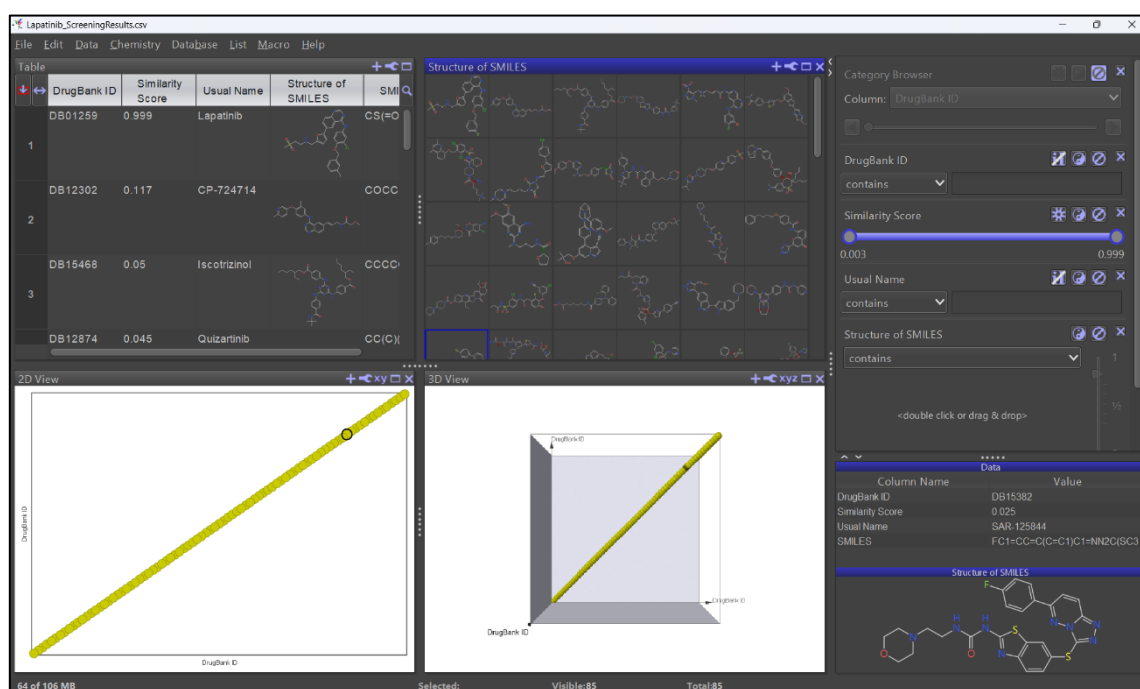


Figure 4: Representation of data in DataWarrior

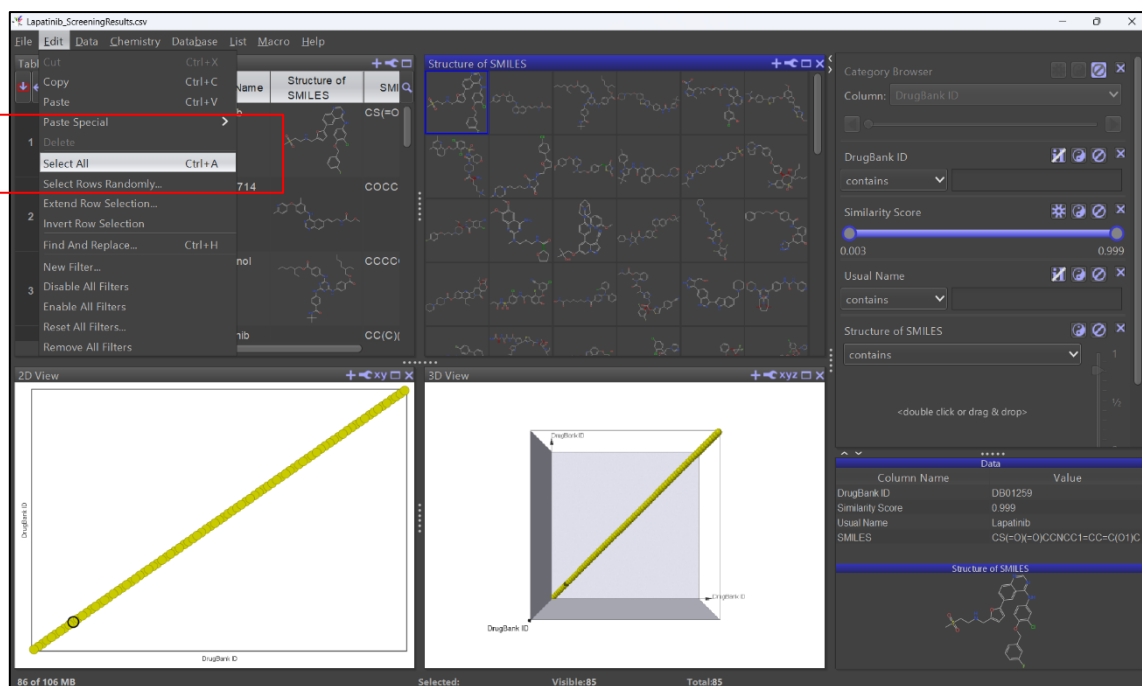


Figure 5: Click on “Select all” option

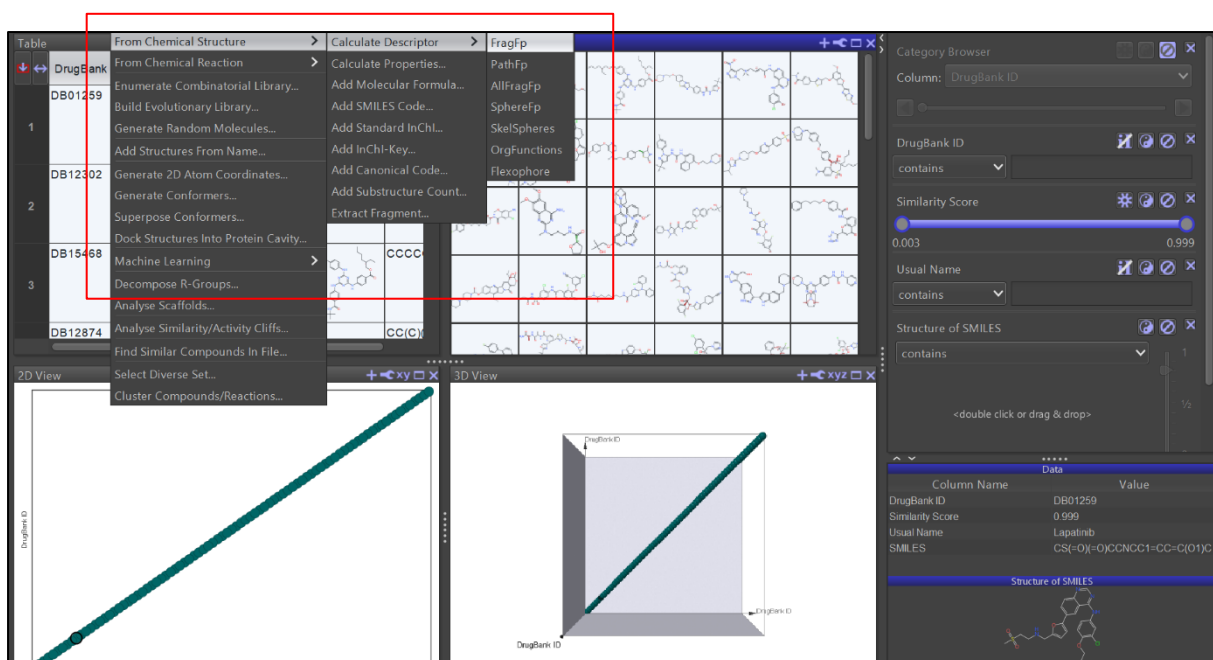


Figure 6: Various filters available in DataWarrior

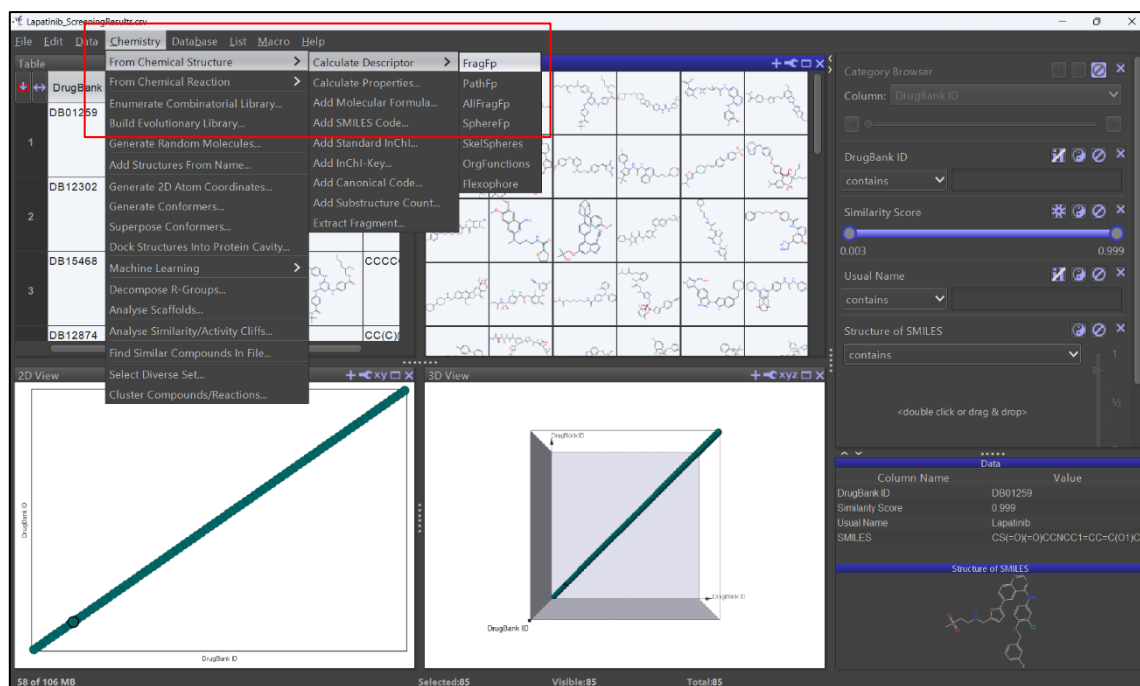


Figure 7: Click on “Chemistry” to calculate properties

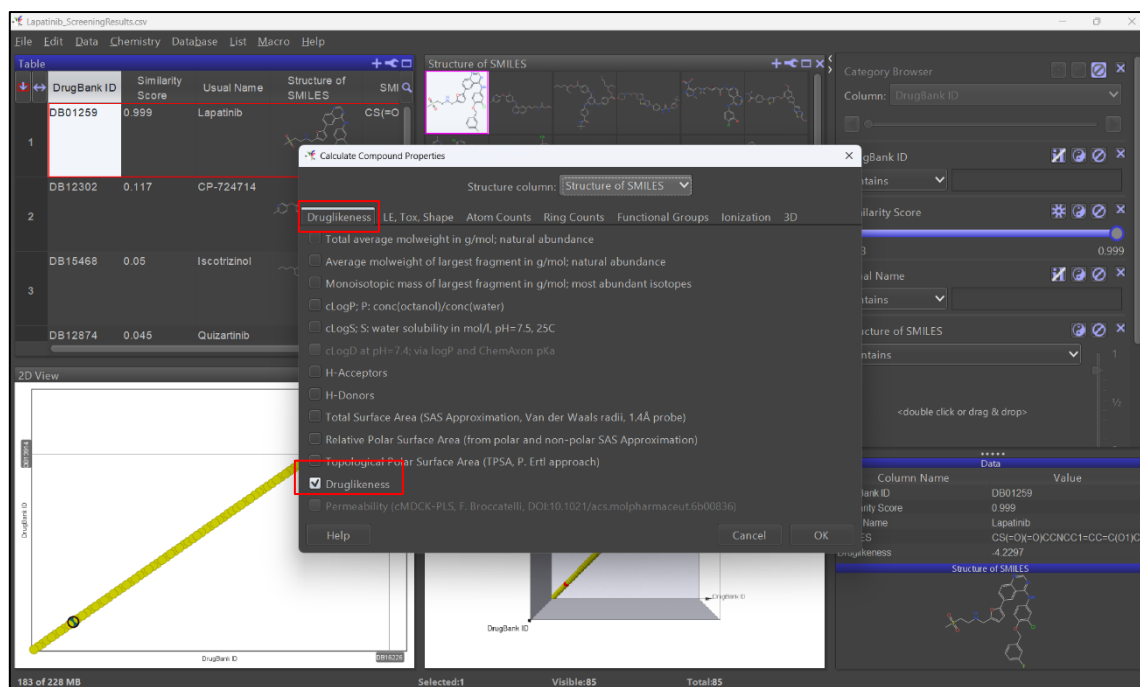


Figure 8: Tick on “Druglikeness” for check the Druglikeness

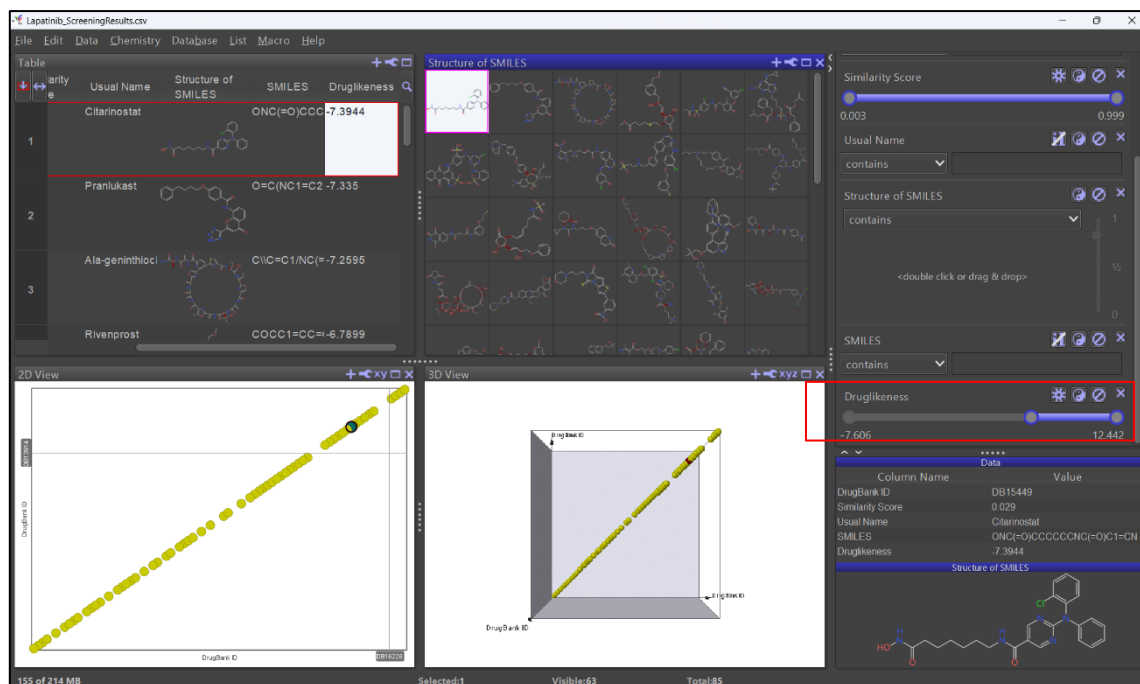


Figure 9: Setting the threshold for druglikeness

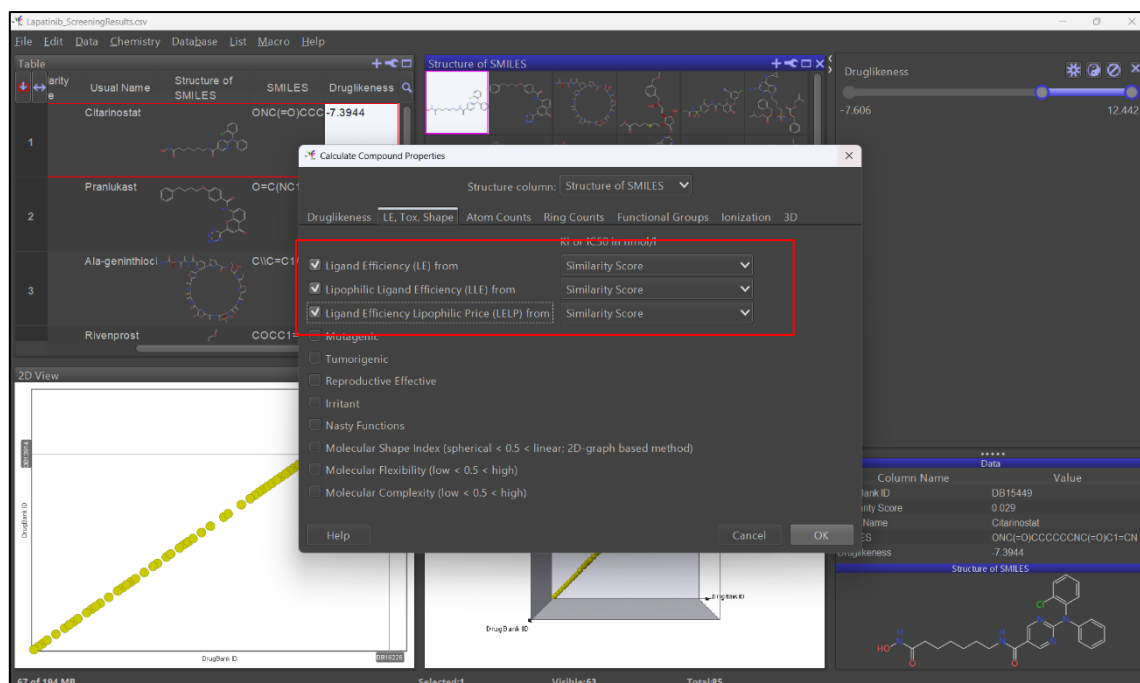


Figure 10: Tick on “Ligand Efficiency” for Ligand Efficiency calculation

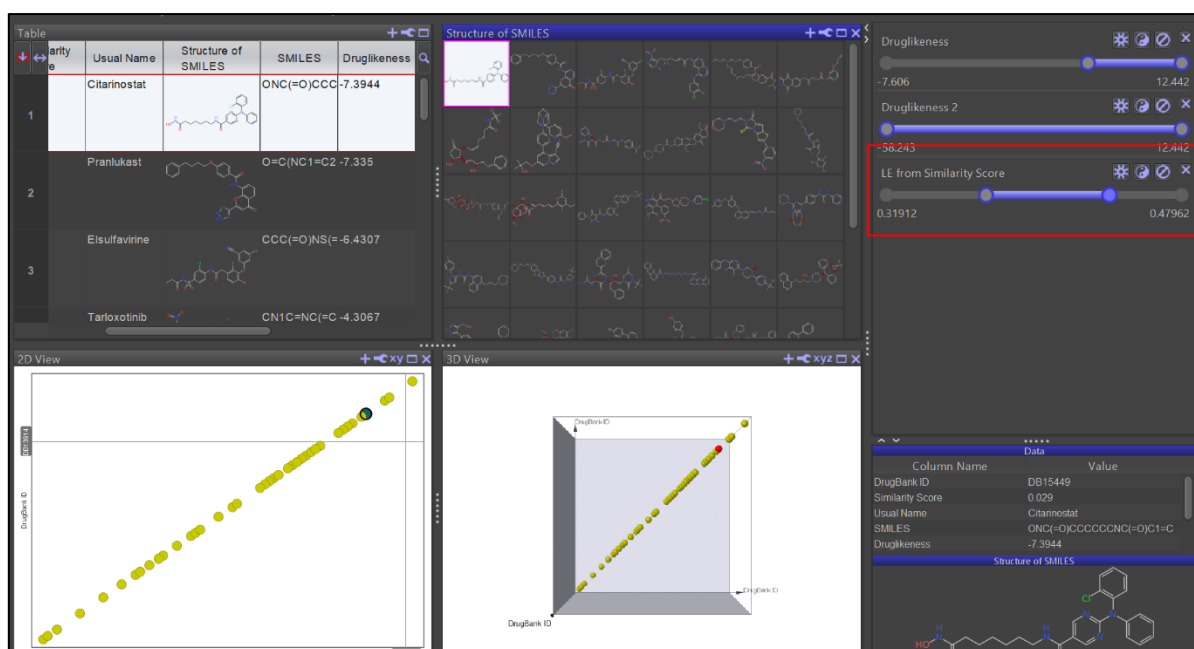


Figure 11: Setting the threshold for ligand efficiency

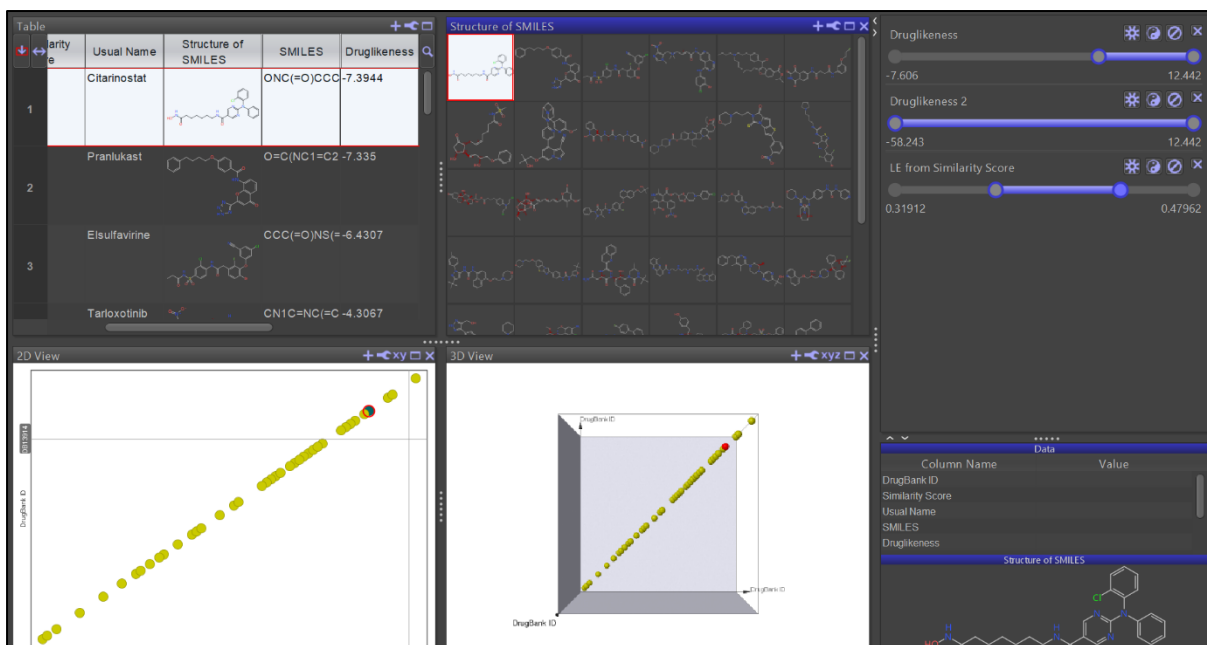


Figure 12: Optimization of leads according to the threshold values

RESULTS:

DataWarrior tool was explored to screen the similar compounds of 'Lapatinib', the dataset contained 339 compounds, which were filtered down to 85 using various physicochemical and drug-likeness criteria. The applied filters included calculated descriptors, calculated properties, drug-likeness, lipophilicity (logP), ligand efficiency (LE), lipophilic ligand efficiency (LLE), and ligand efficiency-dependent lipophilicity (LELP). The selected range for druglikeness was from -7.9 to 12.4, effectively narrowing down the dataset to compounds that meet desirable drug-like characteristics.

CONCLUSION:

DataWarrior proved to be an effective tool for filtering and analyzing a large chemical dataset, helping to prioritize compounds with favourable drug-like properties. The applied filters significantly reduced the dataset while retaining compounds that meet specific ADME and lipophilicity criteria. The ability to modify, select, and manage data columns further enhanced the workflow, making it easier to extract meaningful insights. This approach is highly valuable in drug discovery and cheminformatics, allowing researchers to efficiently screen and refine potential candidates for further study.

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-

WEBLEM: 3
Introduction to SwissDock Server

AIM:

Introduction to Optimizing hits and leads studies using computational approach.

INTRODUCTION:

Docking refers to the computational modeling process that predicts how two molecules, typically a protein and a ligand (such as a drug), interact with each other. The goal of docking is to determine the preferred binding orientation and affinity of the ligand to the protein's active site or another relevant binding pocket. This process is crucial in drug discovery and molecular biology, as it helps researchers identify potential drug candidates and understand molecular interactions at an atomic level. Ligand-target binding refers to the interaction between a ligand (a small molecule, peptide, or ion) and its biological target, which is usually a protein, enzyme, or receptor. This binding is essential in biological processes and drug discovery, as it determines how a ligand influences the function of its target. The strength of this interaction is quantified by binding affinity, typically measured in terms of dissociation constants (K_d) or inhibition constants (K_i), with lower values indicating stronger binding. Ligand optimization modifying a ligand's structure to enhance its binding affinity, specificity, stability, and drug-like properties. This optimization is crucial in drug design, where initial hit compounds are iteratively improved to develop potent and selective drugs.

SwissDock is a docking web server that addresses the limitations described above. The structure of the target protein, as well as that of the ligand, can be automatically prepared for docking. In addition, the cumbersome syntax of the docking engine is hidden behind a clean web interface providing reasonable alternative sets of parameters as well as sample input files. All calculations are performed on the server side, so that docking runs do not require any computational power from the user. The interpretation of docking results and their integration into existing research pipelines is greatly facilitated by the seamless visualization of docking predictions in the UCSF Chimera molecular viewer, which can be launched directly from the web browser. SwissDock is an online molecular docking tool developed by the Molecular Modeling Group at the Swiss Institute of Bioinformatics (SIB). It is designed to predict the binding modes of small molecules (ligands) to target proteins, making it a valuable resource for drug discovery, structural biology, and computational chemistry research. SwissDock is based on the EADock DSS docking algorithm and utilizes the CHARMM (Chemistry at HARvard Macromolecular Mechanics) force field to ensure accurate energy calculations and molecular interactions.

One of SwissDock's primary advantages is its accessibility as a web-based tool, eliminating the need for high-end computational resources or complex software installations. The tool allows for two docking modes: blind docking and focused docking. Blind docking explores the entire surface of the protein to identify potential binding sites, which is useful when the binding site is unknown. Focused docking, on the other hand, restricts the docking process to a specific region of the protein, allowing for more targeted analysis when a known binding site is available. To use SwissDock, users must provide input files including the target protein structure (typically in PDB format) and the ligand (small molecule) structure. The docking process generates multiple binding conformations of the ligand, ranking them based on their

estimated binding energy. Lower energy values indicate more stable interactions between the ligand and the protein. The results are provided in a visual format, often compatible with molecular visualization tools like Chimera, allowing researchers to analyze docking poses in detail.

SwissDock is widely used in pharmaceutical research, particularly for virtual screening and lead optimization in drug development. It helps identify promising candidate molecules for further experimental validation. Its accuracy, free accessibility, and user-friendly interface make it a popular choice among scientists working in molecular modeling and drug discovery.

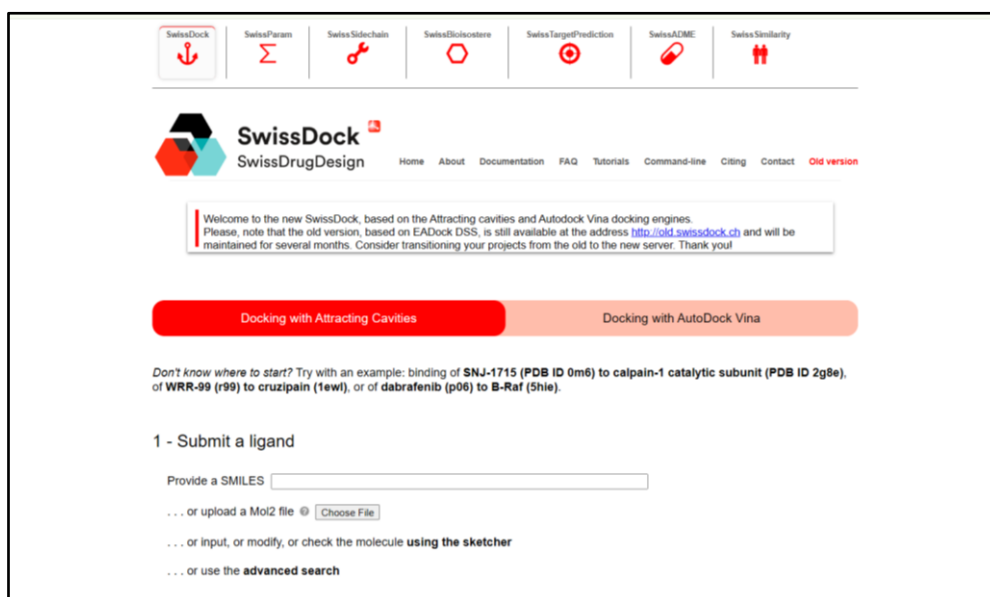


Figure 1: Homepage of SwissDock Tool

REFERENCES:

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3. Grosdidier, A., Zoete, V., & Michielin, O. (2011b). SwissDock, a protein-small molecule docking web service based on EADock DSS. *Nucleic Acids Research*, 39(suppl), W270–W277. <https://doi.org/10.1093/nar/gkr366>
4. Marine Bugnon, Ute F Röhrig, Mathilde Goullieux, Marta A S Perez, Antoine Daina, Olivier Michielin, Vincent Zoete, SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina, *Nucleic Acids Research*, Volume 52, Issue W1, 5 July 2024, Pages W324–W332, <https://doi.org/10.1093/nar/gkae300>

Date: 20/02/2025

WEBLEM: 3(A)
SwissDock Server
(URL: <https://www.swissdock.ch/>)

AIM:

To identify potential drug candidates for target query 'Colchicine' (PDB ID: 5NKN) using SwissDock Server.

INTRODUCTION:

Docking refers to the computational modeling process that predicts how two molecules, typically a protein and a ligand (such as a drug), interact with each other. The goal of docking is to determine the preferred binding orientation and affinity of the ligand to the protein's active site or another relevant binding pocket. This process is crucial in drug discovery and molecular biology, as it helps researchers identify potential drug candidates and understand molecular interactions at an atomic level.

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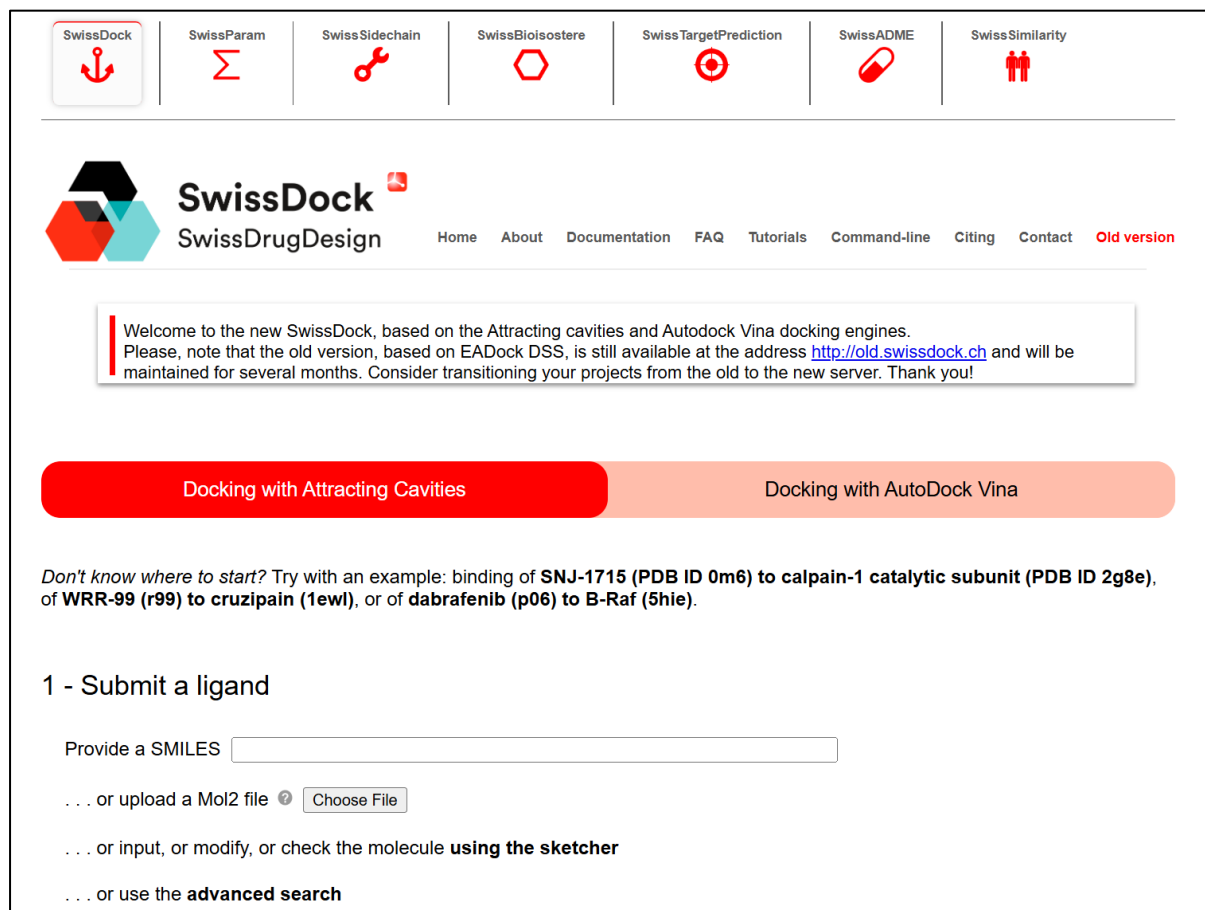
Colchicine:

Colchicine is a medication primarily used to treat and prevent gout attacks, manage familial Mediterranean fever (FMF), and reduce the risk of myocardial infarction. Derived from the *Colchicum autumnale* plant, it works by inhibiting the immune response to urate crystals, reducing inflammation and pain. Colchicine disrupts microtubule polymerization, affecting neutrophil motility and activation, and interferes with the inflammasome complex to reduce inflammatory mediators. It is administered orally and has a narrow therapeutic index, making overdosing a risk. Common side effects include gastrointestinal issues, while severe effects can include pancytopenia and rhabdomyolysis. Colchicine is rapidly absorbed but has limited brain uptake due to P-glycoprotein binding. It is important to avoid grapefruit and grapefruit juice while taking colchicine to prevent increased drug levels.

METHODOLOGY:

1. Open PubChem (<https://pubchem.ncbi.nlm.nih.gov/>) and search for (drug molecule) Colchicine (PDB ID:3P6G).
2. Copy the SMILES representation of the most relevant structure (based on similarity or relevance).
3. Paste the SMILES string into the SwissDock tool (<http://www.swissdock.ch/>).
4. Open the Protein Data Bank (PDB) (<https://www.rcsb.org/>) and search for your target protein.
5. Copy the PDB ID of the appropriate protein structure.
6. Paste the PDB ID of the selected target protein into the SwissDock tool.
7. Choose the appropriate docking parameters (default settings are recommended for beginners).
8. Enter your email ID to receive the docking results.
9. Click Start Docking and wait for the results to be processed.

OBSERVATIONS:



The screenshot shows the SwissDock homepage. At the top, there is a navigation bar with icons and labels for various tools: SwissDock (anchor icon), SwissParam (Sigma icon), Swiss Sidechain (wrench icon), SwissBioisostere (hexagon icon), SwissTargetPrediction (target icon), SwissADME (pill icon), and SwissSimilarity (people icon). Below this is the main header with the SwissDock logo and the text 'SwissDrugDesign'. A navigation menu includes links for Home, About, Documentation, FAQ, Tutorials, Command-line, Citing, Contact, and an 'Old version' link. A welcome message states: 'Welcome to the new SwissDock, based on the Attracting cavities and Autodock Vina docking engines. Please, note that the old version, based on EADock DSS, is still available at the address <http://old.swissdock.ch> and will be maintained for several months. Consider transitioning your projects from the old to the new server. Thank you!'. Below this, there are two large buttons: 'Docking with Attracting Cavities' (red) and 'Docking with AutoDock Vina' (orange). A text block provides an example: 'Don't know where to start? Try with an example: binding of SNJ-1715 (PDB ID 0m6) to calpain-1 catalytic subunit (PDB ID 2g8e), of WRR-99 (r99) to cruzipain (1ewl), or of dabrafenib (p06) to B-Raf (5hie)'. The section '1 - Submit a ligand' contains a form with a 'Provide a SMILES' input field, a link to '... or upload a Mol2 file' with a 'Choose File' button, a link to '... or input, or modify, or check the molecule using the sketcher', and a link to '... or use the advanced search'.

Figure 1: Homepage of SwissDock Tool

1 - Submit a ligand

Provide a SMILES

... or upload a Mol2 file

... or input, or modify, or check the molecule **using the sketcher**

... or use the **advanced search**



Prepare ligand

Reset ligand

Figure 2: Submit ligand of Colchicine in SMILE format from PubChem

2 - Submit a target

Provide a PDB id (e.g. 5hie)

... or upload a PDB file 5nkn.pdb

... or use the **advanced search**



Prepare target

Reset target

Figure 3: Submit target of PDB ID:3P6G from PDB database

3 - Define search space

Search box center Å

Search box size Å

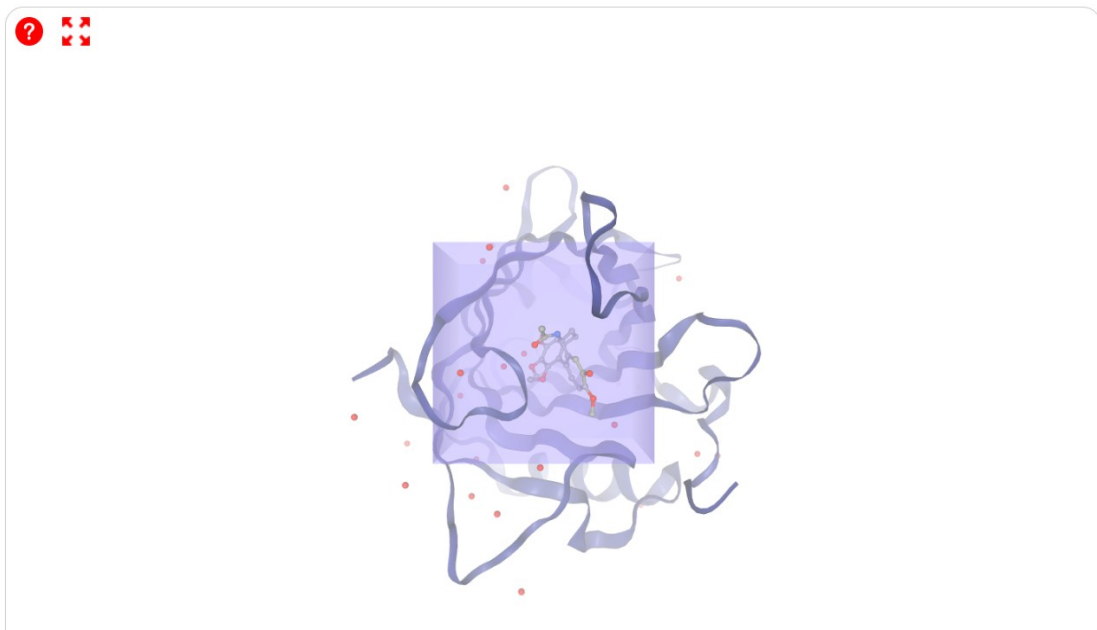


Figure 4: Define search space

4 - Select parameters

Number of RIC

Show extra parameters ☐

☒

This job is estimated to take 0:01:33 (h:mm:ss)

5 - Start docking

Enter an email (optional)

Enter a docking name (optional)

Figure 5: Select Parameter and click start docking

Parameters:

Submission date: March 15, 2025, 4:44 am UTC

Docking name: colchicine

Ligand: CC(=O)N[C@H]1CCC2=CC(=C(C(=C2C3=CC=C(C(=O)C=C13)OC)OC)OC)OC

Target: 5nkn.pdb

Box center: -10 - 15 - -13

Box size: 20 - 20 - 20

Docking method: AC

Sampling exhaustivity: medium

Cavity prioritization: buried

Number of RIC: 1

This job is estimated to take 0:01:33 (h:mm:ss)

You can close this page, you will receive an email at kothariparthi@gmail.com at the end of your docking.

Bookmark [this link](#) to retrieve your results.

Calculation currently running. Run time: 0:01

[STOP JOB](#)

Figure 6: Loading results for query Colchicine

Query

Ligand CC(=O)N[C@H]1CCC2=CC(=C(C(=C2C3=CC=C(C(=O)C=C13)OC)OC)OC)OC
Target 5nkn.pdb
Method Attracting Cavities 2.0
Date March 15, 2025, 4:44 am UTC

Parameters:

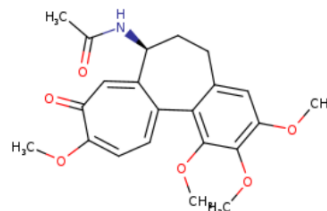
Box center: -10 - 15 - -13 Sampling exhaustivity: medium Number of RIC: 1
 Box size: 20 - 20 - 20 Cavity prioritization: buried

If you publish these results, please, cite the following papers:

Bugnon M, Röhrig UF, Goullieux M, Perez MAS, Daina A, Michielin O, Zoete V. SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. *Nucleic Acids Res.* **2024**

Röhrig UF, Goullieux M, Bugnon M, Zoete V. Attracting Cavities 2.0: improving the flexibility and robustness for small-molecule docking. *J. Chem. Inf. Model.*, **2023**

Ligand



Σ # @ * O ☺

Results

Export your results:  

Figure 7: Details of query and ligand

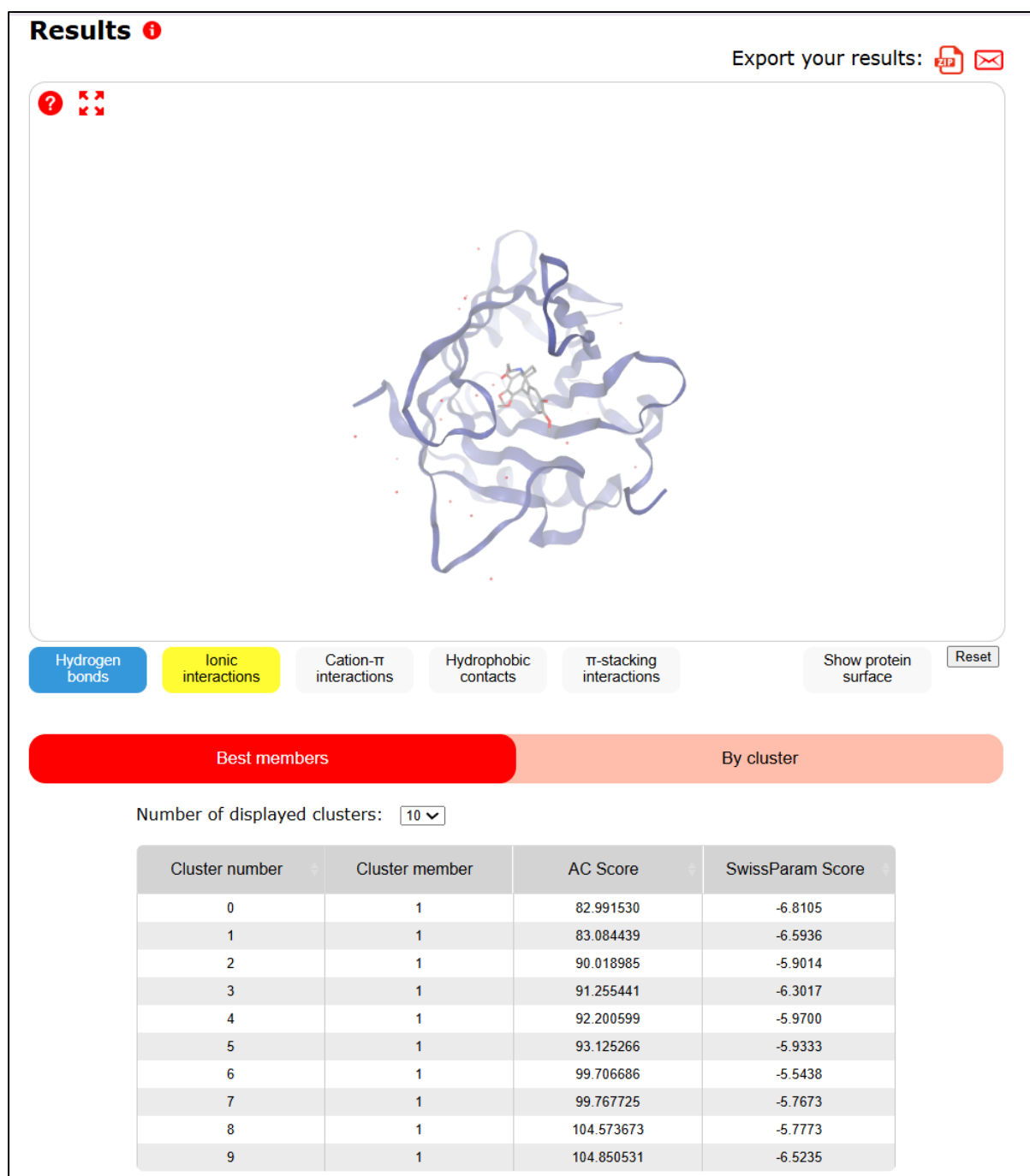


Figure 8: Results for target query ‘Colchicine’ with potential ligand

RESULTS:

SwissDock analysis was performed to identify potential drug candidates targeting Colchicine (PDB ID: 5NKN). The docking simulations generated ten distinct clusters, each representing a potential binding mode of the ligand. Cluster 0 exhibited the lowest AC Score (82.991530 kcal/mol), indicating the strongest predicted binding affinity to Colchicine (PDB ID:5NKN).

CONCLUSION:

SwissDock, a robust and user-friendly online molecular docking tool, was utilized to identify potential drug candidates targeting Colchicine (PDB ID: 3NKN). Leveraging the EADock DSS docking algorithm and the CHARMM force field, SwissDock effectively predicted potential binding modes and corresponding free energy scores for a library of compounds.

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 5. Marine Bugnon, Ute F Röhrig, Mathilde Goullieux, Marta A S Perez, Antoine Daina, Olivier Michielin, Vincent Zoete, SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina, *Nucleic Acids Research*, Volume 52, Issue W1, 5 July 2024, Pages W324–W332, <https://doi.org/10.1093/nar/gkae300>
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-

WEBLEM: 4

Physiologically Based Pharmacokinetic (PBPK)

AIM:

Introduction to Pharmacological modeling to optimize drug development process using computational approach

INTRODUCTION:

Physiologically Based Pharmacokinetic (PBPK) modeling is a mathematical approach used to predict the absorption, distribution, metabolism, and excretion (ADME) of drugs and chemicals in biological systems. By integrating physiological, biochemical, and physicochemical data, PBPK models provide a mechanistic understanding of drug behavior across different tissues and organs. These models play a crucial role in drug development, clinical pharmacology, and regulatory decision-making by optimizing dosing strategies, evaluating drug-drug interactions (DDIs), and assessing pharmacokinetic variability due to age, ethnicity, genetics, and disease conditions.

PBPK modeling reduces reliance on in vivo studies, providing a cost-effective and ethical alternative for predicting human pharmacokinetics. By incorporating in vitro and in silico data, PBPK models align with the 3Rs principles (Replacement, Reduction, and Refinement) in preclinical research. Regulatory agencies, including the FDA and EMA, have recognized PBPK modeling as a critical component of model-informed drug development (MIDD), using it to support dose selection, clinical study waivers, and drug safety evaluations.

A widely used tool for PBPK modeling is PK-Sim®, a software platform designed for whole-body PBPK simulations. PK-Sim® offers rapid access to physiological and anatomical data for humans and various laboratory animals, enabling efficient model development. The software accounts for passive distribution (e.g., blood flow) and active processes (e.g., enzyme-mediated metabolism), providing robust predictions of drug kinetics. Unlike many PBPK modeling tools, PK-Sim® offers structural flexibility for small and large molecules and is fully compatible with MoBi®, an advanced modeling tool that allows for extensive model customization and integration with systems pharmacology approaches.

PK-Sim® utilizes a modular building block approach, categorizing components such as Individuals, Populations, Compounds, Formulations, and Administration Protocols. These reusable building blocks allow for efficient model adaptation; for instance, a PBPK model developed for an intravenous dose in animals can be easily modified for human simulations, enabling first-in-human predictions and formulation optimization.

PBPK modeling is increasingly used in drug discovery, regulatory science, and precision medicine. It supports bioavailability predictions, special population studies, and toxicology assessments while reducing experimental burdens. With continued advancements in PBPK tools like PK-Sim®,

these models will remain integral in pharmaceutical research, streamlining drug development and improving patient outcomes.

Installation:

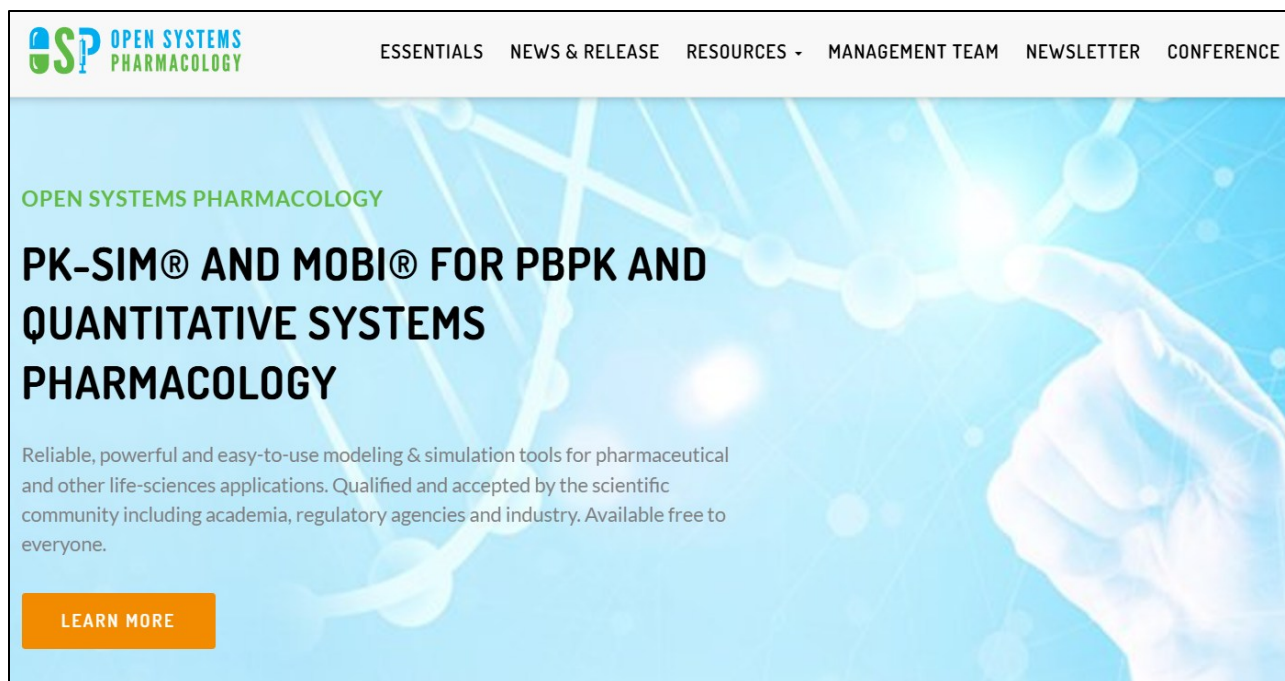


Figure 1: Go to ‘Open System Pharmacology’

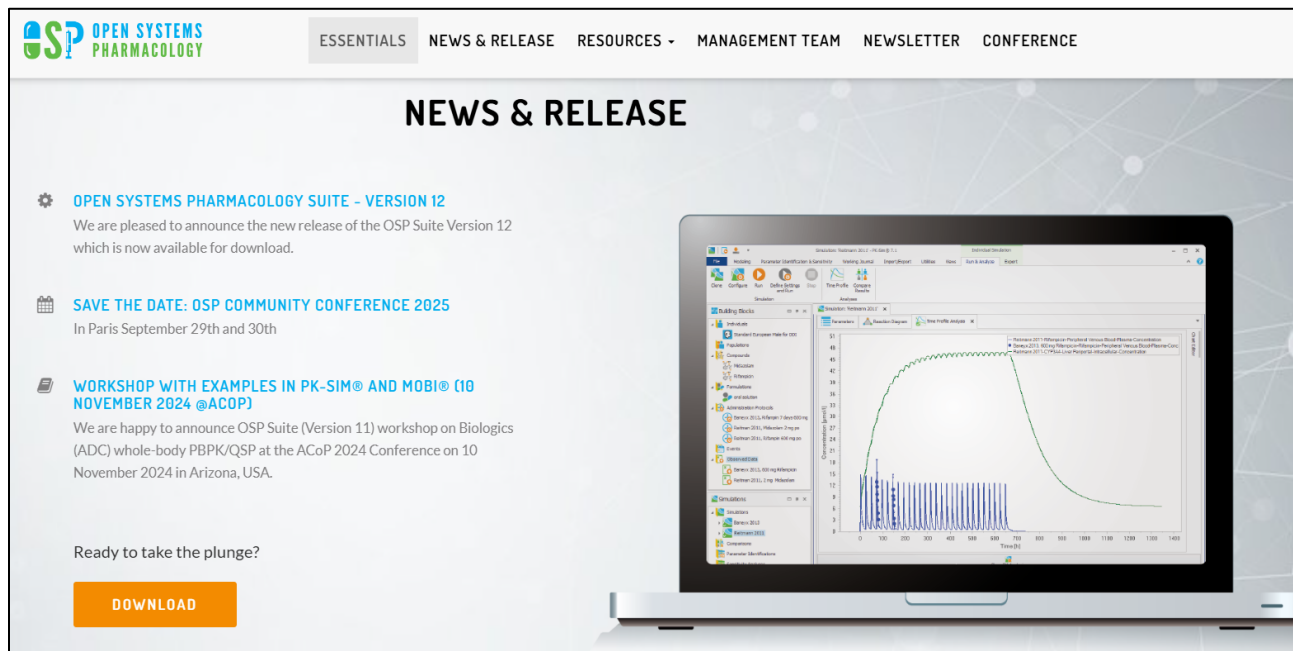


Figure 2: Select ‘News & Release’ and click on Download

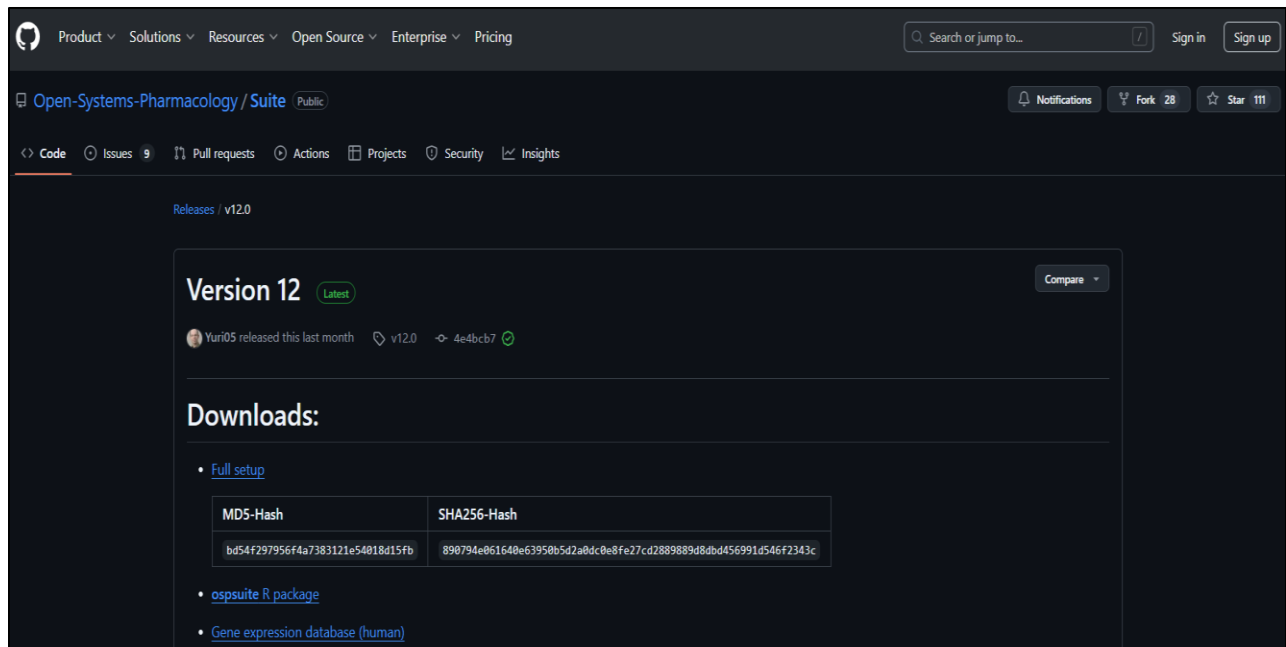


Figure 3: Click on Full Setup File will download

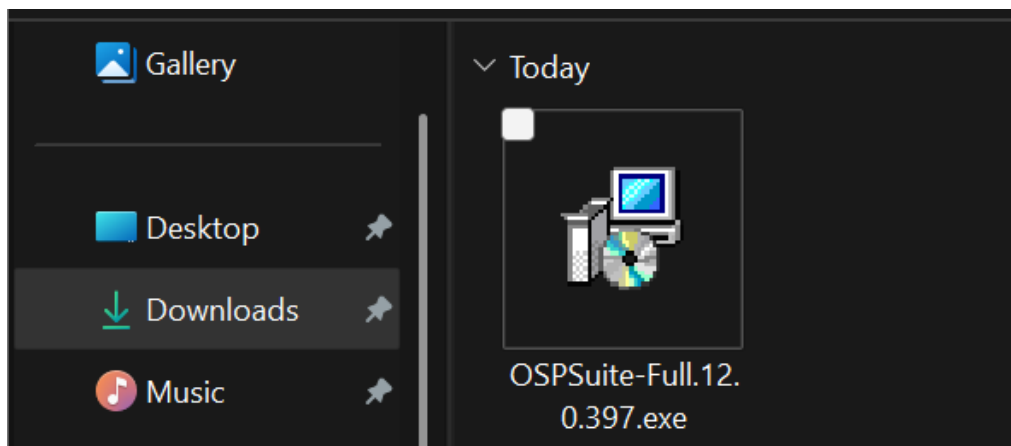
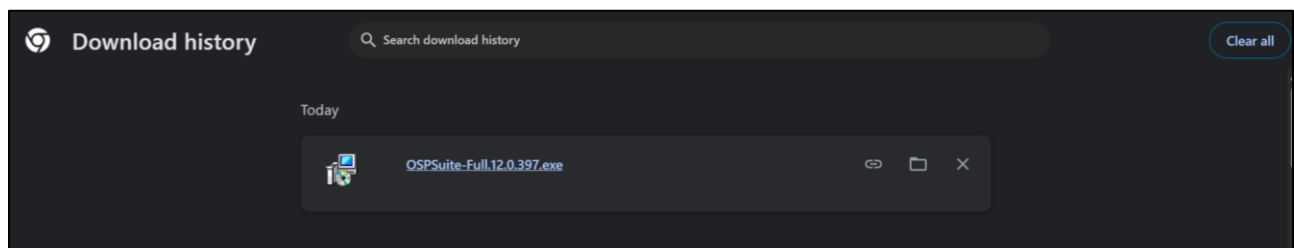


Figure 4: Run the OPSuite.exe as administrator

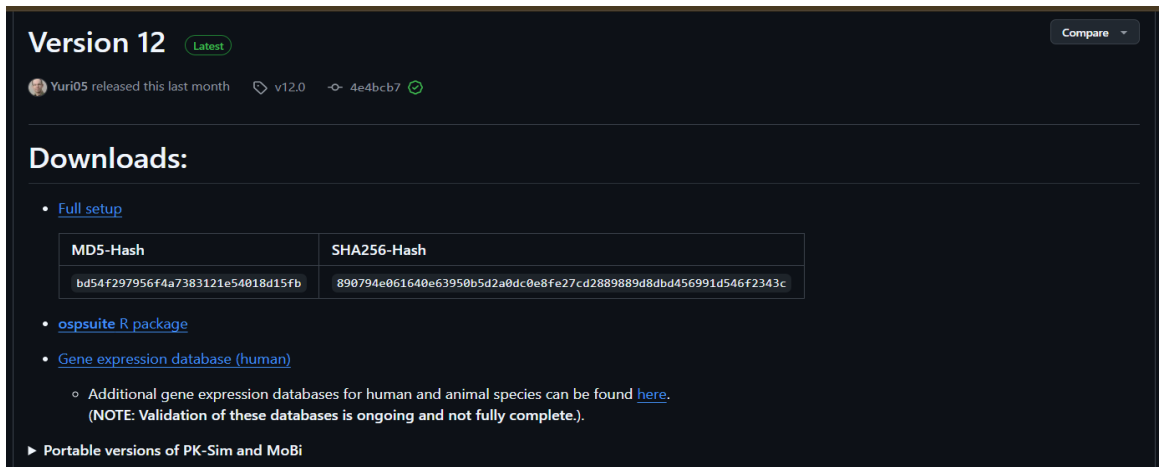


Figure 5: Download Gene expression database(human)

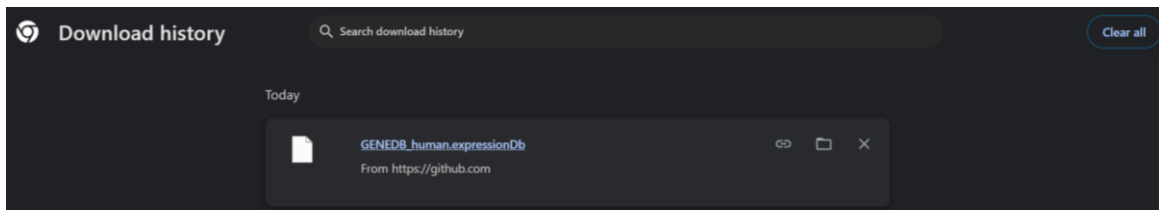


Figure 6: File Downloaded

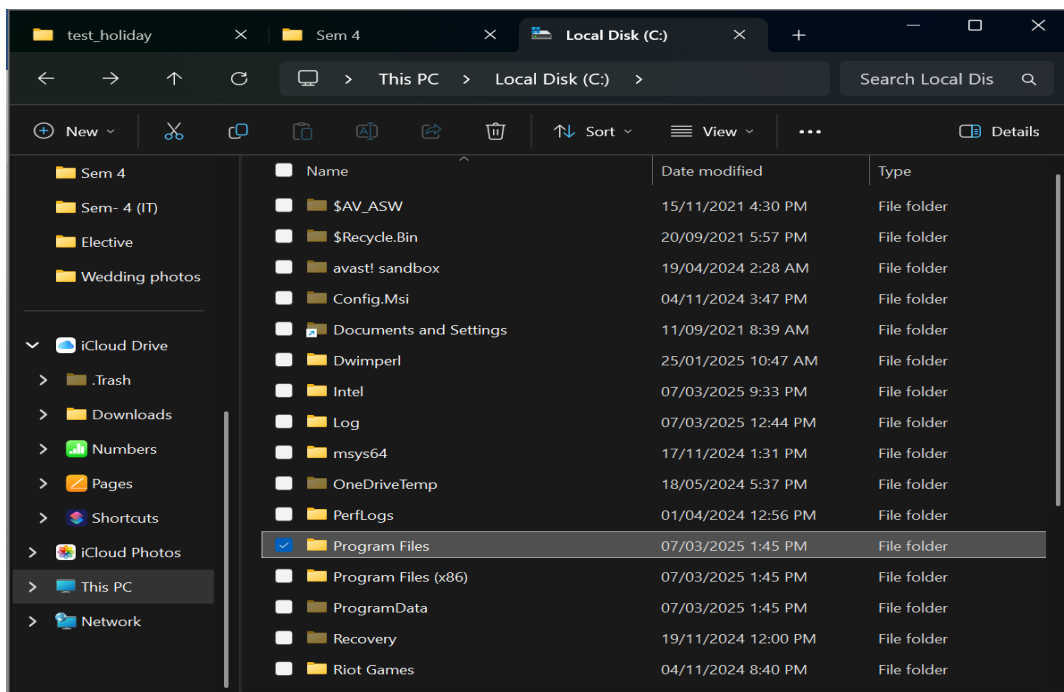


Figure 7: Go to 'C: Drive then Program File'

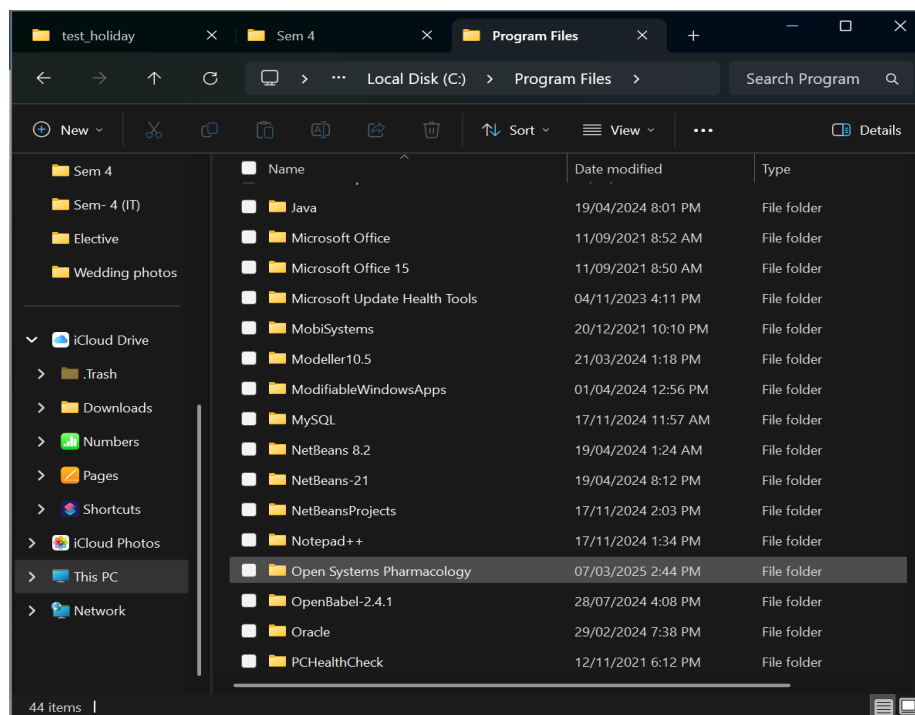


Figure 8: Click on 'Open Systems Pharmacology'

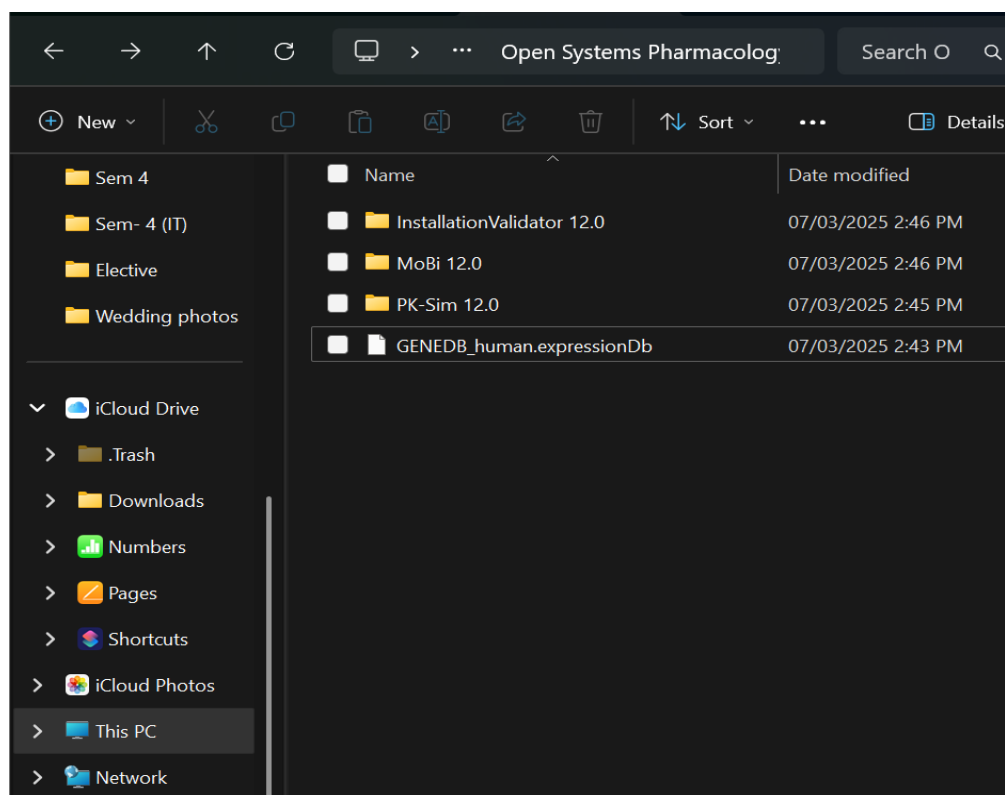


Figure 9: Paste the file 'Gene expression database (human)' in Open Systems Pharmacology

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 2. Jones, H., & Rowland-Yeo, K. (2013). Basic Concepts in Physiologically Based Pharmacokinetic Modeling in Drug Discovery and Development. *CPT: Pharmacometrics & Systems Pharmacology*, 2(8), e63. <https://doi.org/10.1038/psp.2013.41>
 3. Deepika, D., & Kumar, V. (2023). The Role of “Physiologically Based Pharmacokinetic Model (PBPK)” New Approach Methodology (NAM) in Pharmaceuticals and Environmental Chemical Risk Assessment. *International Journal of Environmental Research and Public Health*, 20(4), 3473. <https://doi.org/10.3390/ijerph20043473>
 4. Open Systems Pharmacology Community. (2025). Open Systems Pharmacology. Open-Systems-Pharmacology.org. <https://www.open-systems-pharmacology.org/>
-

WEBLEM: 4(A)

Physiologically Based Pharmacokinetic (PBPK) Simulation

AIM:

To build a PBPK model drug “Ibuprofen” and simulate its pharmacokinetics in the human body, predicting drug concentrations over time using PBPK principles.

INTRODUCTION:

Physiologically Based Pharmacokinetic (PBPK) simulation is a computational approach used to predict how drugs move through the body. It models drug absorption, distribution, metabolism, and excretion (ADME) by using physiological, biochemical, and anatomical data. Unlike traditional pharmacokinetic models, PBPK models incorporate organ-specific compartments, blood flow rates, and enzyme activity to provide a more detailed understanding of drug kinetics.

PBPK simulations are widely used in drug development, clinical pharmacology, and regulatory decision-making. They help predict drug behavior in different populations, such as children, elderly individuals, or patients with liver or kidney diseases. PBPK models also play a crucial role in evaluating drug-drug interactions (DDIs) and optimizing dosing strategies.

One of the most used tools for PBPK modeling is PK-Sim, which allows users to simulate drug behavior in humans and animals. PK-Sim provides pre-built physiological data, making it easier to create realistic models. These simulations help researchers reduce reliance on animal studies, improve drug safety assessments, and support first-in-human dose predictions. In this practical, we will use PK-Sim to build a PBPK model for Ibuprofen and analyze its plasma concentration-time profile.

Ibuprofen

Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) derived from propionic acid and it is considered the first of the propionics. The formula of ibuprofen is 2-(4-isobutylphenyl) propionic acid and its initial development was in 1960 while researching for a safer alternative for aspirin. Ibuprofen was finally patented in 1961 and this drug was first launched against rheumatoid arthritis in the UK in 1969 and USA in 1974. It was the first available over-the-counter NSAID.

On the available products, ibuprofen is administered as a racemic mixture. Once administered, the R-enantiomer undergoes extensive interconversion to the S-enantiomer *in vivo* by the activity of the alpha-methylacyl-CoA racemase. In particular, it is generally proposed that the S-enantiomer is capable of eliciting stronger pharmacological activity than the R-enantiomer.²

METHODOLOGY:

Step 1: Setup and Installation

1. Download and install Open Systems Pharmacology (OSP) Suite.
2. Install the Gene Expression Database if required.

Step 2: Initializing the PBPK Model

1. Open PK-Sim in OSP Suite.
2. Create a new project and import Gene Expression Data if applicable.

Step 3: Defining the Individual Model Parameters

1. Set up an individual profile with:
 - a. Demographics (age, gender, population type).
 - b. Biometrics (weight, height, BMI).

Step 4: Anatomy & Physiology Setup

1. Configure organ systems and physiological parameters, including:
 - a. Gastrointestinal tract (GIT) for absorption.
 - b. Glomerular Filtration Rate (GFR) for renal clearance.

Step 5: Metabolic Enzyme Expression

1. Define metabolizing enzymes (e.g., CYP1A2, CYP3A4).
2. Assign expression levels based on population or experimental data.

Step 6: Compound Definition

1. Select the compound (drug) in OSP Suite.
2. Define physicochemical properties (molecular weight, LogP, pKa, solubility).
3. Set ADME parameters:
 - a. Absorption: Permeability, bioavailability.
 - b. Distribution: Volume of distribution (Vd), plasma protein binding.
 - c. Metabolism: Enzymes involved (CYPs, UGTs).
 - d. Excretion: Renal and hepatic clearance.
4. Formulation & administration: Select route (oral, IV), dose, and schedule.

Step 7: Observer, Create, and Simulation

1. Observer: Define observation points to track parameters (e.g., plasma drug concentration, tissue exposure).
2. Create: Finalize model components (individual, drug, metabolism, administration).
3. Simulation: Run the PBPK simulation and generate plasma concentration-time profiles.

Step 8: Data Interpretation and Validation

1. Compare simulated results with experimental/literature data.
2. Adjust model parameters to refine predictions.

Step 9: Utilities and Applications

1. Predict drug behavior in various populations (e.g., pediatric, elderly, disease states).
2. Assess drug-drug interactions (DDIs).
3. Optimize dosing regimens and formulations.

OBSERVATIONS:

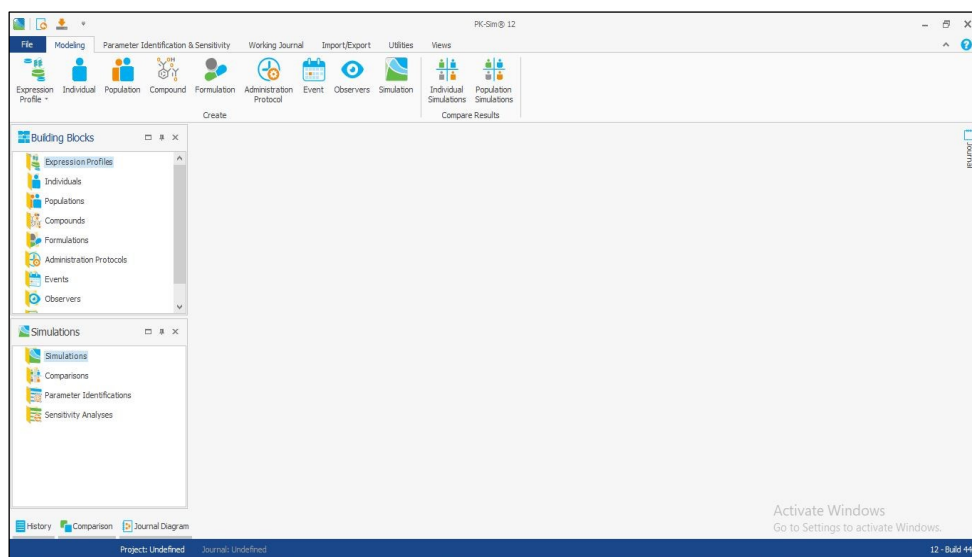


Figure 1: PK-Sim Homepage – Selecting “Create New Project” to Start a PBPK Simulation

Figure 2: Creating a New Individual – Selecting “Human” as Species and naming it “Bioinformatics”

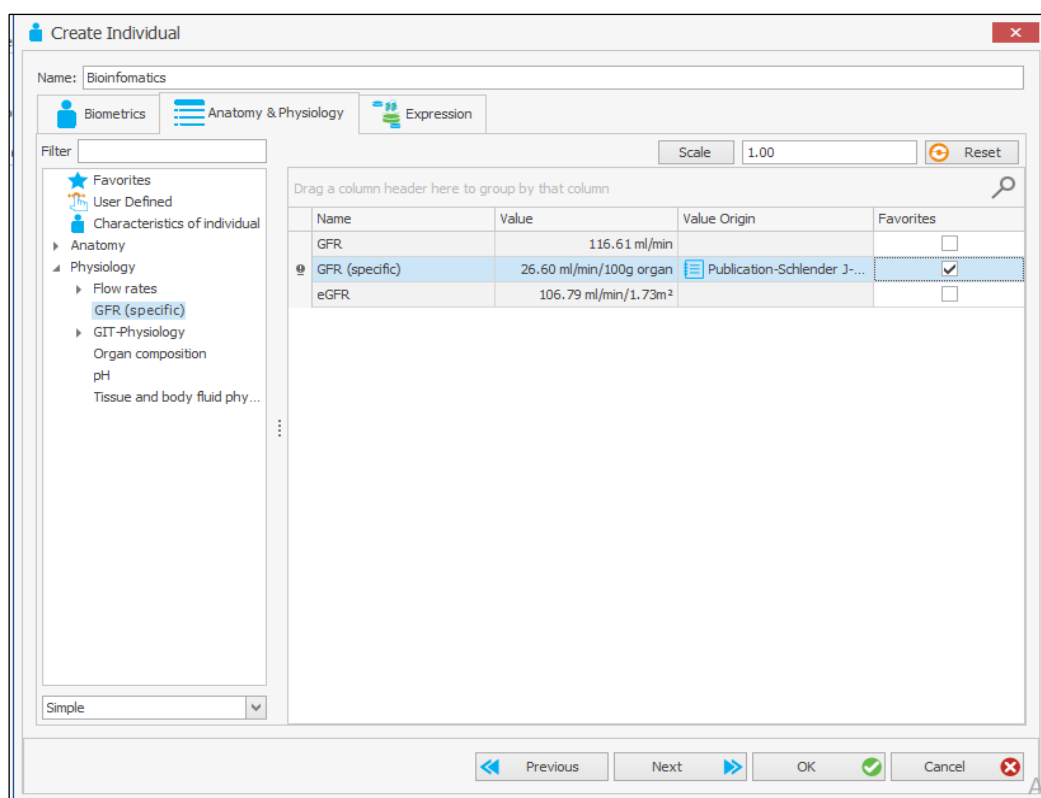


Figure 3: Anatomy & Physiology Setup – Enabling Gastrointestinal Tract (GIT) for Absorption and Glomerular Filtration Rate (GFR) for Renal Clearance

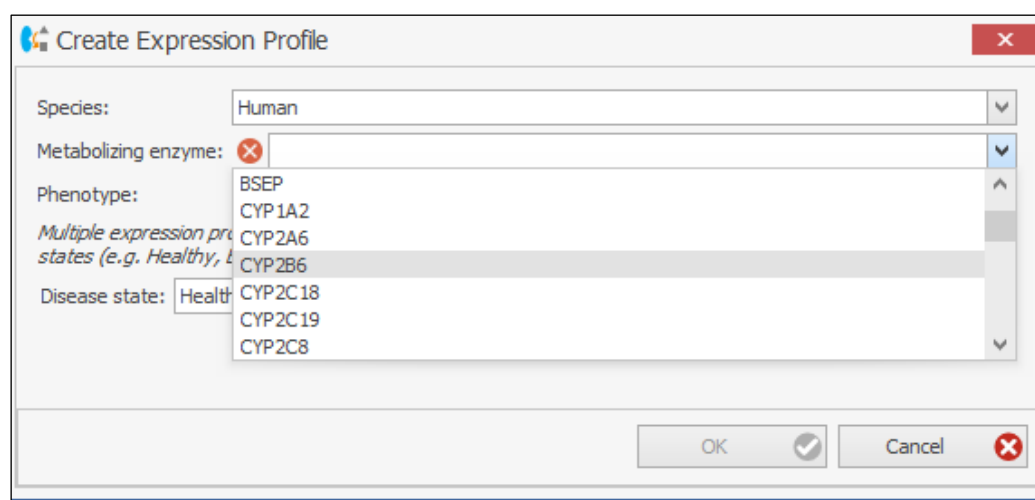


Figure 4: Adding Metabolic Enzymes – Selecting CYP1A2 for Drug Metabolism and Assigning Expression Levels

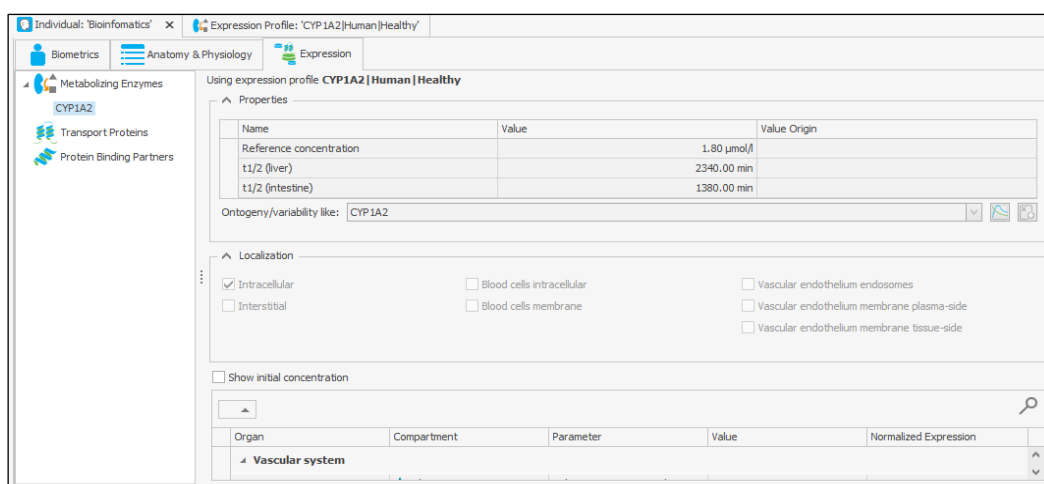


Figure 5: Expression Setup – Selecting CYP1A2 in the Expression Tab

Create Population

Name: Asian

Demographics Expression User Defined Variability Distribution

Based on individual: Bioinformatics

This individual will represent the mean individual of the created population

Population: European (ICRP, 2002)

Population Properties

Number of individuals: 100

Proportion of females [%]: 50

Population Parameters Ranges

Parameter	from	to	Unit
Age	0	100.00	year(s)
Weight			kg
Height			cm
BMI			kg/m ²

Previous Next Cancel

Figure 6: Creating a Population – Defining Population Characteristics and Variability

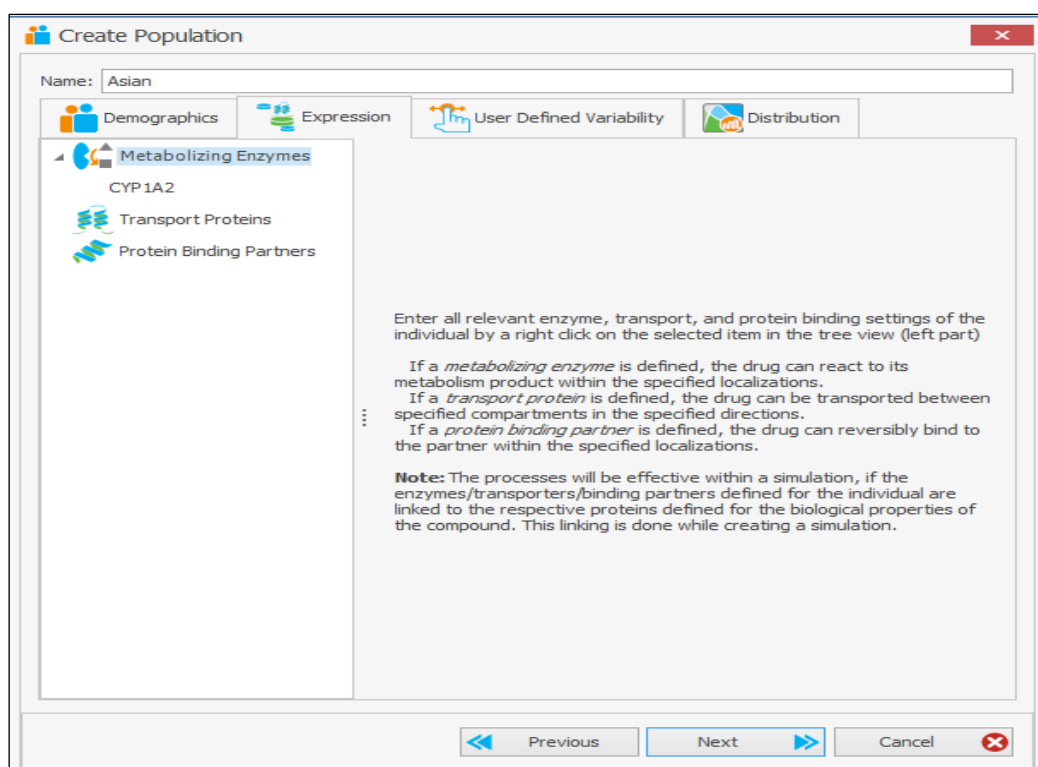


Figure 7: Expression Setup – Selecting CYP1A2 in the Expression Tab to Assign Metabolic Enzyme Activity

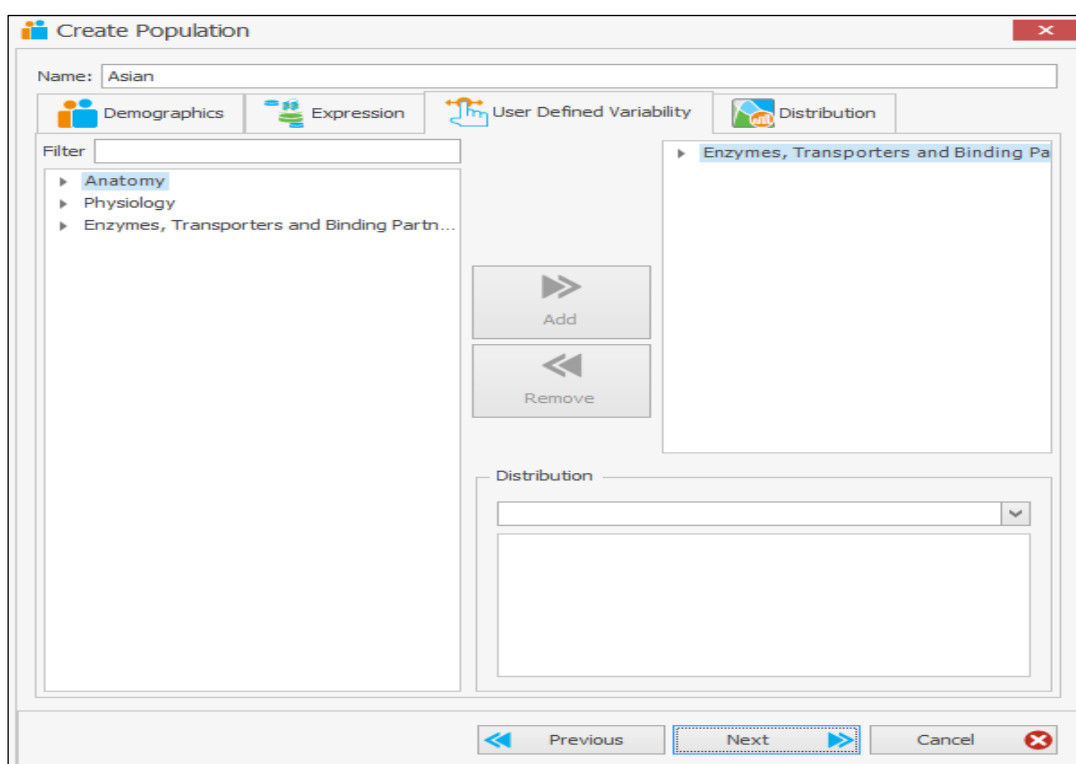


Figure 8: User-Defined Variability – Customizing Variability Parameters for Enzyme Expression and Physiological Factors

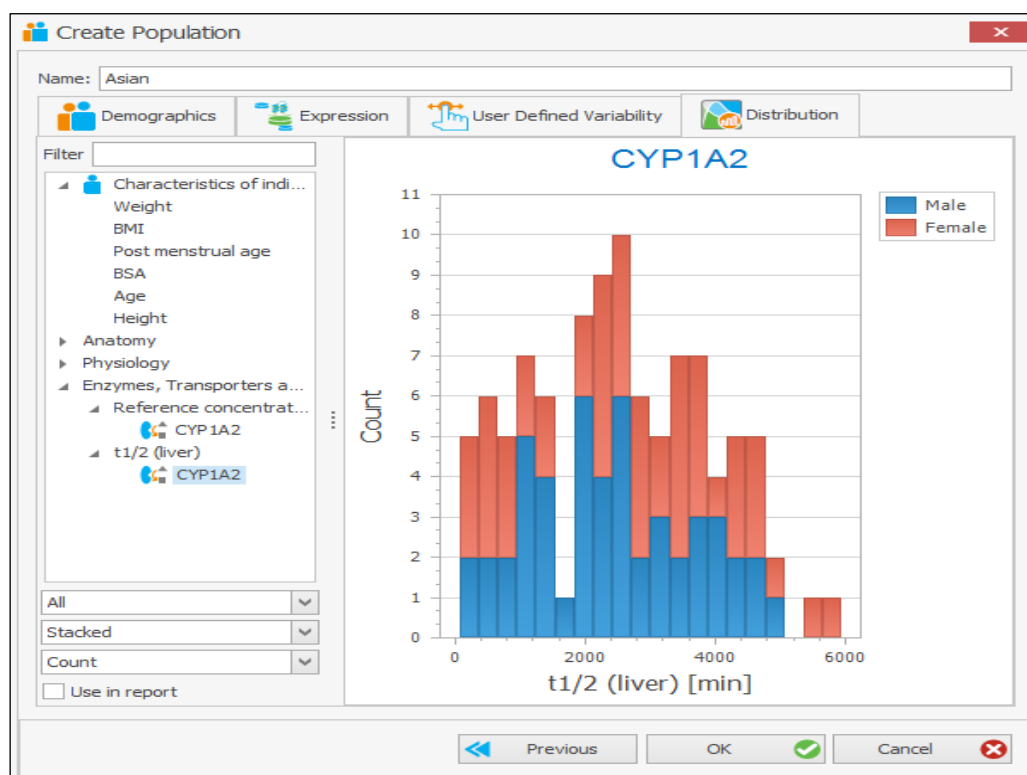


Figure 9: Analysis of Half-Life ($T_{1/2}$) in Liver Based on CYP1A2 Expression – Comparison of Male and Female Populations

The 'Create Compound' dialog box is shown with the 'Basic Physico-chemistry' tab selected. The compound name is 'ibuprofen'. The 'Lipophilicity' is set to 3.50 Log Units. The 'Fraction unbound (plasma, reference value)' is set to 0.04, with 'Albumin' selected as the binding protein and 'Human' as the species. The 'Molweight' section shows Molecular weight, Has halogens, and Effective molecular weight, all set to 206.28 g/mol. The 'Compound type / pKa' section shows Acid, Neutral, and Neutral, with pKa values of 4.45, <None>, and <None> respectively.

Figure 10: Entering Drug Physicochemical Properties – Defining Molecular Weight, LogP, pKa, and Solubility for Absorption Prediction

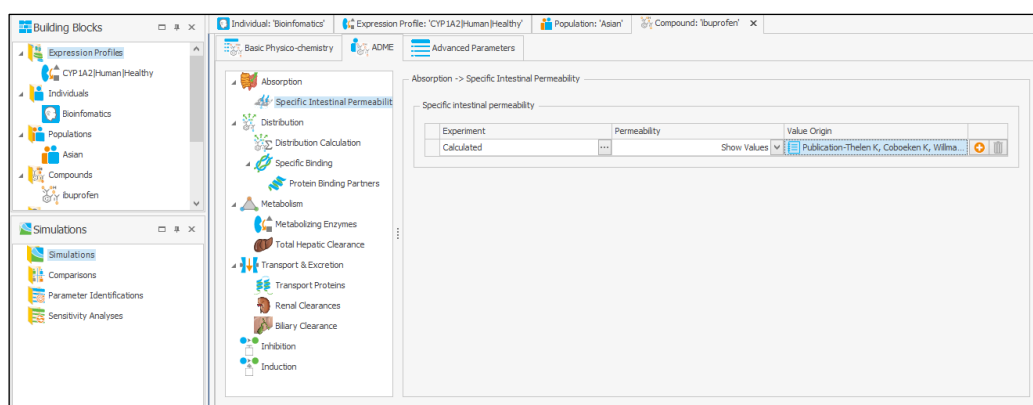


Figure 11: Viewing Absorption Properties – Displaying Permeability, Regional Absorption, and Bioavailability

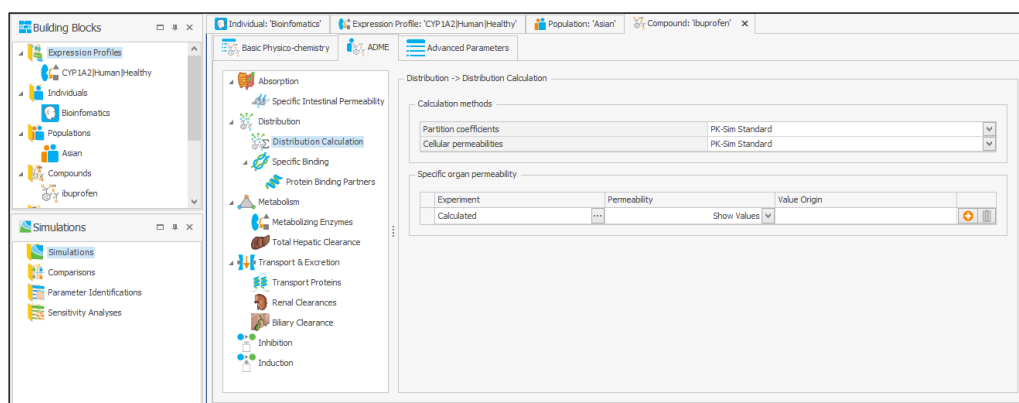


Figure 12: Viewing Distribution Parameters – Displaying Plasma Protein Binding and Partition Coefficients for Tissue Penetration Modeling

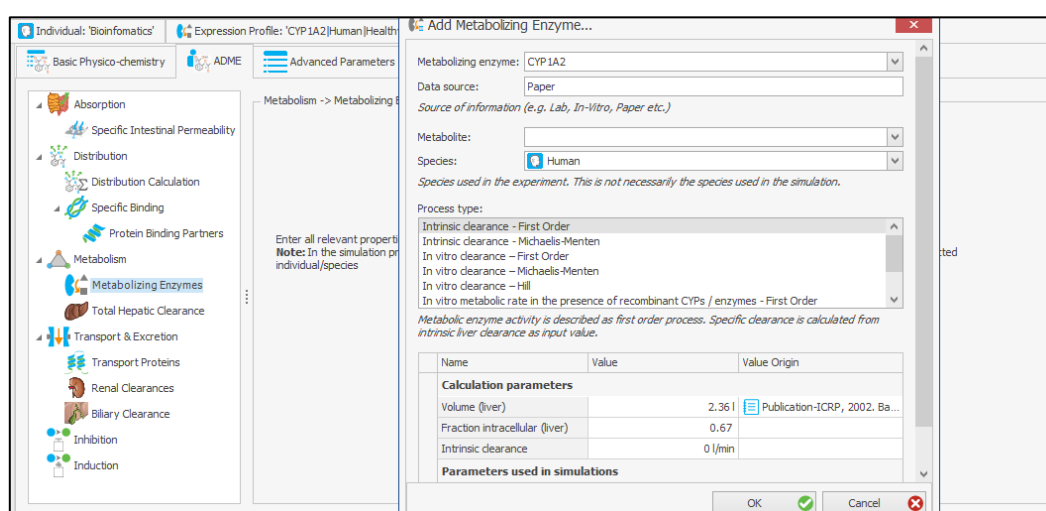


Figure 13: Assigning Drug Metabolism Pathways – Specifying CYP1A2-Mediated Clearance, Reaction Kinetics in the Liver, and Literature References (Paper Source)

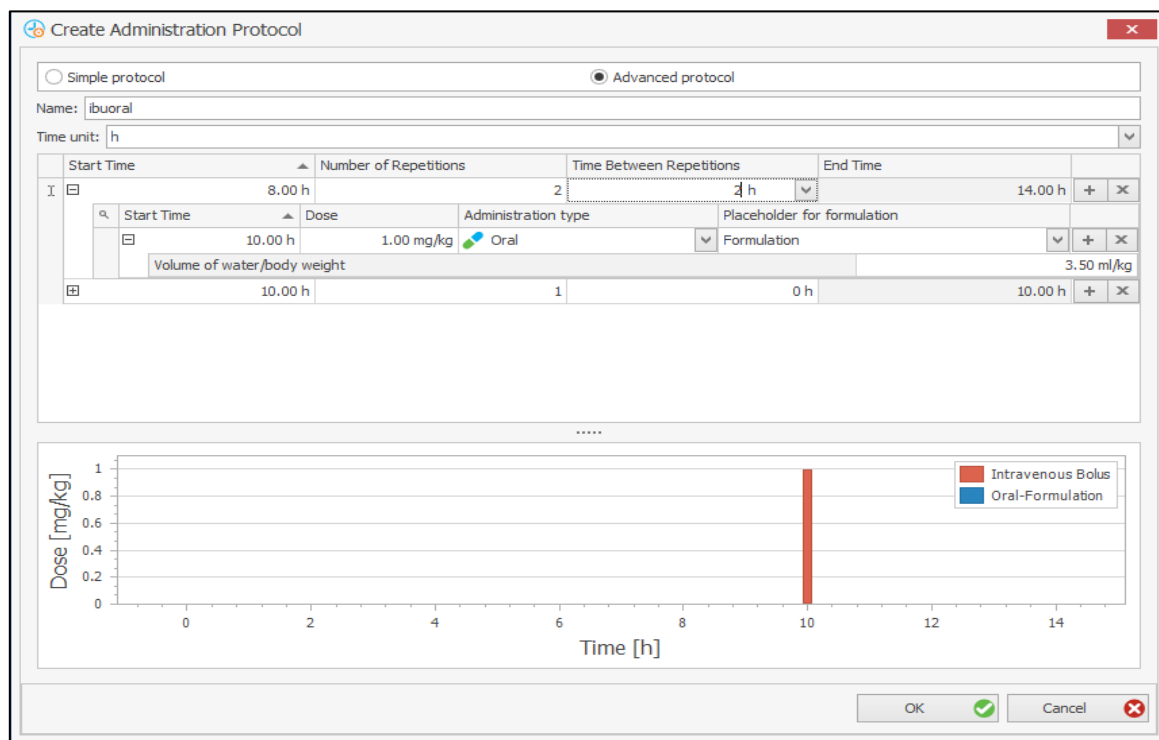


Figure 14: Defining Drug Administration Protocol – Selecting Route (Oral, IV, etc.), Setting Dose (mg), Frequency (hours), and Number of Doses

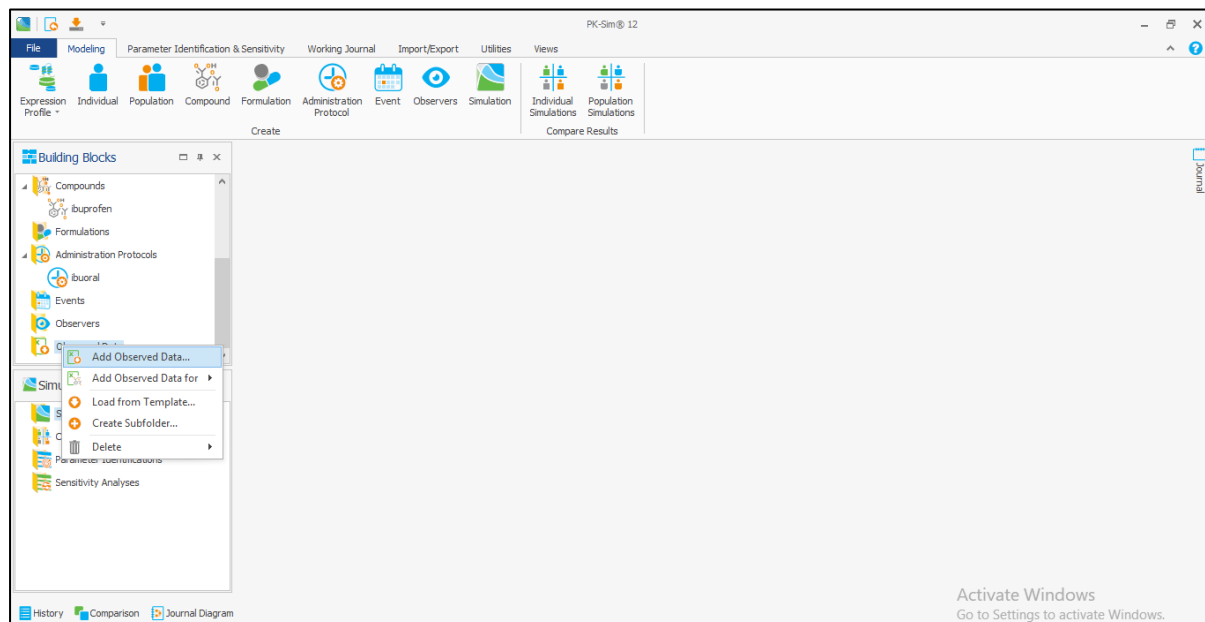


Figure 15: PK-Sim 12 interface with the "Observed Data" menu open

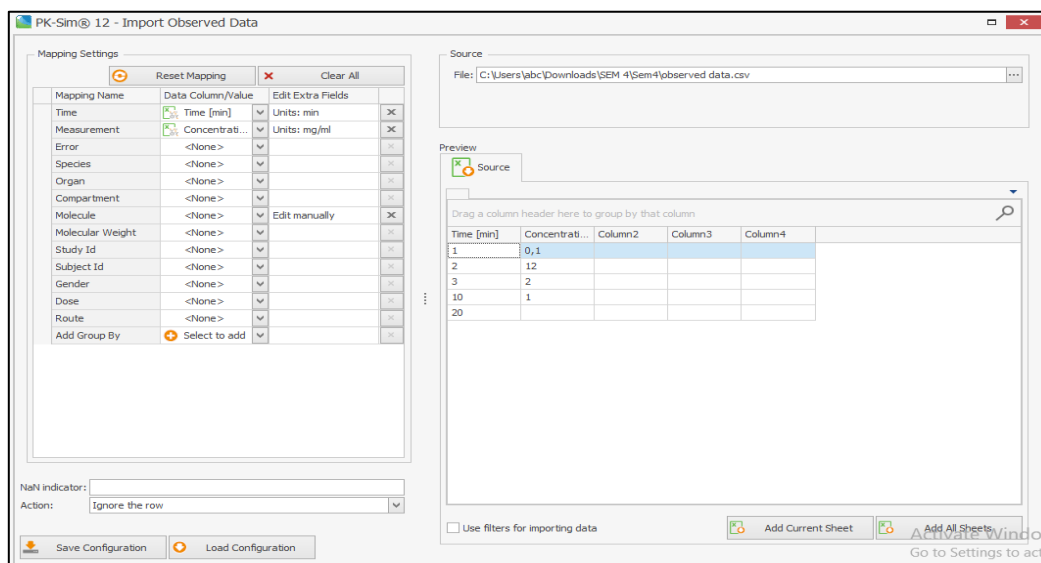


Figure 16: Importing observed data into PK-Sim® 12 from a CSV file

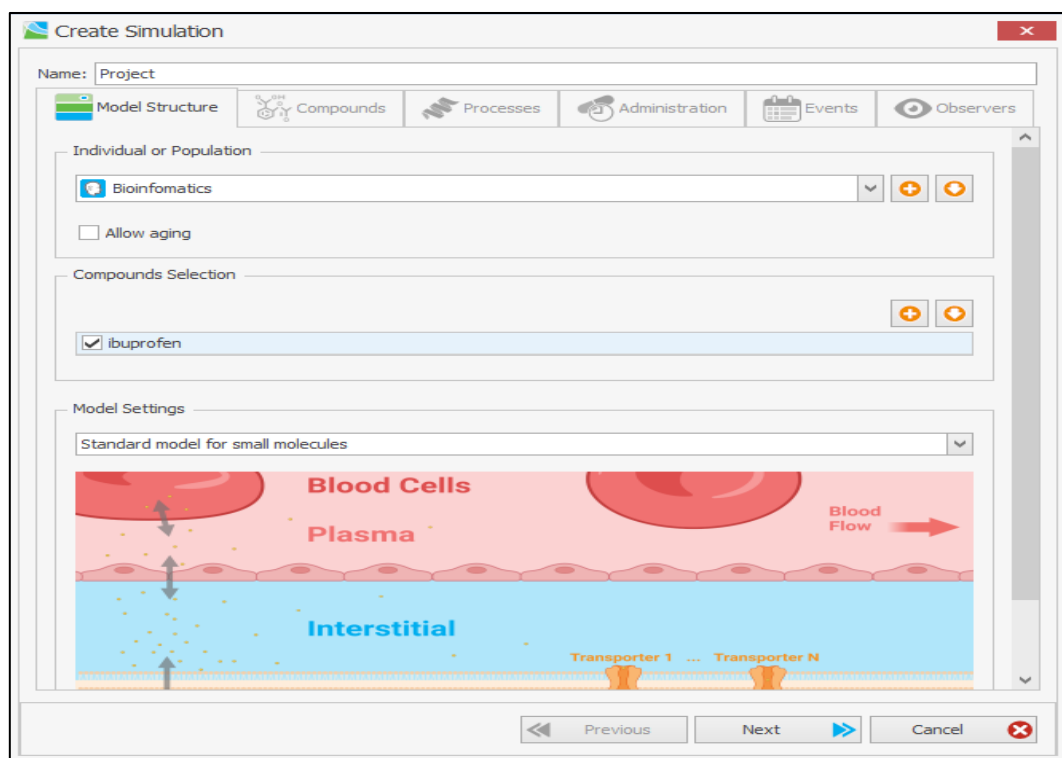


Figure 17: Setting up a simulation with ibuprofen and bioinformatics-based modeling

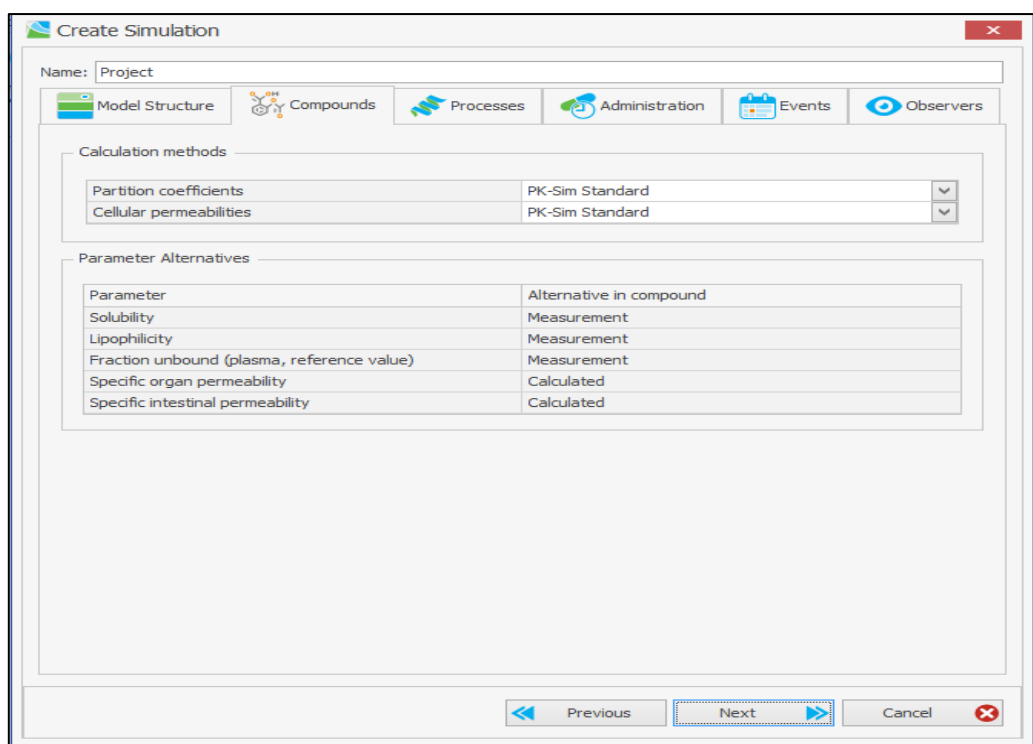


Figure 18: Configuring calculation methods and parameter alternatives for the simulation

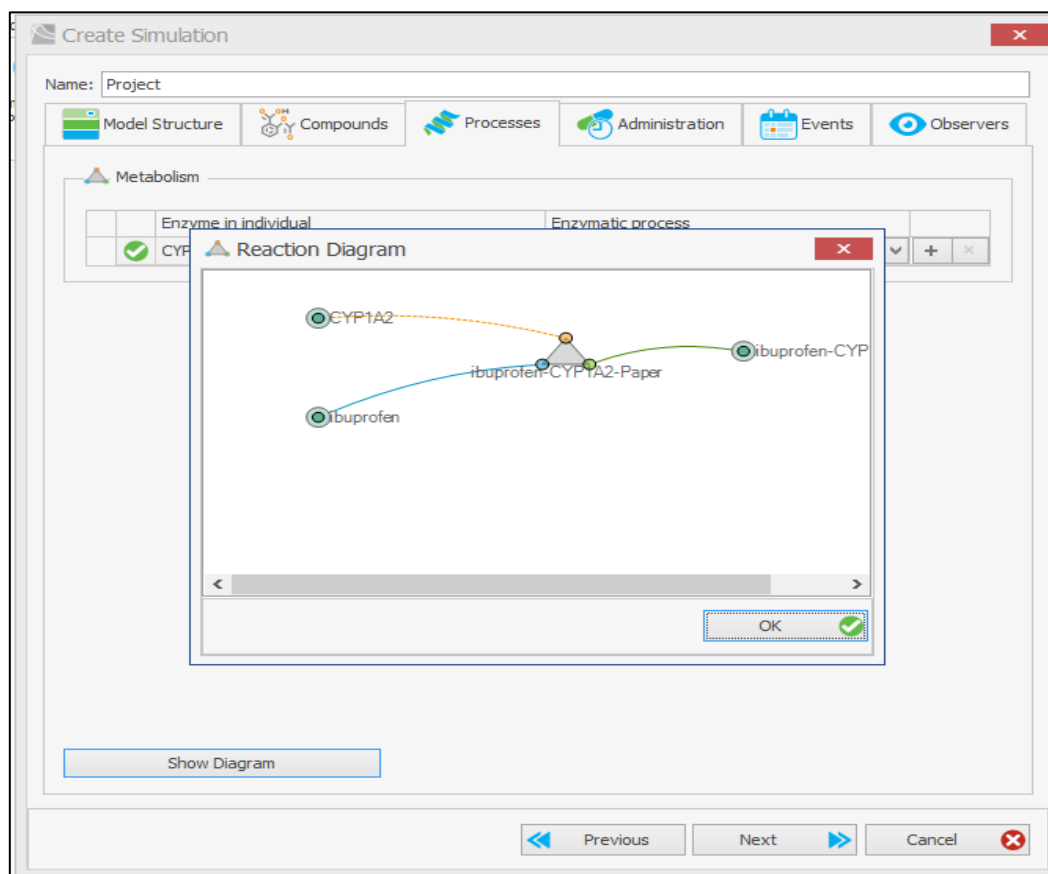


Figure 19: Reaction diagram showing the enzymatic metabolism of ibuprofen

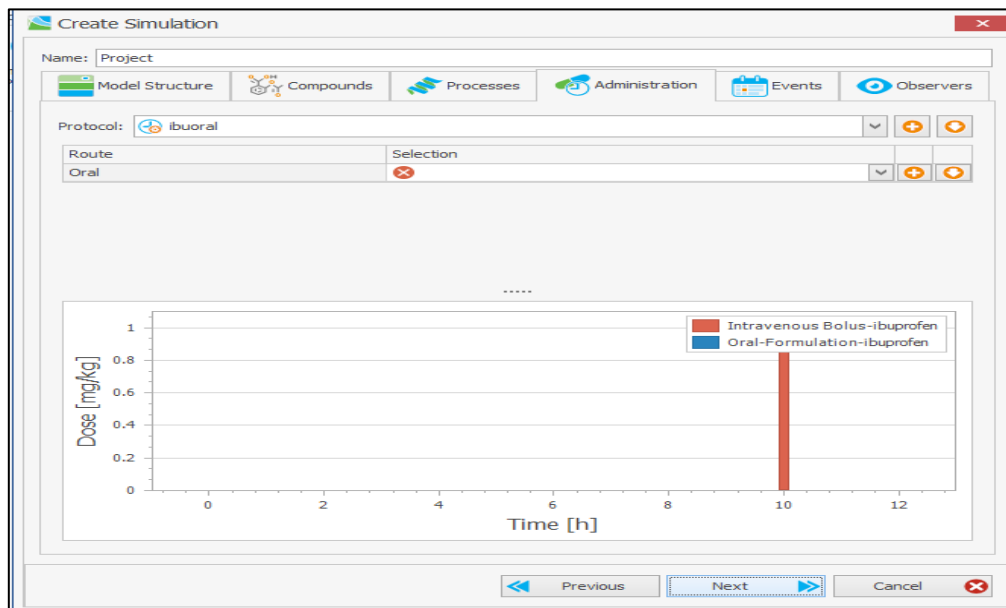


Figure 20: Simulates ibuprofen dosage (mg/kg) over time (hours) for intravenous bolus (red) and oral formulation (blue) administration

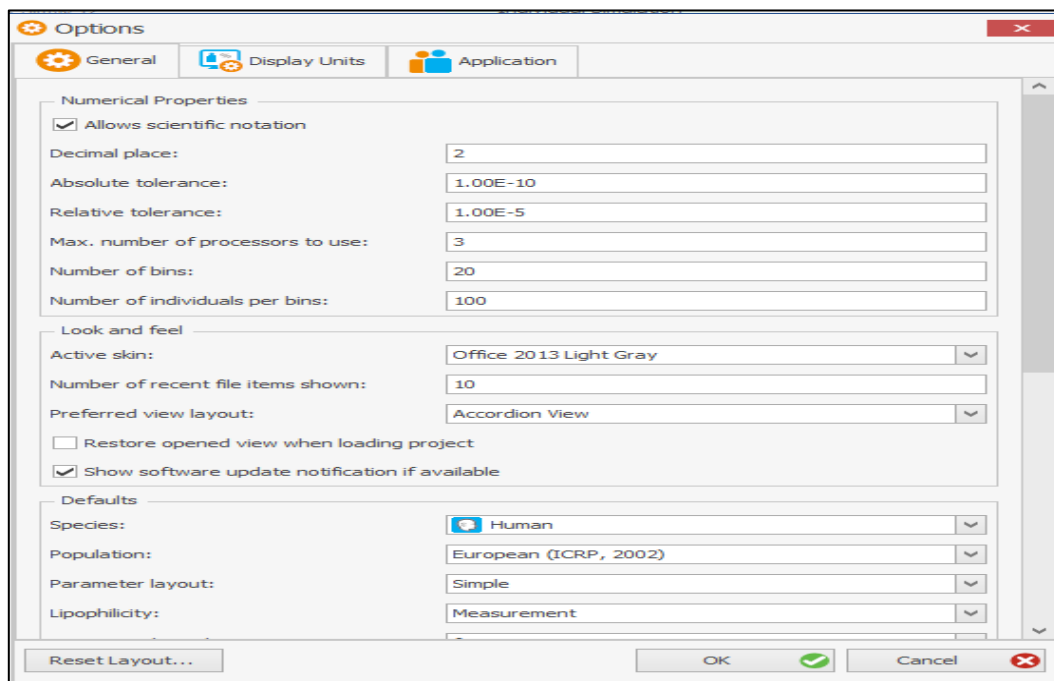


Figure 21: The image displays the "Options" window of a software

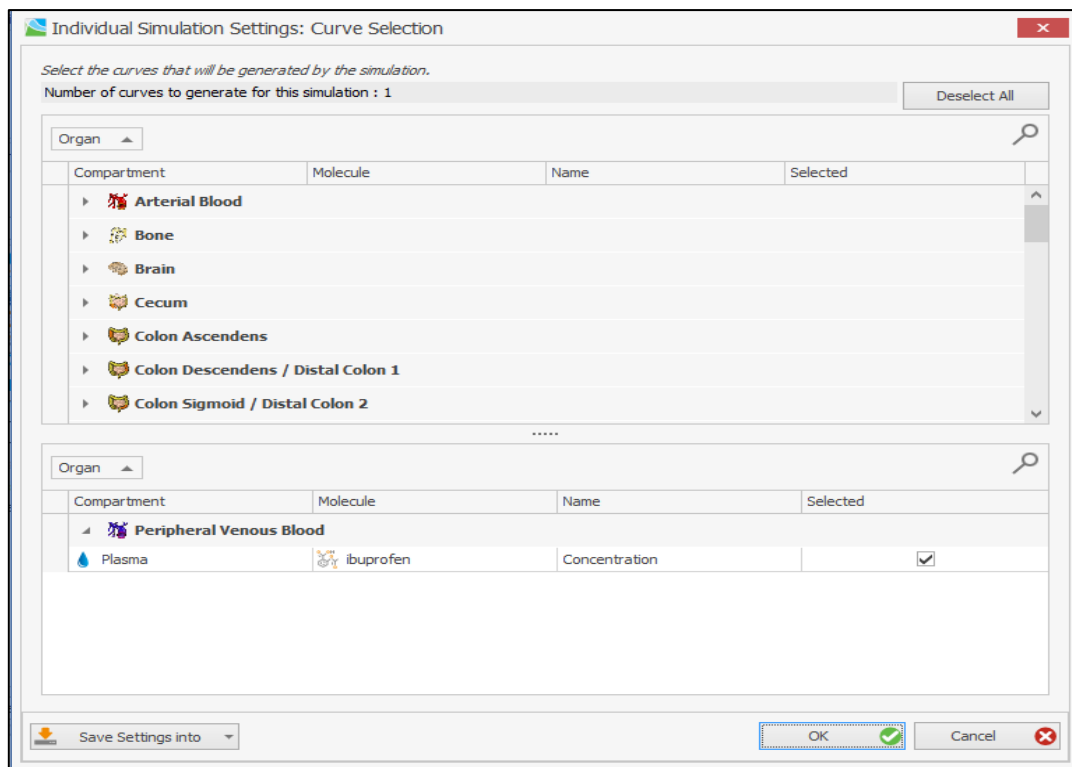


Figure 22: Individual Simulation Settings - Curve Selection

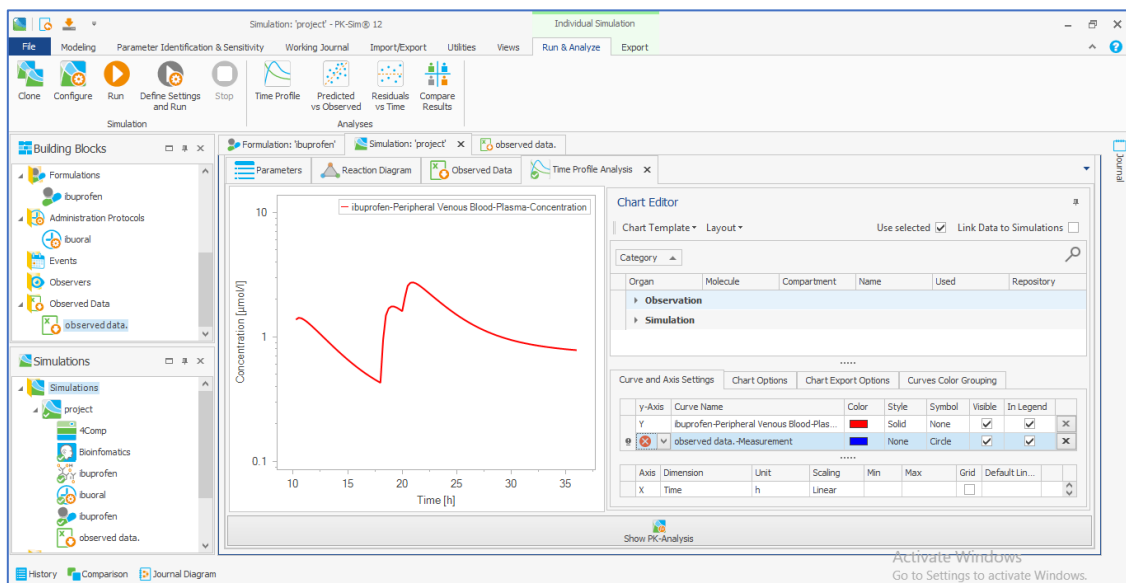


Figure 23: Time Profile Analysis of Ibuprofen Plasma Concentration

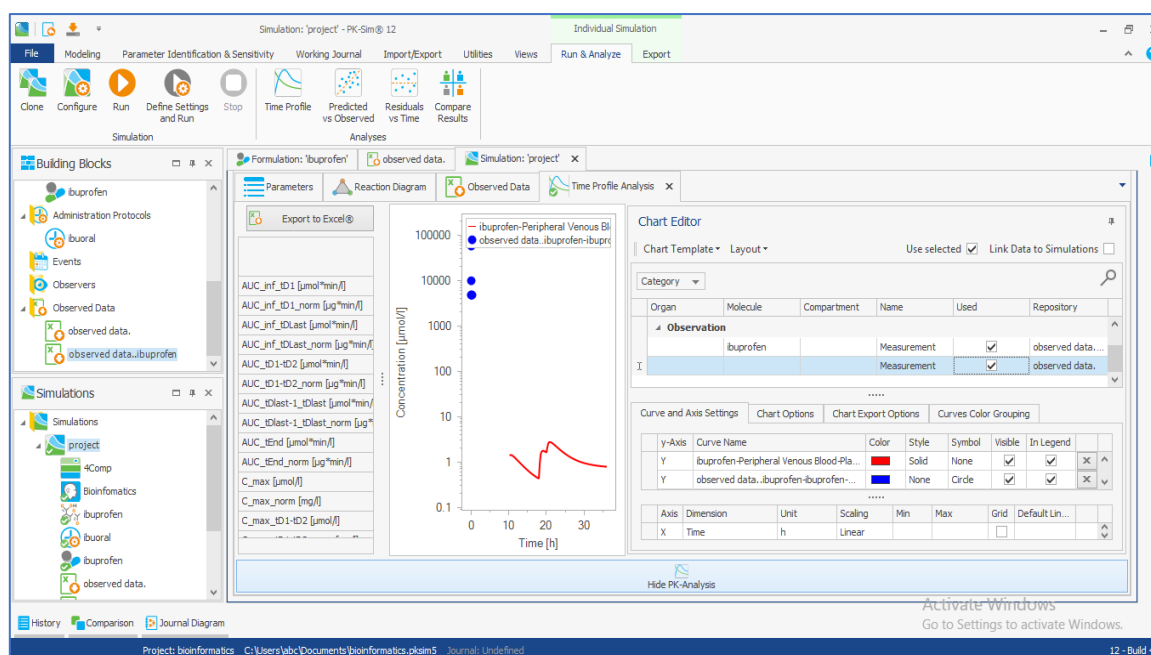


Figure 24: Comparison of Simulated and Observed Ibuprofen Plasma Concentration

RESULTS:

Simulation Overview

The PK-Sim simulation was conducted to analyze the pharmacokinetic profile of Ibuprofen in peripheral venous blood plasma over time. The generated concentration-time profile provides insights into the drug's absorption, distribution, and elimination phases.

Observed vs. Predicted Data

1. The concentration-time graph shows an initial increase, peaking at 21 hours, followed by a gradual decline, representing metabolism and elimination.
2. The observed data points (blue dots) align well with the simulated curve (red line), indicating that the model provides a reasonable fit for Ibuprofen's pharmacokinetic behavior.
3. A slower T_{max} suggests delayed absorption or redistribution compared to conventional ibuprofen formulations.

Pharmacokinetic Parameters

1. Peak Concentration (C_{max}): 2.74 μmol/L (Maximum plasma concentration observed).
2. Time to Peak (T_{max}): 21 hours (Time taken to reach peak concentration).
3. Area Under the Curve (AUC): 31.38 μmol·h/L (Represents total drug exposure in the bloodstream).

CONCLUSION:

The simulation accurately captured the pharmacokinetic profile of Ibuprofen, showing expected trends in absorption, peak concentration, and elimination. T_{max} (21 hours) appears higher than expected, which could be due to extended-release formulation, altered absorption, or physiological variations. The low C_{max} (2.74 μmol/L) suggests that drug accumulation in

plasma is minimal, possibly indicating rapid clearance or low bioavailability. AUC (31.38 $\mu\text{mol}\cdot\text{h/L}$) confirms the extent of drug absorption, which is crucial for dosing decisions. The model provides a reliable approximation of Ibuprofen's behavior, but further refinement (adjusting physiological parameters or additional simulations) could improve accuracy.

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WEBLEM: 5

Introduction To Pharmacokinetics Predictions Using SwissADME Server

INTRODUCTION:

The SwissADME application is an online tool designed to predict the ADME (Absorption, Distribution, Metabolism, and Excretion) properties of small molecules (drugs or compounds) based on their chemical structure. The Swiss ADME tool provides key insights into the pharmacokinetics and drug-likeness of compounds, which are crucial for drug discovery and development. Pool of fast yet robust predictive models for physicochemical properties, pharmacokinetics, drug-likeness and medicinal chemistry friendliness, among which in-house proficient methods such as the BOILED-Egg, iLOGP and Bioavailability Radar.

To be effective as a drug, a potent molecule must reach its target in the body in sufficient concentration, and stay there in a bioactive form long enough for the expected biologic events to occur. Drug development involves assessment of absorption, distribution, metabolism and excretion (ADME). A large variety of *in silico* methods share the objective of predicting ADME parameters from molecular structure. It is based on Lipinski rules which examined orally active compounds to define physicochemical ranges for high probability to be an oral drug (i.e. the drug-likeness). This is so-called Rule-of-five delineated the relationship between pharmacokinetic and physicochemical parameters.

The SwissADME web tool presented here is freely accessible and meant for user-friendly submission and easy analysis of the results, also for non-expert in CADD. Compared to the state-of-the art of free web-based tools for ADME and pharmacokinetics (e.g. pk-CSM and admetSAR) and apart from unique access to proficient methods (e.g. iLOGP or the BOILED-Egg), SwissADME strong points are, non-exhaustively: different input methods, computation for multiple molecules, and the possibility to display, save and share results per individual molecule or through global intuitive and interactive graphs. Finally, SwissADME is integrated in the SwissDrugDesign workspace. One-click interoperability gives access to various CADD tools developed by the Molecular Modelling Group of the SIB Swiss Institute of Bioinformatics, e.g. ligand-based virtual screening (SwissSimilarity), biotarget prediction (SwissTargetPrediction), molecular docking (SwissDock), bioisosteric design (SwissBioisostere), or molecular mechanics (SwissParam).

Here are some features typically offered by Swiss ADME:

1. Drug-Likeness Prediction:

- a. Calculates properties such as Lipinski's rule of five (which predicts oral bioavailability) and other criteria for drug-likeness.
- b. **Lipinski's Rule of Five (Ro5):** This is a widely used set of rules that predict the drug-likeness of a molecule based on its properties. SwissADME evaluates if the molecule adheres to these rules, which assess:
 1. Molecular weight ≤ 500 g/mol
 2. LogP (octanol-water partition coefficient) ≤ 5
 3. Hydrogen bond donors ≤ 5
 4. Hydrogen bond acceptors ≤ 10

2. Lipophilicity:

- a. Predicts the partition coefficient (logP) of the compound, which provides insights into how the compound behaves in lipophilic or hydrophilic environments (important for absorption and distribution).
- b. **Topological Polar Surface Area (TPSA):** Provides insights into a compound's ability to permeate biological membranes, influencing absorption and blood-brain barrier penetration.

3. Solubility:

- a. Calculates the solubility of the compound in water, which impacts the absorption and bioavailability of the drug.

4. Permeability:

- a. Provides information about the permeability of the compound through biological membranes (important for absorption).

5. Metabolism:

- a. Provides information about potential metabolic pathways and the likelihood of metabolism by specific cytochrome P450 enzymes.

6. Toxicity Prediction:

- a. It may also offer predictions for certain types of toxicity, which is important in the early stages of drug development.
- b. Although it doesn't replace formal toxicity testing, SwissADME also provides **preliminary toxicity assessments** based on structural features. For example, it can predict whether a compound might have toxic effects related to **mutagenicity** or **carcinogenicity**.

7. PK Predictions:

- a. The tool provides predictions on **pharmacokinetic** properties, including potential **oral bioavailability** and compound transport behaviour through the gastrointestinal tract.

The Bioavailability Radar is a feature within the SwissADME tool that helps visualize and evaluate the oral bioavailability of a compound. It provides a comprehensive graphical representation of a molecule's drug-likeness and its potential for oral absorption based on key pharmacokinetic properties. This radar plot helps researchers quickly assess how favourable a compound is for becoming an oral drug.

Bioavailability Radar is displayed for a rapid appraisal of drug-likeness, six physicochemical properties are considered: lipophilicity, size, polarity, solubility, flexibility and saturation. A physicochemical range on each axis was defined by descriptors adapted, and depicted as a pink area in which the radar plot of the molecule has to fall entirely to be considered drug-like. The pink area represents the optimal range for each property.

Ideal Ranges for Oral Bioavailability

1. **Molecular Weight:** 150–500 g/mol (lower molecular weight compounds are generally easier to absorb).
2. **LogP:** lipophilicity: XLOGP3 between –0.7 and +5.0 (suggesting optimal balance between hydrophilicity and lipophilicity for absorption and distribution).
3. **Solubility:** log *S* not higher than 6
4. **TPSA:** 20–130 Å² (values outside this range can affect permeability across biological membranes).
5. **saturation:** fraction of carbons in the sp³ hybridization not less than 0.25
6. **Flexibility:** no more than 9 rotatable bonds.

7. **Molar refractivity:** should be 40 to 130 for good absorption and bioavailability
8. Skin permeation: 0 to 10

BOILED Egg:

The BOILED-Egg, an intuitive method to predict simultaneously two key ADME parameters, i.e. the passive gastrointestinal absorption (HIA) and brain access (BBB). Although conceptually very simple as it relies on two physicochemical descriptors only (WLOGP and TPSA, for lipophilicity and apparent polarity), this classification model was built with extreme care regarding statistical significance and robustness. The egg-shaped classification plot includes the yolk (i.e. the physicochemical space for highly probable BBB permeation) and the white (i.e. the physicochemical space for highly probable HIA absorption). Both compartments are not mutually exclusive and the outside **Grey** region stands for molecules with properties implying predicted low absorption and limited brain penetration. Whereas the predictive power of the BOILED-Egg is broad in term of chemical space, it is restricted to passive penetration through gastro-intestinal wall and BBB. We took benefit of its implementation within SwissADME to enrich the graphical output with the prediction of P-gp substrate, which is the most important active efflux mechanism involved in those biological barriers. As a result, the user conveniently obtains on the same graph a global evaluation about passive absorption (inside/outside the white), passive brain access (inside/outside the yolk) and active efflux from the CNS or to the gastrointestinal lumen by colour-coding: **BLUE** dots for P-gp substrates (PGP+) and **RED** dots for P-gp non-substrate (PGP-).

How to Use Swiss ADME:

1. **Input Compound:** You can input a compound structure using SMILES (Simplified Molecular Input Line Entry System) notation, a file (like SDF or MOL), or even the compound's name.
2. **Generate Results:** The tool then computes and displays the ADME-related properties, often with a graphical representation and key data points for easy interpretation.
3. **Interpret the Data:** Based on the calculated properties, users can make decisions about the compound's potential as a drug candidate.

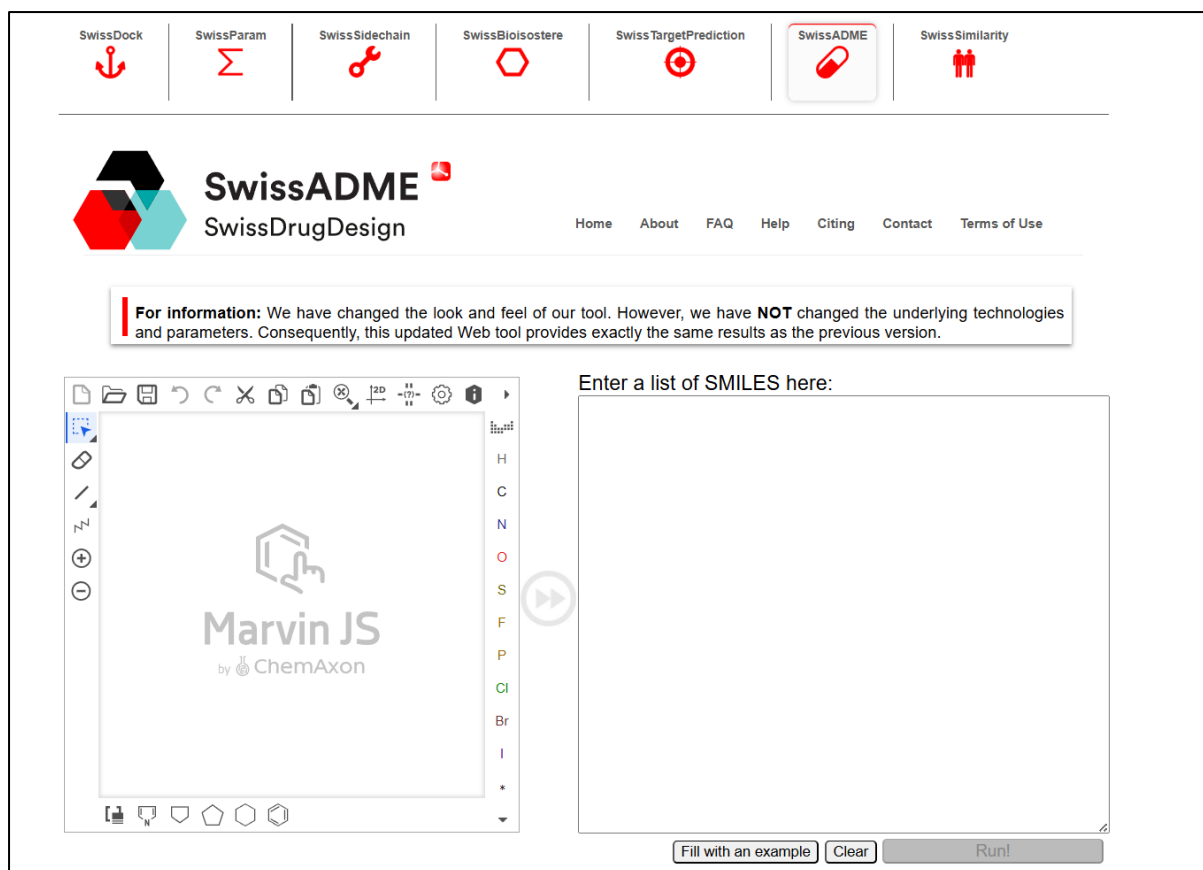


Figure 1: Homepage of SwissADME

REFERENCES:

1. Daina, A., Michielin, O. & Zoete, V. SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Sci Rep* 7, 42717 (2017). <https://doi.org/10.1038/srep42717>
2. SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Sci. Rep.* (2017) 7:42717. <https://www.nature.com/articles/srep42717>

Date: 27/02/2025

WEBLEM:5(A)
SwissADME Server
(URL: <http://www.swissadme.ch/>)

AIM :

To predict pharmacokinetics, drug-likeness for query Lapatinib (PubChem CID: 208908) using SwissADME.

INTRODUCTION:

SwissADME is a comprehensive, free-to-use web tool designed to assist in the early stages of drug discovery by predicting key pharmacokinetic properties, drug-likeness, and physicochemical characteristics of small molecules. Developed by the SIB Swiss Institute of Bioinformatics, the tool integrates multiple advanced, robust predictive models, including BOILED-Egg, iLOGP, and Bioavailability Radar, to provide valuable insights for optimizing molecular designs.

In drug development, assessing a molecule's absorption, distribution, metabolism, and excretion (ADME) properties is crucial for identifying compounds that have the potential to become effective, safe medicines. Traditionally, these properties are evaluated through experimental methods, but with the large volume of molecules being assessed, computational models offer a more efficient alternative. SwissADME is designed to streamline this process by enabling researchers to quickly generate and interpret predictions, even without specialized knowledge in cheminformatics.

The tool's user-friendly interface allows for easy input and fast results, making it accessible to both experts and non-experts. Users can predict a variety of key parameters for individual molecules or large collections and visualize the results through interactive graphs. SwissADME also offers unique features, such as the ability to compute multiple properties simultaneously and access to specialized algorithms and methods that are not typically available in other free tools.

Lapatinib

Lapatinib, known by the brand name Tykerb, is a dual tyrosine kinase inhibitor targeting the epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (HER2). It is used to treat HER2-positive advanced or metastatic breast cancer, often in combination with other medications like capecitabine or letrozole. Lapatinib works by inhibiting tumor cell growth through the blockade of EGFR and HER2 signaling pathways. Common side effects include rash and gastrointestinal issues. It is administered orally and requires caution with certain drug interactions, particularly with strong CYP3A4 inhibitors.

METHODOLOGY:

1. Search for your query in the PubChem database.
2. Retrieve canonical SMILES of the best match.
3. Open the SwissADME web tool at <http://www.swissadme.ch/>
4. Input the molecular structure in SMILES format.

5. Click on the "Run" or "Submit" button to initiate the analysis.
6. Review the results, including key predictions like physicochemical properties, solubility, permeability, pharmacokinetics, and drug-likeness.
7. Analyze the output, including visual representations like the Bioavailability Radar and BOILED-Egg plot for absorption and brain penetration predictions.

OBSERVATIONS:

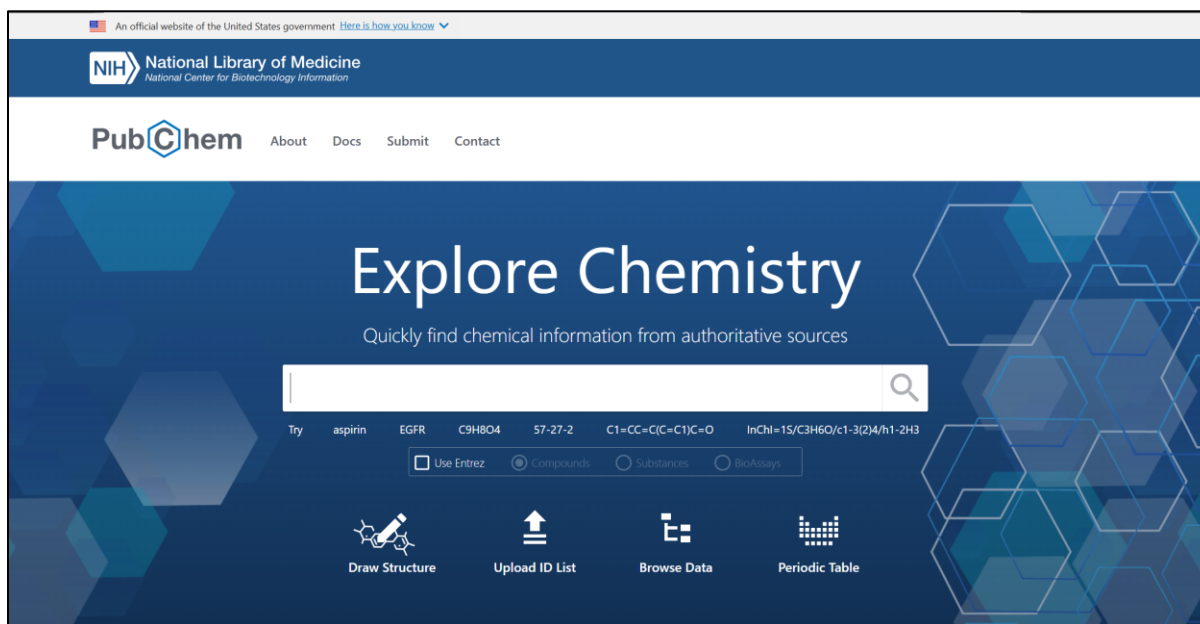


Figure 1: Homepage for PubChem

An official website of the United States government [Here is how you know](#)

NIH National Library of Medicine
National Center for Biotechnology Information

PubChem About Docs Submit Contact

SEARCH FOR
Lapatinib

Treating this as a text search.

BEST MATCH

Lapatinib; 231277-92-2; Tykerb; GW572016; Lapatinib [INN]; GW 572016; Lapatinib (INN); Tyverb; ...

 Compound CID: 208908
MF: C₂₉H₂₆ClFN₄O₄S MW: 581.1g/mol
IUPAC Name: N-[3-chloro-4-[(3-fluorophenyl)methoxy]phenyl]-6-[5-[(2-methylsulfonyl)ethylamino)methyl]furan-2-yl]quinazolin-4-amine
SMILES: CS(=O)(=O)CCNCC1=CC=C(O1)C2=CC3=C(C=C2)N=CN=C3NC4=CC(=C(C=C4)OCC5=CC(=CC=C5)F)Cl
InChIKey: BCFGMOOMADDAQU-UHFFFAOYSA-N
InChI: InChI=1S/C29H26ClFN4O4S/c1-40(36,37)12-11-32-16-23-7-10-27(39-23)20-5-8-26-24(14-20)29(34-18-33-26)35-22-6-9-28(25(30)15-22)38-17-19-3-2-4-21(31)13-19/h2-10,13-15,18,32H,11-12,16-17H2,1H3,(H,33,34,35)
Create Date: 2005-06-24

[Summary](#) [Similar Structures Search](#) [Related Records](#) [PubMed \(MeSH Keyword\)](#)

Figure 2: Best Match for the Query- Metformin (CID:4091)

PubChem Lapatinib (Compound)

2.1.3 InChIKey

BCFGMOOMADDAQU-UHFFFAOYSA-N
Computed by InChI 1.0.6 (PubChem release 2021.10.14)
[PubChem](#)

2.1.4 SMILES

CS(=O)(=O)CCNCC1=CC=C(O1)C2=CC3=C(C=C2)N=CN=C3NC4=CC(=C(C=C4)OCC5=CC(=CC=C5)F)Cl
Computed by OEChem 2.3.0 (PubChem release 2024.12.12)
[PubChem](#)

2.2 Molecular Formula

C₂₉H₂₆ClFN₄O₄S
Computed by PubChem 2.2 (PubChem release 2021.10.14)
[PubChem](#)

2.3 Other Identifiers

2.3.1 CAS

231277-92-2
[CAS Common Chemistry](#); [ChemIDplus](#); [DrugBank](#); [EPA DSSTox](#); [European Chemicals Agency \(ECHA\)](#); [FDA Global Substance Registrati...](#)

CONTENTS

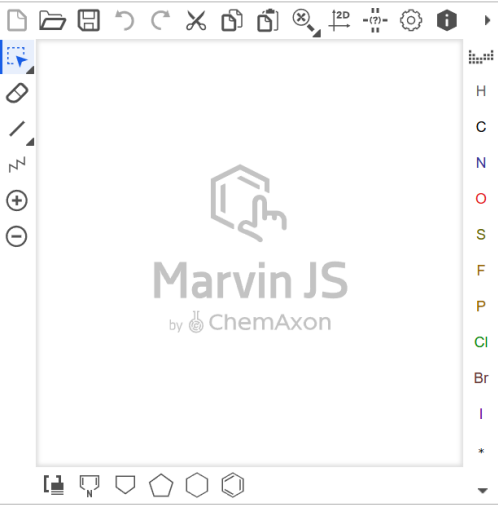
- Title and Summary
- 1 Structures
- 2 Names and Identifiers**
- 3 Chemical and Physical Properties
- 4 Spectral Information
- 5 Related Records
- 6 Chemical Vendors
- 7 Drug and Medication Information
- 8 Pharmacology and Biochemistry
- 9 Use and Manufacturing
- 10 Identification
- 11 Safety and Hazards
- 12 Toxicity
- 13 Associated Disorders and Diseases
- 14 Literature
- 15 Patents
- 16 Interactions and Pathways
- 17 Biological Test Results
- 18 Classification
- 19 Information Sources

Figure 3: SMILES format of the query

SwissADME
SwissDrugDesign

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For information: We have changed the look and feel of our tool. However, we have **NOT** changed the underlying technologies and parameters. Consequently, this updated Web tool provides exactly the same results as the previous version.



Enter a list of SMILES here:

1=CC=C(O1)C2=CC3=C(C=C2)N=CN=C3NC4=CC(=C(C=C4)OCC5=CC(=CC=C5)F)C1

Fill with an example Clear Run!

Figure 5: Input canonical SMILES list of the query

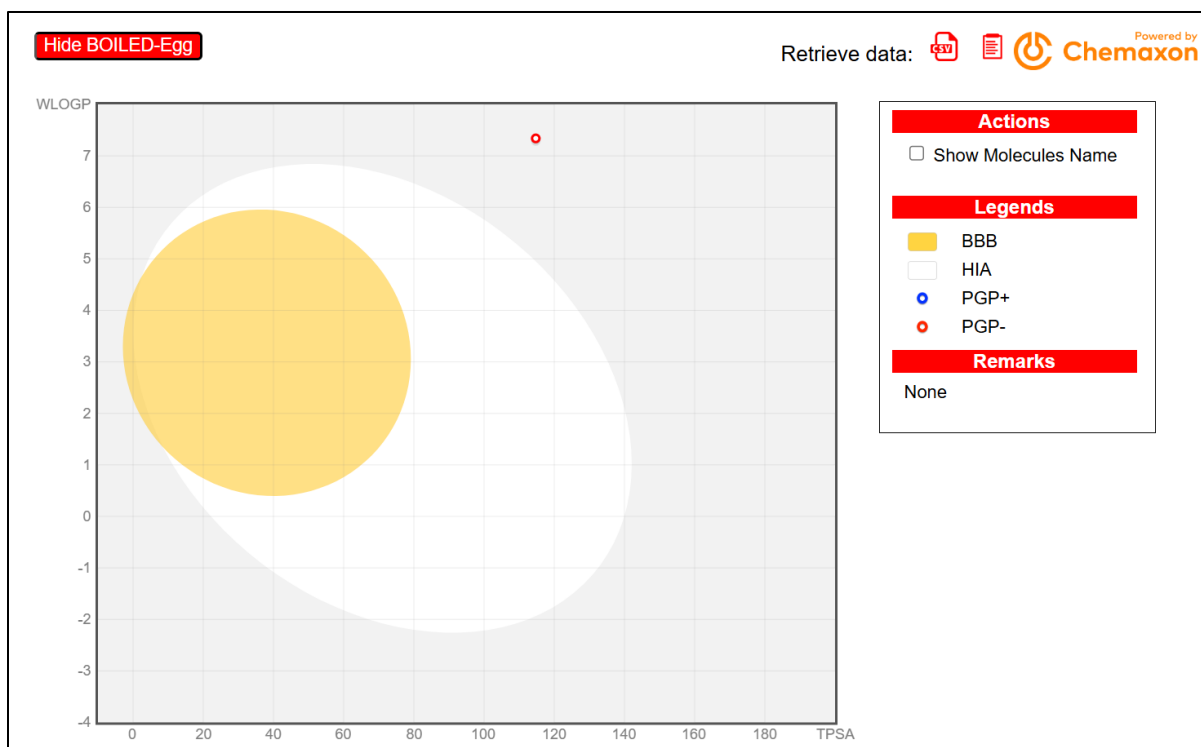


Figure 6: BOILED-Egg plot for the query Metformin

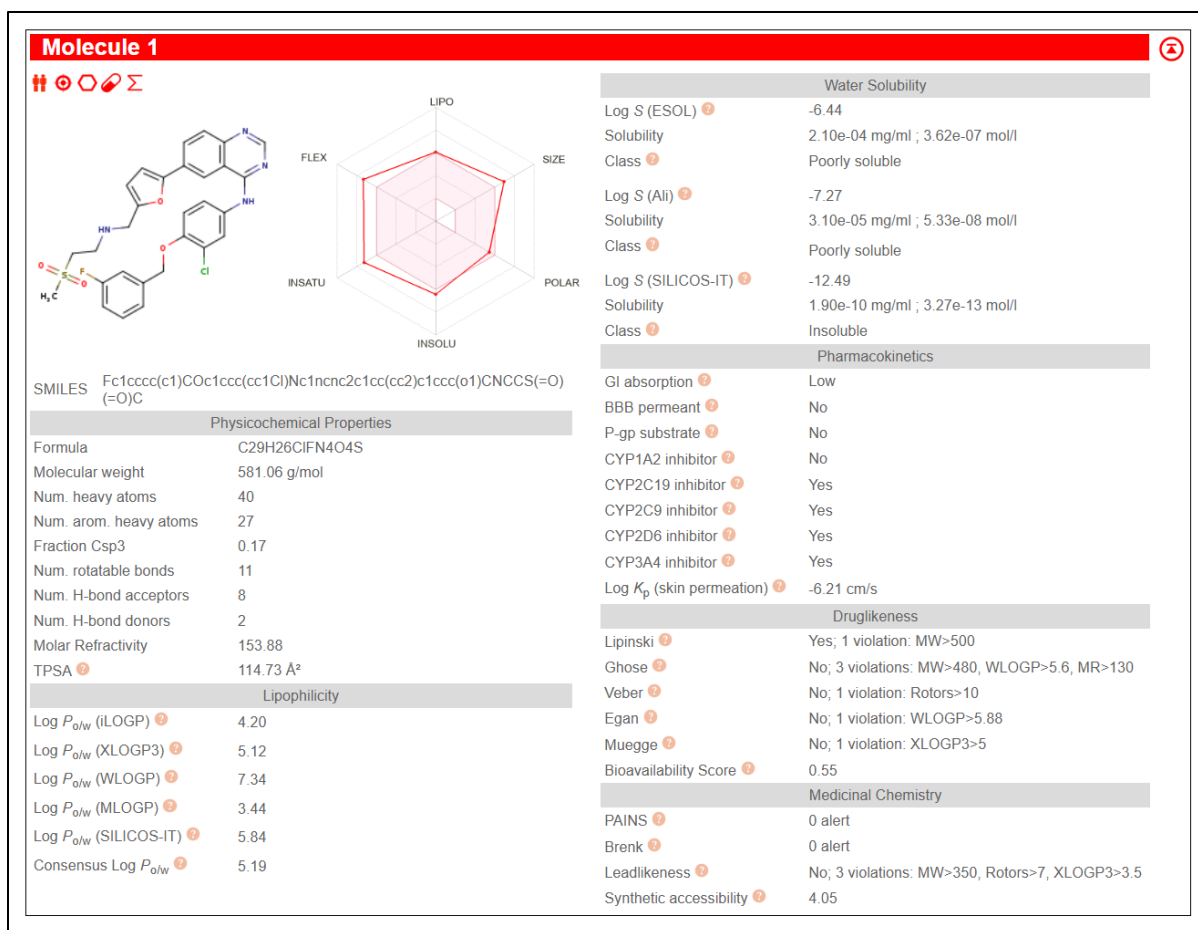


Figure 7: SwissADME Output Panel for the query Lapatinib (Displays 2D structure, SMILES notation, physicochemical properties, lipophilicity, solubility, pharmacokinetics, drug-likeness, and medicinal chemistry evaluation.)

RESULTS:

To evaluate pharmacokinetics and drug-likeness, SwissADME server Lapatinib (PubChem CID-208908) molecule was used. SwissADME output file comprises various sections. The two dimensional chemical structure with canonical SMILES were shown in the first section. The first section shows the bioavailability radar which empowers preliminary glimpse at the drug likeness of the molecules where the pink area represents the optimal range for each property. The query Lapatinib molecule is predicted safe for oral bioavailability.

Second Section compiles simple molecular and physicochemical descriptors such as molecular formula, molecular weight, number of heavy atoms, number of aromatic heavy atoms, fraction csp³, number of rotatable bonds, number of H-bond acceptors, number of H-bond donors, molar refractivity, TPSA respectively. The values drawn for the query molecule are all in acceptable range except including Fraction csp³ (solubility) score.

Third section is Lipophilicity where the partition coefficient between n-octanol and water (log *P*_{o/w}) is calculated and it's a paramount parameter in drug discovery and design. Many

computational methods for log Po/w estimation were developed with diverse performance on various chemical sets which increase the prediction accuracy. Out of all these, iLOGP was considered as a benchmark to compute the drug-like out of all other methods. The LogP value should be <5 and ideally between 1.35-1.8 for good oral and intestinal absorption and the available value for query is 0.34 which is considered this molecule good for oral bioavailability.

Next Section is Water Solubility where a soluble molecule greatly facilitates many drug development activities, primarily the ease of handling and formulation. A drug meant for parenteral usage has to be highly soluble in water to deliver a sufficient quantity of active ingredient in the small volume of such pharmaceutical dosage. Two topological methods to predict Water Solubility are included in SwissADME. The first one is an implementation of the ESOL model (value is 0.29, drug is highly soluble) and the second one is adapted from Ali et al (-0.51, drug is very soluble) where both the method differ from each other w.r.t to general solubility equation. The third method is SILICOS-IT (0.58, drug is soluble) which is developed by SWISSADME. All the values estimated by model tells us that the drug is very soluble.

Fifth Section describe about Pharmacokinetics which evaluate individual ADME behaviors of the molecule under investigation. The predictions for passive human gastrointestinal absorption (HIA) and blood-brain barrier (BBB) permeation both consist in the readout of the BOILED-Egg model which is an intuitive graphical classification model and displayed on the SwissADME result page. The knowledge about compounds being substrate or non-substrate of the permeability glycoprotein (P-gp) is key to appraise active efflux through biological membranes, i.e. from the gastrointestinal wall to the lumen or from the brain. It is also important to know the interaction of molecules with cytochromes P450 (CYP) because CYP and P-gp can process small molecules synergistically to improve protection of tissues and organisms and thus various models are being studied by SWISSADME. The results for the query observed in the form of "YES" or "NO" indicate whether the molecule under investigation has higher probability to be substrate or non-substrate of P-gp.

Sixth section is Drug likeness which assesses qualitatively the chance for a molecule to become an oral drug with respect to bioavailability. Drug-likeness was established from structural or physicochemical inspections of development compounds advanced enough to be considered as oral drug-candidates. This SwissADME section gives access to five different rule-based filters such as Lipinski, Ghose, Veber, Egan and Muegge with diverse ranges of properties inside of which the molecule is defined as drug-like. Also bioavailability score is similar but seeks to predict the probability of a compound to have at least 10% oral bioavailability in rat. In spite of all these filters, Lipinski filter is considered as one of the key parameter in order to access the drug likeness and thus it states that the query molecule doesn't violate any rule, the drug can be considered for oral bioavailability.

Seventh section is Medicinal Chemistry which serves the purpose of supporting medicinal chemists in their daily drug discovery endeavors. Two pattern recognition methods called PAINS & Brenk are implemented for structural alertness and SwissADME returns warnings if any moieties/molecule under evaluation to be considered as promiscuous compounds or not. For query molecules there is 0 alert which indicates that drug can be considered ahead for study. Also the

parameter leadlikeness is similar to drug-likeness, yet focusing on physicochemical boundaries defining a good lead, i.e. a molecular entity suitable for optimization where there is violation of rule for molecular weight and hence it helps chemist to judge whether a given molecule is suitable to initiate lead optimization, in addition to structural filters. Synthetic accessibility (SA) is a major factor to consider in this selection process of drug where SA Score ranges from 1 (very easy to synthesize) to 10 (very difficult for synthesis) and helps in prioritizing molecules to synthesize.

BOILED EGG:

BOILED-Egg, which is a graphical output which is an intuitive method to predict simultaneously two key ADME parameters, i.e. the passive gastrointestinal absorption (HIA) and brain access (BBB). The egg-shaped classification plot includes the yolk (i.e. the physicochemical space for highly probable BBB permeation) and the white (i.e. the physicochemical space for highly probable HIA absorption). Both compartments are not mutually exclusive and the outside grey region stands for molecules with properties implying predicted low absorption and limited brain penetration. In addition the points are coloured in blue if predicted as actively effluxed by P-gp (PGP+) and in red if predicted as non-substrate of P-gp (PGP-). Cetirizine is predicted as brain-penetrant (in the yolk) and subject to active efflux (blue dot).

CONCLUSION:

SwissADME was utilized to compute key physicochemical, pharmacokinetic, and drug-like parameters for a set of molecules. It provides valuable insights into molecular properties such as solubility, permeability, and potential toxicity, which are crucial for drug development. Its detailed outputs help streamline the selection of lead compounds for further research and optimization.

REFERENCES:

1. Daina, A., Michielin, O., & Zoete, V. (2017). SwissADME: a free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules. *Scientific Reports*, 7(1). <https://doi.org/10.1038/srep42717>
 2. Opdam, F. L., Guchelaar, H. J., Beijnen, J. H., & Schellens, J. H. (2012). Lapatinib for advanced or metastatic breast cancer. *The oncologist*, 17(4), 536–542. <https://doi.org/10.1634/theoncologist.2011-0461>
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